Basic Principles of Urology
Rapid Revision

Campbell-Walsh-Wein · Chapters 1–23

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 8h 30min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 17* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
1	Personal & Professional Development of the Urologist	12 min
2	Evaluation of the Urologic Patient	26 min
3	Laboratory Testing	24 min
4	Urologic Ultrasonography	26 min
5	CT & Radiographic Imaging	30 min
6	Magnetic Resonance Imaging	16 min
7	Nuclear Medicine	16 min
8	Haematuria	27 min
9	Occupational Health in Urology	12 min
10	Ethics & Informed Consent	8 min
11	Perioperative Care	25 min
12	Intraoperative Patient Safety	12 min
13	Incisions, Exposure & Drains	11 min
14	Sutures & Suturing	9 min
15	Lower Urinary Tract Catheterisation	11 min
16	Upper Urinary Tract Drainage	19 min
17	Principles of Urologic Endoscopy	39 min
18	Laparoscopic & Robotic Surgery Fundamentals	35 min
19	Energy Modalities in Urology	54 min
20	Complications of Urologic Surgery	28 min
21	Urologic Considerations in Pregnancy	27 min
22	Intraoperative Consultations	15 min
23	Assessment of Urologic & Surgical Outcomes	22 min
★	Putting it together: everyday decisions	6 min

Chapter	Topic	Slow reader time
1	Personal and Professional Development of the Urologist	
	Total (35718 words, at 70 words/min)	8h 30min

Reading time ≈ 12 min (slow pace)

1 · WHY PROFESSIONALISM IS CORE CONTENT

- Professional attitudes are now taught alongside knowledge and skills; development is **lifelong** and needs ongoing reflection.
- **Unprofessional behaviour as a medical student** is one of the strongest predictors of later disciplinary action by medical boards.

2 · PERSONAL DEVELOPMENT AND THE CAREER CYCLE

- **Stimulus → reflex reaction → chosen response**: the reaction may be hard-wired, but the response is under personal control — this is what allows deliberate development.
- Method: define core values and goals → prioritise activities serving them → build habits → review periodically.
- **Cellular model of life**: life is a cell; its cycles (career, physical, mental, social) are organelles that affect each other. Overinvesting in the career starves the rest.

Phase	Timing	Priorities	Hazard
Early career	≈ 10–12 yr	Say yes; mentors; reputation; learning curves	Workload; burnout possibly highest
Mid career	Mid-40s to mid-50s	Peak surgical performance; weigh each opportunity against time	Burnout; selective acceptance protects
Late career	≈ 10–12 yr	Leadership, teaching; hobbies	—
Retirement	Life cycle outlives career	Transition planning	Loss of identity and purpose if career = life

> **KEY POINT** Mentorship and peer groups at every stage add collegiality and guard against isolation and burnout.

3 · BURNOUT

- **Definition**: emotional exhaustion, **depersonalisation** and diminished personal accomplishment, in response to chronic interpersonal stressors. Distinct from depression, fatigue or job dissatisfaction.
- Prevalence ≈ **40%** of physicians; urologists approach **50%**; rising, especially in women and those <45.
- **Physician suicide**: ≈ one US physician/day; relative rate **1.4× men, 2.7× women**; fear of losing the licence blocks help-seeking.
- Effects: more medical error, lower-quality care.
- **Contributors**: loss of autonomy, purpose and community; electronic health record (EHR) time; workload; administrative burden. When these block good care → **moral injury**.
- **20% rule**: <**20%** of effort on the most meaningful activity → ≈ 3× burnout risk.

Prevention strategy	Effect
Institutional awareness seminars	Minimally effective
Practice interventions (scribes, removing frustrations)	More effective
Restoring autonomy over schedule and environment	Significantly reduces burnout
Mindfulness-based stress reduction, communication training, exercise	Evidence-based reduction

> **KEY POINT** Interventions that restore autonomy, purpose or community work; awareness alone does not.

4 · MORALE, CHANGE AND COMPLICATIONS

- **Morale**: reconnect daily work to service; a "routine" visit may be the most important event in a patient's week.
- **Broaden the portfolio** before a crisis: resilience options exist only if mental, physical and social cycles were already built. A new habit forms in ≈ **3 weeks**.
- **Complications**: "the only way to avoid them is to stop operating". Most arise from avoidable events within a **system**, not one person. Learnable **self-compassion** converts them into growth.

5 · FOUR DOMAINS OF LEARNING

Domain	Framework	Assessment
Knowledge	Bloom's taxonomy	Easiest; certification exams
Psychomotor	Stages of technical skill	Courses, proctorship, fellowships, simulation
Communication	Team Strategies and Tools to Enhance Performance and Patient Safety (TeamSTEPPS)	Often untaught
Affective	Krathwohl's taxonomy	Formative assessment

- **Bloom (2001)**, low → high: remember, understand, apply | analyse, evaluate, **create**.
- **Psychomotor**: proctorship often needed for credentialing (robotic, laser). Simulation for acquisition, maintenance, warm-up.

Stage	Level	Element converted
Imitation	Novice	—
Manipulation	Competent	Recollection → situational
Precision	Proficient	Recognition → holistic
Articulation	Expert	Decision → intuitive
Naturalisation	Master	Awareness → absorbed

- **Communication** errors contribute to ≈ **70%** of medical errors and 27% of malpractice claims. TeamSTEPPS trains it as a structured team skill.
- Tools: **brief** (before), **huddle** (regroup), **debrief** (after); **check-back** (closing the loop); **SBAR** = Situation, Background, Assessment, Recommendation; **CUS** = Concerned, Uncomfortable, Safety issue; **two-challenge rule** = voice a concern at least twice. Situation awareness = individual; shared mental model = team.
- **Krathwohl**, ascending: receiving → responding → valuing → organisation → characterisation by a value set.

> **KEY POINT** Certification tests the easiest domain (knowledge), but harm arises mainly in communication, affective and psychomotor domains — hence deliberate training there.

6 · RESPONSIBILITY AND ORGANISATIONAL SUPPORT

- Hospitals credential; boards assess mainly knowledge; peer review and surgical logs cover other domains. Only the **individual urologist** is accountable across all domains.
- **Patient Advocacy Reporting System (PARS)**: patient complaints → trained peer messengers → less malpractice exposure. **Coworker Reporting System (CORS)**: co-worker reports → most individuals never reported again.

- Surgeons in the highest quartile of patient complaints have **more surgical complications** — unprofessional behaviour is a measurable safety signal.

7 · MCQ ONE-LINERS

1. Strongest predictor of later board discipline: unprofessional behaviour in medical school.
2. Burnout triad: emotional exhaustion, depersonalisation, low personal accomplishment.
3. Physician suicide relative to population: 1.4× men, 2.7× women.
4. <20% effort on meaningful work → ≈ 3× burnout.
5. Awareness seminars alone are minimally effective against burnout.
6. Restoring autonomy significantly reduces burnout.
7. Habit formation ≈ 3 weeks.
8. Bloom's apex: creating.
9. Knowledge domain is easiest to assess.
0. Affective domain taxonomy: Krathwohl.
11. Highest Krathwohl level: characterisation by a value set.
12. Master stage: naturalisation — awareness absorbed.
13. Expert is intuitive but still monitoring.
14. Communication errors ≈ 70% of medical errors.
15. CUS: Concerned, Uncomfortable, Safety issue.
16. Two-challenge rule: voice concern at least twice.
17. Shared mental model = team outcome of situation monitoring.
18. Robotic credentialing often needs proctorship.
19. PARS uses patient complaints; CORS uses co-worker reports.
20. Highest complaint quartile surgeons have more complications.

2 Evaluation of the Urologic Patient: History and Physical Examination

Reading time ≈ 26 min (slow pace)

1 · PRINCIPLES AND SETTING

- History and examination are the **primary diagnostic test**; laboratory and imaging tests are chosen on their findings.
- **Confirmation bias**: relying on prior records without examining the patient misses findings (penile tumour in a "haematuria" referral).
- Sitting (not standing) improves satisfaction and reduces litigation. **Teach-back**: patient restates information in own words.
- Always address the **chief complaint**, even when something more serious (e.g. incidental renal mass) is found.

2 · PAIN

- **Pain vs tenderness**: pain is reported, tenderness elicited. Pain without tenderness = **referred** pain — a non-tender painful testis may reflect an ipsilateral ureteric stone, not orchitis.

Feature	Obstructive (colic)	Parenchymal
Cause	Distension of hollow organ (pelviureteric junction stone)	Capsular stretch — inflammation, infection, subcapsular bleed (pyelonephritis)
Pattern	Intermittent	Constant; costovertebral angle tenderness (CVAT)
Behaviour	Restless, seeks position	Lies still

- **Renal pain**: costovertebral angle (below 12th rib, lateral to spine); radiates to flank, ipsilateral scrotum/labium. Nausea/vomiting because the **coeliac plexus** supplies kidney and foregut.
- **Ureteric colic mechanism**: distension and hyperperistalsis → prostaglandins → spasm and lactic acid → A-delta/C fibres → **T11–L1** dorsal root ganglia. Hence non-steroidal anti-inflammatory drugs (NSAIDs), which block prostaglandin synthesis, work. Gradual obstruction may be painless. Ureterovesical junction stone → urgency, frequency.
- **Vesical**: suprapubic; inflammatory pain worst when full, eased by voiding. **Strangury** = stabbing pain at end of micturition (inflamed detrusor). Acute retention painful; atonic bladder retains large volumes painlessly.
- **Penile** pain: flaccid — balanitis, paraphimosis, ulcer, referred cystitis; erect — Peyronie disease, priapism.
- **Scrotal**: epididymitis eased by elevation; varicocele ache worse by evening, eased supine.
- Chronic pain without urological cause (central sensitisation) → pain specialist, no opioids.

3 · HAEMATURIA

- Up to **one-quarter** have urological cancer. Commonest cause of gross haematuria over 50 years: **bladder cancer**.
- Risk factors: age, smoking, industrial chemicals, irritative symptoms, alkylating chemotherapy, analgesic abuse, chronic foreign body.
- Exclude urinary tract infection (UTI), nephrological disease, trauma first. Confirm red blood cells (RBCs) by microscopy — dipstick alone is insufficient (pseudohaematuria).
- **Microscopic haematuria** (American Urological Association, AUA): **≥3 RBC/high-power field (HPF)** on a properly collected midstream specimen. Patients on anticoagulants/antiplatelets are evaluated the same way.

Clue	Suggests
Initial haematuria	Urethra/prostate
Total	Bladder or above
Terminal	Bladder neck (squeezed at end of void)
Vermiform clots / cuboid clots	Upper tract / bladder
Clears with irrigation but recurs / clears once catheter placed	Upper tract / prostate

AUA risk group (microhaematuria)	Evaluation
Low	Shared decision: repeat urinalysis (positive → intermediate) or cystoscopy + renal ultrasound (US)
Intermediate	Cystoscopy + renal US
High	Cystoscopy + multiphasic computed tomography (CT) urogram

> **KEY POINT** Cystoscopy is constant for anyone investigated; only the intensity of upper-tract imaging escalates with risk. → *Ch 3*

4 · LOWER URINARY TRACT SYMPTOMS (LUTS)

- No single post-void residual (PVR) defines retention — judge PVR trend with bother and sequelae (stones, infection, renal injury).
- Chronic bladder outlet obstruction (BOO) itself causes detrusor irritability. A weak stream may be **outlet or detrusor** (diabetic hypocontractility) failure.
- **Bladder diary** (usually 48 h): functional capacity (normal **300–400 mL**), nocturia, polyuria.
- **Frequency** >5–6 voids/day. Irritative symptoms may be the only sign of **carcinoma in situ (CIS)** — low threshold for cystoscopy in smokers over 50.
- **Nocturia**: daytime frequency without nocturia → psychogenic; nocturia without daytime frequency → **nocturnal polyuria** (evening fluids, oedema mobilised supine, age-related loss of concentration).
- Obstruction: benign prostatic hyperplasia (BPH), cancer, stricture, dyssynergia, meatal stenosis. Post-void dribble is often an early BPH symptom.

International Prostate Symptom Score (IPSS)

- AUA Symptom Score: **7 items**, each 0–5 — incomplete emptying, frequency, intermittency, urgency, weak stream, straining, nocturia. IPSS = AUA score + quality-of-life (QoL) question (0 delighted → 6 terrible), outside the total.
- Total 0–35: **mild 0–7, moderate 8–19, severe 20–35**. Non-specific; value is before/after comparison.

5 · URINARY INCONTINENCE

Type	Mechanism	Hallmarks
Continuous	Outlet bypassed — fistula, ectopic ureter	Constant wetness; prior pelvic surgery, radiation
Overflow	Volume exceeds capacity	Often nocturnal; long-standing BOO; cured by relieving obstruction
Stress (SUI)	Sphincter weakness	Cough/sneeze; multiparity; after radical prostatectomy or transurethral resection of the prostate (TURP)
Urge	Leak with urgency	Overactive bladder, cystitis, neurogenic bladder, occult cancer
Mixed	Stress + urge	Treating one alone may worsen symptoms
Functional	Intact outlet; cannot reach toilet	Mobility, cognition

- **Ectopic ureter**: dysplastic upper-pole moiety of a duplex kidney opens beyond the sphincter (vagina/perineum) → girl wet continuously by day despite normal voiding; less at night because urine pools in the saccular upper pole when supine.
- SUI → raise outlet resistance (pelvic floor exercises, sling, sphincter). Urge → relax bladder (anticholinergics, botulinum toxin).
- **Enuresis**: normal to 3 years; ~15% at 5 y, ~1% at 15 y. Primary enuresis beyond 6 years → urological evaluation. Secondary enuresis → consider abuse or bullying.

6 · SEXUAL AND OTHER SYMPTOMS

- **Erectile dysfunction (ED)** (National Institutes of Health 1993): inability to attain/maintain erection sufficient for satisfactory intercourse. History types it: vasculogenic, neurogenic, psychogenic, endocrine, drug-induced. Situational ED with preserved nocturnal erections → psychogenic. **ED heralds occult coronary artery disease** → cardiology.
- **Low libido + ED** → early-morning testosterone on **two occasions ≥2 weeks apart**.
- **Premature ejaculation**: AUA/Sexual Medicine Society — within ~**2 min**; International Society for Sexual Medicine — lifelong within ~1 min, acquired ≤~3 min; plus inability to delay and distress. First-line: topical anaesthetic, stop–start, counselling, or selective serotonin reuptake inhibitor.
- **Anejaculation**: androgen deficiency, retroperitoneal surgery, tamsulosin, prostate surgery, midline cyst.
- **Haematospermia**: usually benign, self-limiting; persistent → digital rectal examination (DRE), prostate-specific antigen (PSA), cystoscopy, cytology (tuberculosis, cancer).
- **Pneumaturia**: commonest cause **enterovesical fistula** (diverticulitis, Crohn disease); rarely gas-forming infection.
- Bloody urethral discharge → consider urethral carcinoma.
- **Bother decides** treatment in benign disease — unless objective sequelae (renal injury, UTI, stones) override it.

7 · MEDICAL, SURGICAL, DRUG AND SOCIAL HISTORY

- Record comorbidity **severity and control**; hunt occult coronary disease and sleep apnoea.
- **Performance status**: activities of daily living (ADLs) and instrumental ADLs. ADL deficiency predicted complications after percutaneous nephrolithotomy (PCNL) better than Charlson index or American Society of Anesthesiologists (ASA) score.
- Prior abdominal surgery → Hasson (open) laparoscopic entry. **Read the operative note**: a "TURP" may have been a channel resection only.

Side effect	Drugs
ED	Thiazides, propranolol
Ejaculatory dysfunction	α-blockers, methyldopa, phenothiazines
Priapism	**Trazodone**, phenothiazines, hydralazine, prazosin
Retention	Anticholinergics, calcium-channel blockers, α-agonists (pseudoephedrine), antihistamines
Reduced spermatogenesis	Alkylating agents, marijuana
Gynaecomastia	Digoxin, cimetidine, verapamil

> **KEY POINT** Retention comes from either side of the balance: α-agonists raise outlet tone; anticholinergics, calcium-channel blockers and antihistamines weaken the detrusor. α-blockade links ejaculatory dysfunction and priapism.

Antithrombotic	Action	Half-life
Warfarin	Vitamin K antagonist	36–42 h
Dabigatran	Direct thrombin inhibitor	8–17 h
Rivaroxaban / apixaban	Factor Xa inhibitors	5–12 h / 12 h
Clopidogrel / ticagrelor	P2Y12 blockade, irreversible / reversible	6 h / 6–8 h

- Half-life ≠ duration: clopidogrel acts for the **life of the platelet**.
- Procedural bleeding risk — **low**: cystoscopy, ureteroscopy, laser lithotripsy; **intermediate**: TURP, transurethral resection of bladder tumour (TURBT), robotic pyeloplasty/reimplantation; **high**: PCNL, artificial urinary sphincter, robotic nephrectomy. Decide with the cardiologist; coronary stent vs primary prevention changes it.
- **Contrast allergy** → US or magnetic resonance imaging (MRI), or premedicate (corticosteroid + antihistamine). Shellfish allergy is **not** contrast cross-reactivity. Retrograde/antegrade pyelography is safe even after prior contrast anaphylaxis.
- **Family**: polycystic kidney disease, tuberous sclerosis, von Hippel–Lindau, renal tubular acidosis, cystinuria; clustering in prostate cancer and stones — risk rises with **number** of affected relatives.

8 · PHYSICAL EXAMINATION — GENERAL

- Offer chaperone; abnormal vital signs → immediate triage.
- Stigmata: buffalo hump (Cushing), fibrofolliculomas (**Birt–Hogg–Dubé** → renal tumours).
- **Frailty**: gait speed, grip strength; low albumin/prealbumin predict poor surgical outcome.

9 · KIDNEY, BLADDER, PROSTATE

- **Kidney**: rarely palpable; right lower (liver). Bimanual ballottement: posterior hand at costovertebral angle, anterior hand palpates on deep inspiration. Neonates: thumb anterior, fingers posterior; **neonatal flank mass is usually renal**. Transillumination: cyst/hydronephrosis glows; solid does not. Flank hyperaesthesia before vesicles → **herpes zoster**.
- **Bladder**: palpable above **~150 mL**, visible ~500 mL; percuss upward from symphysis. **Bimanual examination under anaesthesia after TURBT** is standard of care — independently predicts pT3 even in the CT/MRI era.
- **DRE**: bent forward ~90° or lateral decubitus; inspect anus; gentle lubricated finger. Normal prostate chestnut-sized, rubbery like the **contracted thenar eminence**; hard nodule like a **flexed knuckle** → biopsy.

10 · MALE GENITAL EXAMINATION

- **Penis**: retract foreskin (most penile cancers on prepuce/glans). Ulcer → infection or squamous carcinoma. Meatus: hypospadias (ventral; record site, calibre), epispadias (dorsal, with exstrophy). Dorsal fascial plaque → Peyronie disease.
- **Scrotal skin**: **Fournier gangrene** — black necrotic skin, "dishwater" discharge, crepitus, pain out of proportion.
- **Lie**: normally vertical. Horizontal lie, high testis, short cord + pain → torsion. Absent **cremasteric reflex** highly sensitive for torsion in children; a present reflex argues against but does not exclude it.
- Normal testis **4–5 × 2–3 cm**; small → Klinefelter, hypogonadism.
- **Hard testicular mass = cancer until proven otherwise**; epididymal mass almost always benign. Hydrocele transilluminates.
- Varicocele ("bag of worms"): examine standing with Valsalva. Paediatric hernia: "silk glove" sign.

11 · FEMALE PELVIC EXAMINATION

- Lithotomy → inspect urethra (caruncle, prolapse) → full-bladder cough for SUI → half-speculum per compartment → palpate urethra (mass/expressed pus → **urethral diverticulum**) → bimanual.
- **Q-tip test**: swab to bladder neck, Valsalva; deflection **≥30°** = urethral hypermobility (not required for SUI diagnosis).
- Prolapse: withdraw speculum during Valsalva; document by Pelvic Organ Prolapse Quantification (POP-Q).

12 · NEURO-UROLOGICAL EXAMINATION

- Saddle sensory loss → sacral root lesion.
- **Bulbocavernosus reflex** (**S2–S4**): squeeze glans/clitoris or tug Foley catheter → anal sphincter contraction felt on DRE.
- † Anal tone and voluntary squeeze test the same sacral segments.

> **KEY POINT** S2–S4 serve perineal sensation, the bulbocavernosus reflex and continence together — sacral sensory loss plus absent reflex localise the same lesion (e.g. cauda equina).

13 · SPECIAL POPULATIONS

- **Obesity**: body mass index (BMI) class 3 ≥40 kg/m²; delay elective surgery for weight loss.
- **Transgender**: record gender identity and sex assigned at birth; examine genitalia only for a purpose.
- **Telemedicine**: better access; cannot replace examination, PVR or cystoscopy.

14 · MCQ ONE-LINERS

1. Pain without tenderness suggests referred pain.
2. Ureteric colic is perceived in T11–L1 segments.
3. Strangury: stabbing pain at the end of micturition.
4. Commonest cause of gross haematuria over 50 years: bladder cancer.
5. AUA microhaematuria: ≥3 RBC/HPF on microscopy.
6. Terminal haematuria: bladder neck source.
7. Vermiform clots indicate upper-tract bleeding.
8. High-risk microhaematuria: cystoscopy plus CT urogram.
9. IPSS severe: 20–35; QoL question scored 0–6.
0. Continuous daytime wetting with normal voiding in a girl: ectopic ureter.
1. Primary enuresis persisting after 6 years warrants evaluation.
2. ED may herald occult coronary artery disease.
13. Testosterone: early morning, two samples ≥2 weeks apart.
14. Commonest cause of pneumaturia: enterovesical fistula.
15. Trazodone causes priapism.
16. Clopidogrel effect lasts the platelet lifespan despite 6-h half-life.
17. PCNL is a high-bleeding-risk procedure.
18. Shellfish allergy does not predict iodinated contrast reaction.
19. Bimanual examination after TURBT independently predicts pT3.
20. Absent cremasteric reflex: highly sensitive for torsion in children.
21. Hard testicular mass is cancer until proven otherwise.
22. Bulbocavernosus reflex tests S2–S4.

3 Indications and Interpretation of Laboratory Testing

Reading time ≈ 24 min (slow pace)

1 · URINALYSIS — COLLECTION AND GROSS EXAMINATION

- Complete urinalysis = **gross → dipstick (biochemical) → sediment microscopy**. Microscopy indicated for abnormal dipstick, stones, acute kidney injury (AKI), initial chronic kidney disease (CKD) work-up.
- Midstream clean catch (retract foreskin; separate labia). Examine within **2 h**, else refrigerate at 2–8 °C. Delay → bacterial overgrowth, pH change, lysis of red blood cell (RBC) and white blood cell (WBC) casts; refrigeration precipitates crystals.
- Children: suprapubic aspiration (SPA) has lowest contamination; catheterisation intermediate and often preferred; **bag specimen — never for culture** (chemistry only).
- **Colour**: red not always blood — beetroot, rifampicin, phenothiazines; orange — phenazopyridine; green-blue — methylene blue, amitriptyline, triamterene; brown-black — alkaptonuria, melanin, metronidazole, nitrofurantoin; brown/cola → glomerular disease. Colour prompts microscopy.
- **Cloudy urine**: commonest phosphaturia (alkaline, post-prandial) — **clears with acetic acid**; pyuria does not. Rare: chyluria (lymphatic–urinary fistula).

2 · DIPSTICK CHEMISTRY

- Dip fully, shed excess on rim, hold **horizontally** (vertical → reagents run between pads); outdated strips are a common source of error.

Interferent	False negative	False positive
Ascorbic acid	Glucose, bilirubin, blood, leucocyte esterase	—
Phenazopyridine / dyes	Leucocyte esterase	Nitrite, bilirubin, ketones
Alkaline urine	—	Protein
High specific gravity	Leucocyte esterase, glucose	—
Semen, menses, exercise	—	Blood
Mesna, levodopa	—	Ketones
Contamination	—	Leucocyte esterase and nitrite (commonest)

- **Specific gravity** 1.001–1.035; **fixed at 1.010** in renal insufficiency. Raised by glucose, contrast (>1.035), dehydration, inappropriate antidiuretic hormone; lowered by diuretics, diabetes insipidus. Osmolality 50–1200 mOsm/L.
- **pH** 4.5–8.0 (mean 5.5–6.5); urine follows serum pH except in **renal tubular acidosis (RTA)** — acidaemia with alkaline urine (cannot acidify below **5.5**); risk of calcium phosphate stones. Type I (distal, H^+ secretion defect) — urine always alkaline; type II (proximal, bicarbonate loss) — can acidify when severe.
- pH **>7.5** with infection → urease-splitter (*Proteus*): urea → ammonia → struvite + calcium phosphate → staghorn. Uric acid and cystine stones form in acid urine → alkalinise.
- **Glucose**: renal threshold ≈ serum **180 mg/dL**; glucose-oxidase method is glucose-specific; sodium–glucose cotransporter 2 (SGLT2) inhibitors cause glycosuria. **Ketones** (nitroprusside) detect acetoacetate only — not β-hydroxybutyrate.
- **Bilirubin**: only conjugated appears in urine (water-soluble) — hepatic disease or biliary obstruction. **Urobilinogen** ↑ haemolysis, liver disease; ↓ biliary obstruction, antibiotics.

3 · HAEMATURIA

- Normal **<3 RBC/high-power field (HPF)**. Haem pad (peroxidase) detects RBCs, haemoglobin and myoglobin: intact RBCs give **speckles**, free pigment a uniform colour; detects ~5 RBC/µL. High sensitivity, lower specificity.
- **A positive dipstick must be confirmed by microscopy** before imaging or cystoscopy (American Urological Association [AUA] 2020).
- **Asymptomatic microscopic haematuria (AMH)**: ≥3 RBC/HPF on a properly collected specimen without obvious benign cause. → *Ch 2* for risk-stratified work-up.

Red urine	Sediment	Supernatant	Plasma
Haematuria	RBCs	Clear	—
Haemoglobinuria	No RBCs	Red, haem +	**Red**
Myoglobinuria	No RBCs	Red, haem +	**Clear**
Dyes, porphyria	No RBCs	Red, haem −	—

> **KEY POINT** Why plasma differs: haemoglobin is large and haptoglobin-bound, so it reaches urine only after haptoglobin saturates (plasma red); myoglobin is small and unbound, so it is filtered at once (plasma clear).

Feature	Glomerular	Non-glomerular medical	Urological
RBC shape	**Dysmorphic**	Round	Round
RBC casts	Present	Absent	Absent
Proteinuria	Significant	Often significant	Minimal even with gross bleeding

- Proteinuria 2+–3+ (100–300 mg/dL) is almost never urological → **nephrology**.
- **Commonest glomerular cause: immunoglobulin A (IgA) nephropathy (Berger)** (~one-third) — young men; haematuria after upper respiratory infection or exercise; mesangial IgA deposits; biopsy-proven; supportive treatment (renin–angiotensin blockade). Once diagnosed, repeat haematuria work-ups are unnecessary.
- Clues: deafness + family history → **Alport**; rash, arthritis, low complement, antinuclear antibody → lupus; haemoptysis → Goodpasture; recent throat/skin infection, raised antistreptolysin O, low C3 → poststreptococcal.
- Non-glomerular medical: papillary necrosis (diabetes, sickle cell, analgesics), renal emboli (atrial fibrillation), nutcracker syndrome, exercise (clears with rest).
- **Therapeutic anticoagulation does not cause haematuria** — urological disease is found as often as in controls → full work-up.

4 · PROTEINURIA

- Normal **80–150 mg/day**; ~40% Tamm–Horsfall protein.

Type	Mechanism	24-h amount	Example
Glomerular (commonest)	Leaky filter; albumin-rich	>1 g suspect, >3 g near-certain	Glomerulonephritis, **diabetes**, amyloid
Tubular	Low-molecular-weight proteins not reabsorbed	Seldom >2–3 g	Fanconi, heavy metals, Balkan nephropathy
Overflow	Excess plasma protein	0.3–2 g	**Myeloma** (Bence Jones light chains)

- Dipstick (tetrabromophenol blue) detects mainly **albumin** (≥20–30 mg/dL); sulfosalicylic acid detects all proteins. **Dipstick negative, sulfosalicylic acid positive → myeloma** → immunofixation for Bence Jones protein.

- Transient (fever, exercise) → repeat. **Orthostatic**: young males, upright only, <1 g/day, benign. Persistent → 24-h quantitation + electrophoresis.

5 · INFECTION — LEUCOCYTE ESTERASE, NITRITE, CULTURE

	Leucocyte esterase	Nitrite
Detects	Host response (neutrophil esterase)	Organism (nitrate reductase of gram-negatives)
Performance	Sensitive, non-specific	**Specific, insensitive**
Classic miss	Glycosuria, proteinuria	**Enterococcus**; low counts

- Only one positive → **microscopy + culture before antibiotics**; esterase alone → consider non-infective inflammation. Dipstick never replaces microscopy and culture.
- Bacteria: 1/HPF ≈ >30,000/mL; **5/HPF ≈ 10^5/mL** — the clean-catch threshold, mainly for women; any bacteria in a proper male specimen → culture.
- † Culture thresholds: clean-catch ≥10^5 colony-forming units (CFU)/mL; catheter specimen ≥10^3 CFU/mL; any growth on SPA is significant.

6 · SEDIMENT MICROSCOPY

- Centrifuge 10–15 mL at 3000 rpm for 5 min → resuspend → coverslip lowered edge-first (avoids bubbles). Scan low power (×100), especially coverslip edges where casts collect; high power (×400) for RBC morphology, crystals, bacteria.
- WBC normal 1–2/HPF (men), ≤5 (women). **Eosinophils** → acute interstitial nephritis. Fresh "glitter cells" → urinary pathology.
- Epithelial: squamous → contamination; transitional — atypia → cytology; **renal tubular cells always mean renal pathology**.
- Yeast mimics RBCs — distinguished by **budding**/hyphae. *Schistosoma haematobium* ova: **terminal spike**. Expressed prostatic secretion: excess/clumped WBCs → prostatitis.

Cast (Tamm–Horsfall matrix)	Meaning
Hyaline	May be normal (exercise, heat)
RBC	Glomerular bleeding — glomerulonephritis
WBC	Pyelonephritis, interstitial nephritis
Muddy-brown granular	**Acute tubular necrosis**
Fatty	Nephrotic syndrome

Crystal	pH	Shape
Uric acid	Acid	Rhomboid / lemon plates
Cystine (always abnormal — diagnostic of cystinuria)	Acid	**Hexagonal**
Struvite	Alkaline	**Coffin-lid** prisms
Calcium oxalate	Either	Envelope

7 · SERUM TESTS AND RENAL FUNCTION

- **Creatinine and estimated glomerular filtration rate (eGFR)** whenever obstruction is suspected — renal function sets urgency of drainage. Check electrolytes in post-obstructive diuresis.
- † Creatinine rises only after ~50% GFR loss and depends on muscle mass; eGFR by CKD Epidemiology Collaboration (CKD-EPI) equation. CKD stages: G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m^2.
- Albumin: marker of debility and recovery, but falsely low in acute inflammation. Complete blood count (CBC) for anaemia in haematuria, infection, oncology follow-up.

- † Coagulation: prothrombin time/international normalised ratio (warfarin, liver disease), activated partial thromboplastin time (heparin) and platelets when a bleeding risk or antithrombotic use is present.

8 · TUMOUR MARKERS

- **Prostate-specific antigen (PSA)** = human kallikrein 3, serine protease from luminal cells; **organ-specific, not cancer-specific** — raised in benign prostatic hyperplasia (BPH) and prostatitis; correlates with volume. Most reliable test for follow-up after any prostate cancer treatment. Interpret with digital rectal examination.

Marker	Half-life	Raised in	Never in	False positives
α-fetoprotein (AFP)	**5–7 d**	Yolk sac, embryonal	**Pure seminoma, choriocarcinoma**	Liver disease, hepatic/gastric cancer, infants
β-human chorionic gonadotropin (βhCG)	**24–36 h**	Choriocarcinoma, embryonal, seminoma	—	Marijuana, hypogonadism (luteinising hormone cross-reacts)
Lactate dehydrogenase (LDH)	24 h	Most often raised; non-specific	—	Many; isolated rise rarely treated

- Measure at diagnosis, **after orchidectomy**, during chemotherapy and in follow-up. Raised AFP in "seminoma" = non-seminomatous germ cell tumour.
- **Urine cytology**: not for screening or initial AMH work-up; highly sensitive and specific for **high-grade** urothelial carcinoma, poor for low-grade. Use for surveillance, persistent haematuria after negative work-up. NMP22, BTA, UroVysion fluorescence in situ hybridisation (FISH): not routine.

9 · STONE, ADRENAL AND ANDROLOGY TESTS

- **Every first stone** (AUA): history, serum chemistries, calcium (± parathyroid hormone, uric acid), urinalysis, **stone analysis at least once**. High-risk or interested patients: **24-h urine** on usual diet — volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine. Calcium-load testing no longer routine.
- **Adrenal incidentaloma**: all — 1 mg dexamethasone suppression (cortisol) and plasma free metanephrines; hypertensives — aldosterone (>15 ng/dL) and aldosterone-to-renin ratio (>20) → adrenal vein sampling.
- **Hypogonadism**: total/free testosterone, luteinising hormone, follicle-stimulating hormone, prolactin, thyroid-stimulating hormone, oestradiol.

Semen parameter † (World Health Organization [WHO] 2021 lower reference limit)	Value
Volume	**1.4 mL**
Concentration / total count	**16 million/mL** / 39 million
Total / progressive motility	42% / 30%
Normal morphology (strict)	**4%**
Vitality	54%

† Abstinence 2–7 days; at least two samples; pH ≥7.2; leucocytes >1 million/mL = leucocytospermia.

10 · OFFICE TESTS AND IMAGING

- **Uroflowmetry** (maximum flow rate, Qmax): >15 mL/s low probability of obstruction; 10–15 equivocal; <10 often obstruction or detrusor failure. Unreliable if voided volume **<100 mL**. Low

flow cannot separate obstruction from weak detrusor → pressure–flow study.

- **Post-void residual** by bladder scan; <100 mL generally acceptable.
- No antibiotic prophylaxis for simple cystoscopy/urodynamics without risk factors (age, steroids, immunodeficiency, anatomical anomaly, external catheter, colonised material, distant infection, long hospitalisation).
- Renal ultrasound: hydronephrosis, cortex brighter than liver → medical renal disease; **not sufficient alone for haematuria**. **Non-contrast computed tomography (CT) = gold standard for stones**; magnetic resonance imaging for soft tissue, not stones. Plain kidneys–ureters–bladder film follows radio-opaque stones (>70%).

11 · MCQ ONE-LINERS

1. Bag urine specimens are unsuitable for culture.
2. Phosphaturia clears with acetic acid; pyuria does not.
3. Specific gravity fixed at 1.010 indicates renal insufficiency.
4. RTA: inability to acidify urine below pH 5.5.
5. Urine pH >7.5 with infection: urease-splitting *Proteus*.
6. Ascorbic acid gives false-negative blood and glucose.
7. Renal glucose threshold ≈ 180 mg/dL.
8. Nitroprusside does not detect β-hydroxybutyrate.
9. Myoglobinuria: red urine, clear plasma, no RBCs.
10. Dysmorphic RBCs and RBC casts mean glomerular bleeding.
11. Commonest glomerular haematuria: IgA nephropathy.
12. Haematuria with deafness: Alport syndrome.
13. Dipstick-negative, sulfosalicylic-positive proteinuria suggests myeloma.
14. Commonest cause of proteinuria without haematuria: diabetes.
15. Nitrite misses enterococci.
16. Muddy-brown casts: acute tubular necrosis.
17. Renal tubular epithelial cells always indicate renal pathology.
18. Cystine crystals are hexagonal and always abnormal.
19. Struvite crystals: coffin-lid shape.
20. *S. haematobium* ova have a terminal spike.
21. PSA is organ-specific, not cancer-specific.
22. AFP is never raised in pure seminoma.
23. AFP half-life 5–7 days; βhCG 24–36 hours.
24. Cytology is specific for high-grade urothelial carcinoma.
25. Every first stone former needs stone analysis once.
26. Non-contrast CT is the gold standard for stones.
27. † WHO 2021 normal morphology lower limit: 4%.

4 Basic Principles of Urologic Ultrasonography

Reading time ≈ 26 min (slow pace)

1 · WHY ULTRASOUND (US) IS THE "UROLOGIST'S STETHOSCOPE"

- Safe, cheap, portable, real-time; assesses fluid and flow; often faster and better than computed tomography (CT) or magnetic resonance imaging (MRI).
- Defining advantage: the **same physician performs and interprets**.
- MCQ Goldberg & Pollack: A-mode first separated solid, cystic and complex renal masses. Perri: first Doppler in acute scrotum.

> **KEY POINT** Oldest Doppler trap: **torsion can give a false-negative** because reactive hyperaemia increases flow. Acute transplant rejection = loss of diastolic flow.

2 · PHYSICAL PRINCIPLES

- Alternating current → piezoelectric crystals → **longitudinal mechanical wave** (compression/rarefaction) → gel → tissue → partial reflection → transducer reconverts echoes (it is a receiver >99% of the time).
- Signal chain: pulse generator → transducer → echoes analysed for amplitude and transit time → time-gain compensation (TGC, boosts deeper echoes) → amplifier → scan converter → monitor.
- **Amplitude** = pixel brightness; **round-trip time** = depth. Urological range **3.5–20 MHz**.
- Velocity = frequency × wavelength; soft tissue **1540 m/s**, so frequency and wavelength vary inversely.

	Axial resolution	Lateral resolution
Separates	Objects one behind the other	Objects side by side
Improved by	Higher frequency	Narrow beam: focal zone at depth of interest

> **KEY POINT** **Resolution–penetration trade-off**: absorption (energy → heat) is proportional to frequency, so high frequency resolves finely but attenuates fast — 7–18 MHz for testis, 3.5–5 MHz for adult kidney.

- Attenuation: reflection (source of image), scattering, interference (organ texture, "speckle"), absorption.
- **Impedance difference at an interface, not density, decides reflection.** Soft tissues cluster (1.4–1.7 $\times 10^6$ kg/m²·s); air (0.0004) and bone (7.8) are extremes → hence coupling gel, bowel-gas obscuration and stone shadowing.
- Kidney and liver have near-identical impedance → hepatorenal interface faint; kidney–fat interface sharp.

Artefacts

Artefact	Mechanism	Appearance / site
Acoustic shadowing	Strong reflection/attenuation	Dark band behind stone; stone looks crescentic → size inaccurate
Increased through-transmission	Less attenuation than surroundings	Bright tissue behind simple cyst
Edging (critical angle)	Beam refracted along curved surface	Shadow at testicular pole, prostatic lobes at urethra
Reverberation	Bounce between two strong reflectors	Equally spaced fading echoes; second at twice depth
Comet tail	Many small reflectors (bowel gas)	Bright tail; may hide lower renal pole

Artefact	Mechanism	Appearance / site
†Twinkling	†Rough reflective surface on colour Doppler	†Colour mosaic behind stone; detects small non-shadowing stones

Fix by changing angle, frequency or focal zone. Shadowing and enhancement are opposite ends of one variable: shadow implies calcium, enhancement implies fluid.

3 · MODES

- **B-mode**: real-time 2D grey-scale. Echogenicity vs reference (liver): hyperechoic (fat), hypoechoic (water), isoechoic, anechoic (cyst).
- **Doppler**: frequency shift depends on velocity, direction and angle. **Resistive index (RI)** = (peak systolic velocity [PSV] − end-diastolic velocity [EDV]) / PSV; reflects downstream resistance (renal artery stenosis, obstruction, penile insufficiency).
- **Colour**: velocity + direction (red toward, blue away — a convention; colour ≠ artery/vein). **Spectral**: waveform, RI. **Power**: amplitude only, **3–5×** more sensitive, less angle-dependent → slow flow (torsion, hypervascular prostate lesions).
- **Harmonic imaging**: receives non-linear multiples of the transmitted frequency → less noise and clutter; used in contrast US. **Spatial compounding**: multi-angle composite → less artefact.
- **Elastography**: same force displaces soft tissue more (Young's modulus E = stress/strain). Strain (manual compression): qualitative, operator-dependent. Shear-wave: machine-generated low-frequency pulses create shear waves (1–10 m/s) whose speed gives absolute stiffness ($E = 3\rho c^2$); less operator-dependent. Hard ≠ malignant: a soft, small vascular testicular nodule proved a Sertoli cell nodule, while a stable hard nodule was benign on surveillance.
- **Multiparametric US** (grey-scale + Doppler + elastography + contrast) parallels multiparametric MRI in real time; microbubbles are safe in renal failure, unlike gadolinium (nephrogenic systemic fibrosis).

4 · CONTRAST-ENHANCED US (CEUS)

- Perflutren **microbubbles**, eliminated via **lungs** → usable in renal insufficiency; avoid in certain cardiopulmonary conditions.
- Renal mass: tumour vascularity, intraoperative cryoablation monitoring, surveillance in chronic kidney disease; less accurate than CT/MRI for initial work-up.
- Bubbles cannot cross the glomerulus → **never opacify the collecting system**. To image urine, instil into urine: voiding urosonography for vesicoureteral reflux (VUR), ureteric patency after percutaneous nephrolithotomy (PCNL).

5 · SAFETY AND ALARA

	Mechanical	Thermal
Nature	Cavitation, streaming	Heating
Greatest	Near gas: lung, bowel	High frequency, focusing, long dwell, Doppler
Index	Mechanical index (MI) = cavitation probability; **≤0.7** low risk; limit time if >0.4 next to lung/bowel	Thermal index (TI) = probability of 1 °C rise; **<2** for urology; rises up to 6 °C harmless unless >60 s

- MI and TI are displayed **estimates, not safety limits**; no confirmed bioeffect in non-fetal scanning within Food and Drug Administration (FDA) limits.
- **ALARA** (As Low As Reasonably Achievable): low power, appropriate mode, short examination, adjusted focus/frequency, cine loop. The **informed operator** is the most important safety factor.
- Documentation: report signed by performing physician plus labelled images (superior to screen left; side label for paired organs); as many images as needed for a complete study.

6 · RENAL US

- **Probes**: curvilinear 3.5–5 MHz; linear 6–10 MHz intraoperatively.
- **Technique**: midaxillary line below costal margin, cranial end to screen left → centre kidney → rotate parallel to ribs to remove shadow → transverse at hilum. Left kidney higher, no liver window → deep inspiration.
- **Normal**: length **10–12 cm**, width 4–5 cm; measure in true midsagittal plane (off-axis underestimates). Cortex slightly hypoechoic to liver; iso/hyperechoic in medical renal disease.
- Echogenic central band = hilar fat and vessels; capsule a bright thin line; Randall plaques = bright foci at calyceal tips.
- Pyramids (hypoechoic, prominent in children) mimic calyces — pyramids lie outside hilar fat, collecting system inside.
- Cortex >**7 mm**; parenchyma >**15 mm**; intrarenal **RI 0.47–0.70**.
- **US-guided PCNL access**: no radiation; shows liver, spleen, colon, lung → wider safe window. Plan puncture from above, below or beside the probe so the hub clears ribs, iliac crest and table; keep the needle tip visible in-plane throughout. Coaxial wires are echogenic, hydrophilic wires poorly visible; wire disappearance under a polyurethane dilator marks dilator depth. Access learning curve is short, US-guided dilation longer. Hard with long skin-to-stone distance, non-dilated system, staghorn.
- **Renal biopsy**: severe hypertension (not number of passes) raises complications; haematuria appears within 6 h → 6-h observation equals admission.
- **Limits**: masses <**2 cm** missed; angiomyolipoma (AML) is hyperechoic **but so are some small renal cell carcinomas (RCC)** → CT.

7 · BLADDER / PELVIC US

- Full bladder is the acoustic window. Uses: post-void residual (PVR), prostate size, wall, stones, tumour, ureterocele, clot, catheter and suprapubic tube guidance.
- **Bladder volume** = W × H × L × **0.625**; **prostate** (ellipsoid) = W × H × L × **0.523**.
- **Bladder wall thickness** predicts bladder outlet obstruction (BOO) better than uroflowmetry, PVR or prostate volume. **Intravesical prostatic protrusion** (perpendicular distance from bladder-base line to tip of protrusion, sagittal view) correlates with urodynamic BOO and helps plan treatment (middle lobe, prostatic urethral length).
- Ureteric jets are visible on colour Doppler, but confirming absence needs up to 10 min; clinical value questioned.
- Automated bladder scanners are **not imaging** → wrong PVR with obesity, ascites, diverticulum, urinoma, lymphocele (fluid outside bladder mimics urine).

8 · SCROTAL US

- Linear **7–18 MHz**; warm room, scrotum supported; copious gel (hair traps air); gentle pressure (compression alters echogenicity and volume). Sagittal sweeps medial → lateral, transverse upper → lower pole, then epididymis.
- Normal: homogeneous, finely granular; mediastinum testis = normal bright line. **Show both testes side by side** — diffuse lymphoma/leukaemia infiltration is visible only by comparison.
- Intratesticular PSV <10 cm/s; RI >**0.6** suggests impaired spermatogenesis.
- Uses: mass, pain, trauma, infertility, impalpable lesion localisation, sperm retrieval guidance.

> **KEY POINT** **Torsion**: hallmark is absent intratesticular flow, but epididymal collaterals can appear within hours. The contralateral testis proves the study can show flow. **US cannot diagnose torsion — only the surgeon can**; strong suspicion → explore.

9 · PENILE AND URETHRAL US

- Linear 12–18 MHz; perineal window for bulbar urethra. **Sonourethrography** (retrograde saline/gel) shows stricture and spongiofibrosis without radiation, even intraoperatively.
- Translabial US: diverticulum, pelvic floor, echogenic midurethral sling.

Penile duplex Doppler (after intracavernosal injection)

Parameter	Threshold	Meaning
PSV	**≥35 cm/s** / 25–35 / **<25**	Normal / equivocal / arteriogenic erectile dysfunction (ED)
Artery dilation	<75%	Arteriogenic ED
Acceleration time	>110 ms	Arteriogenic ED
†EDV (PSV normal)	†>5 cm/s; RI <0.75	†Veno-occlusive dysfunction

PSV is the most accurate measure; arteriogenic ED flags systemic cardiovascular disease. Elastography localises impalpable Peyronie plaques.

10 · TRANSRECTAL US (TRUS)

- 7.5–10 MHz. Digital rectal examination first → survey base to apex incl. seminal vesicles and rectal wall → three diameters → volume and prostate-specific antigen density → Doppler.
- Indications: volume, abnormal DRE, biopsy, abscess drainage, haematospermia, azoospermia with low-volume ejaculate (dilated seminal vesicles, midline cyst — aspiration, refill common).
- Young gland homogeneous with peripheral zone brighter, zones hard to separate; older gland: transition zone enlarges and is distinct; "capsule" = prostate–fat impedance interface.
- **Microultrasound (~29 MHz)**: detection and negative predictive value comparable to multiparametric MRI at lower cost.

11 · SPECIAL GROUPS AND THERAPEUTIC US

- **Children**: first-line; small size lets deep structures be seen.
- **Pregnancy**: first-line, safe in all trimesters; grey-scale poor for ureteric stones → add transvaginal US, RI, ureteric jets; US-guided stent exchange spares fetal radiation.
- **High-intensity focused US (HIFU)**: focused thermal ablation of localised prostate cancer; sequence: transrectal probe → lesions demarcated → serial slices outlined on grid → ablation along grid → treated zone hypoechoic. MRI fusion for US-invisible lesions. Complications: haematuria, haematospermia for weeks, rectal/urethral injury, **delayed rectourethral fistula**. Low-intensity acoustic wave therapy for ED; burst-wave lithotripsy (emerging).

12 · MCQ ONE-LINERS

1. Soft-tissue speed of sound: 1540 m/s.
2. Higher frequency → better axial resolution, less penetration.
3. Lateral resolution is best at the focal zone.
4. Reflection depends on impedance difference, not density.
5. Posterior enhancement = fluid; shadowing = stone/calcification.
6. Reverberation: second echo at twice the true depth.
7. Power Doppler: 3–5× more sensitive, no direction information.
8. Normal renal RI 0.47–0.70.
9. Bladder volume factor 0.625; prostate 0.523.
10. Kidney 10–12 cm; parenchyma >15 mm, cortex >7 mm.
11. Pyramids mimic calyces; pyramids lie outside hilar fat.
12. Intravenous microbubbles are exhaled and never opacify urine.
13. MI and TI are estimates, not safety limits.
14. Cavitation risk is greatest near lung and bowel.
15. Most important safety factor: informed operator.
16. Automated bladder scanner is not an imaging study.
17. Bladder wall thickness predicts BOO better than flow rate.
18. Intratesticular RI >0.6: impaired spermatogenesis.
19. US cannot diagnose torsion; only the surgeon can.

20. Diffuse testicular lymphoma is seen only by side-by-side comparison.
21. Penile PSV <25 cm/s = arteriogenic ED.
22. Microultrasound ≈ 29 MHz.
23. Small hyperechoic renal mass: AML or RCC — CT needed.
24. Late HIFU complication: rectourethral fistula.
25. Shear-wave elastography is quantitative; strain elastography is qualitative.
26. CEUS does not need renal excretion and suits chronic kidney disease.
27. Renal biopsy complications rise with severe hypertension, not number of passes.
28. †Twinkling artefact on colour Doppler suggests stone.

5 Principles of CT and Radiographic Imaging

Reading time ≈ 30 min (slow pace)

1 · RADIATION BIOLOGY AND SAFETY

- Ionising radiation damages DNA directly or via free radicals → apoptosis or attempted repair.
- MCQ Clarence Dally (Edison's assistant): first known radiation death.

	Deterministic	Stochastic
Threshold	Yes	None — linear no-threshold model
Dose increases	Severity	Probability (cumulative)
Mechanism	Cell loss	Mutation in oncogene / tumour suppressor
Examples	Erythema, epilation, sterility, cataract	Cancer (leukaemia, solid tumours)

- **Units**: absorbed dose = gray (Gy); equivalent dose = sievert (Sv; weighting factor 1 for x-rays); **effective dose** (Sv) adds tissue weighting — used for exposure limits.

> **KEY POINT** Because stochastic risk has no threshold, **no dose is safe** → **ALARA** (as low as reasonably achievable): risk–benefit analysis before every ionising study, low-dose protocols, non-ionising alternatives.

Study	Effective dose
Chest x-ray	**0.07 mSv**
Kidney–ureter–bladder (KUB) film / two-view abdomen	**0.7 / 1.4 mSv**
Ultra-low-dose CT abdomen–pelvis	**≈1 mSv** (1–2)
Low-dose CT	<3–4 mSv
Conventional CT abdomen–pelvis	**10–20 mSv**; 10 mSv ≈ 1 in 1000 cancer risk

2 · RADIOGRAPHIC STUDIES

- X-rays: electrons accelerated from cathode to (tungsten) anode in a vacuum tube. Differential tissue absorption creates contrast.
- **KUB**: supine anteroposterior, upper renal pole to inferior pubic rami. Uses: follow-up of radio-opaque stones, before shock-wave lithotripsy (SWL), stent/drain position, ileus, foreign body. Projection hides overlapping pathology → poor sensitivity; CT detects >95% of stones. Upright film is most sensitive for free air.
- †**Intravenous urography (IVU)**: serial films after intravenous contrast (nephrogram → pyelogram → ureters → bladder); now largely replaced by CT urography.

Static cystography

- Shows extravasation, filling defects, reflux, diverticula, fistula, anastomotic leak. American Urological Association (AUA): plain or CT cystography for suspected bladder injury.
- **Technique**: scout → catheter → ≥**300–350 mL** dilute contrast by gravity → clamp → full anteroposterior film → obliques → drain → **post-drainage film** (posterior leak hidden behind dense contrast).
- Avoid in active infection (pyelonephritis, sepsis).

3 · FLUOROSCOPIC STUDIES

Setting	Peak kilovoltage (kVp)	Tube current (mA)
Changes	Photon energy	Photon number
Dose	Rises exponentially	Rises linearly

Setting	Peak kilovoltage (kVp)	Tube current (mA)
Penetration / contrast	↑ / ↓	Unchanged / ↑

Retrograde pyelography

- Chosen when intravenous (IV) contrast is hazardous (allergy, renal dysfunction); maps the collecting system for intrarenal surgery or percutaneous access; shows peristalsis (CT cannot).
- Cystoscopy → catheterise ureteric orifice → gentle injection of ~50% contrast (dense contrast hides small filling defects).
- Over-pressure → pyelovenous, pyelotubular, pyelosinus, pyelolymphatic **backflow** → bacteraemia/sepsis. Air contrast: rare air embolism.
- †**Antegrade pyelography** (via nephrostomy) when retrograde access is impossible; also shows obstruction level from above.

Voiding cystourethrography (VCUG)

- **Gold standard for vesicoureteral reflux (VUR)**; also posterior urethral valves, diverticula, neurogenic bladder.
- Febrile urinary tract infection (UTI) in children: ultrasound first. European Association of Urology (EAU) — VCUG if ultrasound abnormal or age <12 months; American Academy of Pediatrics — VCUG only if ultrasound suggests risk (age 2–24 months).
- Catheter: infant 5 Fr feeding tube, child 6–8 Fr, adult 14–16 Fr. Fill by gravity → obliques → void under fluoroscopy → post-void film; cyclic filling raises sensitivity.
- **Bladder capacity**: <2 y = weight (kg) × 7; **2–14 y = (age × 30) + 30** mL; >14 y = 300–500 mL.
- Requires sterile urine. Spinal injury above T6 → autonomic dysreflexia on distension.

Retrograde urethrography (RUG)

- For anterior urethra: stricture, trauma, diverticulum, fistula; before urethroplasty.
- **Technique**: 45° oblique → catheter tip in fossa navicularis, balloon ~2 mL (overinflation injures) → penis on traction → 20–30 mL contrast. Foley in situ → pericatheter study with feeding tube.
- Prostatic urethra does not distend normally — not a stricture. Posterior urethra needs combined VCUG.

	RUG	VCUG
Segment	Anterior urethra	Posterior urethra, bladder, reflux
Classic use	Stricture, trauma	VUR, posterior urethral valves

Loopography

- Contrast via stoma: refluxing anastomosis shows upper tracts; non-refluxing needs antegrade study. Safe in contrast allergy (minimal absorption).

Fluoroscopic renal access

- Still commonest for percutaneous nephrolithotomy (PCNL); ultrasound harder for upper pole, obesity, staghorn.
- **Bull's-eye**: C-arm tilted 15–30° toward surgeon, needle end-on as a dot over the calyx → rotate away to judge depth. **Triangulation**: 0° to locate calyx, then 30° caudal (upper pole) or cranial (lower pole); two views fix depth.

4 · FLUOROSCOPY DOSE REDUCTION

- **Time**: dose ∝ fluoroscopy time; last image hold; justify each activation; record time (feedback alone reduces it).
- **Collimation**: halving aperture diameter cuts exposed area fourfold; improves contrast.
- **Low-dose mode** halves mA and dose. **Pulsed**: continuous = 30 pulses/s; 8/4/1 pps theoretically save 73/87/97%.
- **Distance**: inverse square law — doubling distance → ¼ dose.

- **Geometry**: source below patient, detector close to patient. Keep instruments out of beam (automatic exposure control raises kVp/mA).
- **Shielding**: 0.25–0.5 mm lead apron, thyroid shield, glasses; legs get most scatter.

> **KEY POINT** Surgeon dose is mostly scatter from the patient, so everything that lowers patient dose protects the team; time, distance and shielding add on top.

5 · CT PRINCIPLES

- Rotating tube; opposite detectors record attenuation profiles → reconstructed 3D volume.
- **Hounsfield unit (HU)** = $[(\mu_{tissue} - \mu_{water}) / \mu_{water}] \times 1000$. Range −1000 to +3000. Not absolute — varies between machines.
- **Windows** rescale a narrow HU range: soft-tissue, lung, bone (best for subtle stones and stones next to stents).

Material	HU
Air	−1000
Macroscopic fat (diagnostic of AML)	**≤ −10** (†fat ≈ −100)
Water / simple cyst	0 / −9 to 20
Lipid-rich adrenal adenoma	**<10**
Hyperdense cyst (Bosniak II)	≥70 unenhanced
Uric acid stone	**200–400**
Calcium oxalate stone	**600–1200**

Image quality and artefacts

- **Noise** falls with higher mAs, thicker slices, smaller patient; measured by signal-to-noise ratio.
- **Beam hardening**: low-energy photons absorbed first → streaks/low central HU; metal prostheses. Photon-counting detectors reduce it.
- **Partial volume averaging → pseudoenhancement** of small cysts (<10–15 mm) from adjacent parenchyma → cyst read as tumour → overtreatment. Use thin slices, central region of interest.

Dose reduction and reconstruction

- Lower tube current (commonest) or voltage; size-adjusted, real-time dose modulation.
- Filtered back projection: fast, noisy, higher dose. Iterative: ~half dose, slower, "waxy" texture. **Deep learning reconstruction**: high quality at ≈1 mSv, fast enough for emergencies.

Dual-energy CT (DECT)

- Two energies (e.g. 80/140 kVp) → iodine maps and **virtual unenhanced** images (saves the true noncontrast phase).
- Uses: hyperdense cyst vs enhancing mass, adrenal nodules, fewer urography phases, **uric acid vs non-uric acid stone** (near-perfect).
- Limits: more dose than low-dose CT; small field of view.

6 · NONCONTRAST CT (NCCT) FOR STONES

- **Gold standard**: all stones visible **except protease-inhibitor (indinavir) stones**. Gives size, site, obstruction signs, composition by HU.
- Low-dose: cutting tube current up to **95%** preserves accuracy. **Ultra-low-dose CT (≈1 mSv) < two-view KUB (1.4 mSv)**.
- Avoid ultra-low dose in body mass index (BMI) >30, very thin patients (phlebolith confusion), metal hardware, suspected infected stone (missing a 1–2 mm obstructing stone is worse than the radiation).

Body	First imaging in suspected colic
AUA	CT in adults; standard-dose if BMI >30; NCCT before PCNL or SWL
EAU	Ultrasound first; CT if inconclusive (randomised trial: less radiation, same outcomes)

CT, low-dose and ultra-low-dose CT are equally accurate (sensitivity ≈95%); ultrasound sensitivity 84%, specificity only 53%; KUB sensitivity 57%.

7 · CONTRAST CT PHASES AND PROTOCOLS

Phase	Timing	Shows / use
Early arterial	20–30 s	Renal arteries, bleeding, aneurysm (CT angiography)
Corticomedullary	**30–70 s**	Cortex only; hypervascular masses, clear cell subtype, vessels, renal vein
Portal venous	70–90 s	Liver, nodes, routine abdomen/staging
Nephrographic	**90–120 s**	Homogeneous parenchyma; **most sensitive for renal masses**, caval thrombus
Excretory	**7–10 min**	Collecting system, filling defects, leaks

- **Renal mass protocol**: noncontrast + corticomedullary + nephrographic ± delayed. Enhancement **≥20 HU** = solid; 10–20 equivocal; <10 = cyst. Clear cell renal cell carcinoma (RCC) enhances early and avidly; papillary, chromophobe, oncocytoma later.
- Fat-poor AML: hyperdense on NCCT, rapid enhancement and washout.
- **Adrenal protocol**: noncontrast, portal venous, 15-min delay. Absolute washout = (PV − delayed)/(PV − unenhanced) × 100 >**60%**; relative = (PV − delayed)/PV × 100 >**40%** → adenoma.
- **Trauma**: noncontrast + arterial + nephrographic + delayed. Living donor: four-phase CT angiography.
- Nodes: prostate cancer cut-off 10 mm (poor sensitivity → PSMA positron emission tomography (PET)-CT); bladder cancer short axis >8 mm pelvic, >10 mm abdominal.

Bosniak 2019 (cystic renal masses)

Class	Key CT features	Malignancy / action
I	Simple fluid −9 to 20 HU, wall ≤2 mm, no septa/calcification	Benign; no follow-up
II	1–3 thin septa, any calcification; or homogeneous ≥70 HU unenhanced, non-enhancing >20 HU, too small to characterise	Benign; no follow-up
IIF	Minimally thickened (3 mm) enhancing wall/septa or ≥4 thin septa	<10%; follow-up 6, 12 months, then yearly × 5 y
III	Thick (≥4 mm) or irregular wall/septa (≤3 mm obtuse protrusion)	~50%; urologist
IV	Enhancing nodule: any acute-angled protrusion, or ≥4 mm obtuse	~90%; urologist

8 · CT UROGRAPHY (CTU) AND CT CYSTOGRAPHY

- Thin slices, always an excretory phase. **Single-bolus** (4 phases, most dose); **split-bolus** (combined nephrographic–excretory, less dose, lower sensitivity for subtle urothelial lesions); **DECT + split-bolus** = all information in one acquisition.
- Upper tract urothelial carcinoma: routine test; filling defect, soft-tissue density lower than stone or clot, moderate enhancement. Poor distal ureteric opacification → **hydration** (simplest, most effective), diuretic, prone position.
- Bladder cancer: CTU for multiple/high-risk/trigonal tumours; do CT **before transurethral resection of bladder tumour (TURBT)** — post-resection change mimics invasion. Detects T3b/T4 but cannot separate muscle-invasive from non-invasive.
- Ureteric injury: excretory extravasation; transection → distal ureter not opacified.
- **CT cystography**: exclude urethral injury → **retrograde** fill ≥300–350 mL (passive filling by excreted contrast misses ruptures); no post-drainage images needed.

Extraperitoneal rupture	Intraperitoneal rupture
Flame-shaped/"molar tooth" contrast in space of Retzius; scrotum if urogenital diaphragm breached	Contrast outlines bowel loops, liver and spleen (dome rupture)

Pitfall: a large extraperitoneal leak prevents distension and hides a coexisting intraperitoneal rupture.

PET-CT: prostate-specific membrane antigen (PSMA) PET-CT detects biochemical recurrence even at prostate-specific antigen (PSA) <0.5 ng/mL; most sensitive for nodal and bone metastases.

9 · IODINATED CONTRAST REACTIONS

Immediate (<1 h–hours)	Delayed (hours–days)
Urticaria, angio-oedema, bronchospasm, hypotension, anaphylaxis	Milder maculopapular rash, urticaria

ALGORITHM
Observe ≥30 min → reaction: stop contrast → anaphylaxis: **intramuscular adrenaline 0.3–0.5 mg** every 5–15 min (max 1 mg) → oxygen + β2-agonist (salbutamol 2 puffs, up to 3×) for bronchospasm → mild pruritus: observe or H1-antihistamine (can progress) → later skin/provocation tests.

- **Premedicate** high-risk (prior reaction, asthma, atopy, drug allergy, cardiac disease): prednisone 50 mg orally at **13, 7 and 1 h** + diphenhydramine 50 mg at 1 h; or methylprednisolone 32 mg at **12 and 2 h** + diphenhydramine. Breakthrough reactions still occur.

10 · CONTRAST-ASSOCIATED ACUTE KIDNEY INJURY (AKI)

- **Contrast-associated AKI** = AKI within 48 h of contrast, whatever the cause; **contrast-induced AKI** = subset caused by contrast (rare but real).
- Kidney Disease: Improving Global Outcomes (KDIGO) AKI: creatinine rise **≥0.3 mg/dL** in 48 h, or ≥1.5× baseline in 7 days, or urine <0.5 mL/kg/h for 6 h.
- †Highest risk: estimated glomerular filtration rate (eGFR) **<30** mL/min/1.73 m² or existing AKI.
- **Prevention**: noncontrast study where possible; IV fluids for high risk; iso-/low-osmolar agent; lower kVp (80) so less iodine is needed; weight-based dose.
- **†Metformin**: hold for 48 h after contrast if eGFR <30 or AKI, restart after renal function rechecked (lactic acidosis risk).

KEY POINT Hypersensitivity is idiosyncratic (premedication, adrenaline); nephrotoxicity is dose-related (hydration, less contrast). Prophylaxis for one does nothing for the other.

11 · MCQ ONE-LINERS

1. Stochastic effects: no threshold, probability rises with dose.
2. Chest x-ray 0.07 mSv; KUB 0.7 mSv; CT abdomen–pelvis 10–20 mSv.
3. Ultra-low-dose CT is below two-view KUB dose.
4. kVp raises dose exponentially; mA linearly.
5. Water 0 HU, air −1000 HU.
6. Uric acid stones 200–400 HU; calcium oxalate 600–1200 HU.
7. Indinavir stones are invisible on NCCT.
8. Nephrographic phase is most sensitive for renal masses.
9. Excretory phase: 7–10 min.
0. Enhancement ≥20 HU = solid enhancing lesion.
1. Pseudoenhancement: partial volume in cysts <10–15 mm.
12. AML: fat ≤ −10 HU on NCCT.
13. Adrenal adenoma: <10 HU, absolute washout >60%, relative >40%.
14. Bosniak IIF follow-up: 6, 12 months, then yearly for 5 years.
15. Bosniak IV: enhancing nodule, ~90% malignant.
16. DECT separates uric acid from non-uric acid stones.
17. Best aid to distal ureteric opacification: hydration.
18. CT before TURBT avoids post-resection pseudo-invasion.
19. CT cystography needs retrograde filling, not passive.
20. Molar-tooth sign = extraperitoneal bladder rupture.
21. VCUG is gold standard for VUR.
22. Bladder capacity 2–14 y: (age × 30) + 30 mL.

23. Retrograde pyelography suits contrast allergy or renal failure.

24. Prednisone premedication: 13, 7, 1 h before.
25. KDIGO AKI: creatinine ≥0.3 mg/dL within 48 h.

6 Magnetic Resonance Imaging

Reading time ≈ 16 min (slow pace)

1 · PHYSICS AND SEQUENCES

- Versus computed tomography (CT), magnetic resonance imaging (MRI) offers superior **soft-tissue contrast** and no ionising radiation.
- **T1** = longitudinal relaxation (z-axis); **T2** = transverse relaxation (x–y plane). Fluid: **dark T1, bright T2**. Kidney: cortex brighter than medulla on T1.
- **Diffusion-weighted imaging (DWI)**: Brownian motion of water. High cellularity restricts diffusion → bright DWI, **dark apparent diffusion coefficient (ADC)** map.
- **Dynamic contrast-enhanced (DCE)** MRI: serial T1 after gadolinium, measuring perfusion and permeability.
- **Chemical shift imaging (CSI)**: fat and water protons precess at different frequencies. In-phase (IP) echo time adds their signals; out-of-phase (OP) cancels them, so a voxel containing both **loses signal on OP** = microscopic fat. Fat suppression nulls **macroscopic** fat.

2 · INDICATIONS, CONTRAST AND SAFETY

- Modality of choice in **pregnancy, renal insufficiency, iodinated contrast allergy**.
- Gadolinium-based contrast agents (GBCAs) shorten relaxation times → enhancement on T1. Chelation makes toxic gadolinium safe.
- **Nephrogenic systemic fibrosis (NSF)**: gadolinium precipitates in skin and muscle → systemic fibrosis; historic concern at glomerular filtration rate (GFR) **<30 mL/min/1.73 m²**.
- American College of Radiology (ACR) **group II agents** carry little or no NSF risk even in end-stage renal disease: gadobenate, gadoteridol, gadoterate, gadobutrol. NSF is now largely historical.
- †Contraindications: non-conditional pacemakers/defibrillators, ferromagnetic aneurysm clips, cochlear implants, metallic orbital foreign bodies.

3 · ADRENAL MRI

- **Adenoma** (usually <3 cm, non-functioning): intracytoplasmic lipid → **OP signal drop**.
- Signal intensity (SI) index = $(SI_{IP} - SI_{OP})/SI_{IP} \times 100$; more specific than visual reading; no agreed cut-off.
- Lipid-poor adenomas (10–30%) are indeterminate → CT washout. Functioning adenoma: lateralise by adrenal vein sampling.

Lesion	Key MRI features
Adenoma	OP drop; CT <10 HU (Hounsfield units), relative washout >40%, absolute >60%
Adrenocortical carcinoma (ACC)	>4 cm (>6 cm: 25% are ACC); T2 high, heterogeneous; no OP drop; venous thrombus; fluorodeoxyglucose (FDG)-avid; ≈ 62% functioning
Myelolipoma	Macroscopic fat: T1 bright, **loses signal on fat suppression**, **India ink artefact**, little OP drop
Metastasis	Large, heterogeneous, T2 high; delayed peak enhancement (≥53 s)
Pheochromocytoma	"Light bulb" T2 bright — but ⅓ are T2-dark; marked early enhancement
Haematoma	T1 bright at 1–7 weeks; dark haemosiderin rim

KEY POINT **Trap**: metastasis from clear cell renal cell carcinoma (RCC), liposarcoma or hepatocellular carcinoma contains lipid and mimics adenoma. T2 brightness is non-specific (pheochromocytoma, ACC, metastasis, cyst).

4 · RENAL MASS

Cysts

- Haemorrhagic/proteinaceous contents: T1 bright; chronic haemorrhage → T2-dark wall ring.
- **Enhancement** is the most important sign of malignancy (better than wall calcification). Calcium is a signal void on MRI, so MRI sees enhancing nodules in calcified cysts better than CT.
- Enhancement = SI rise **≥15%**; peak 2–4 min.
- Absent pseudocapsule (T1/T2 low rim) predicts pT3a.

Subtyping (multiparametric: T1, T2, DCE, DWI, CSI)

Mass	T2	Enhancement	Other
Clear cell RCC	High, heterogeneous	Avid (>200%)	High ADC (≥1281 favours clear cell); OP drop ≥25% diagnostic once angiomyolipoma excluded
Papillary RCC	Low, homogeneous (haemosiderin)	Weak	Lowest ADC
Chromophobe	Central scar bright (≈ 37%)	Intermediate	—
Oncocytoma	Central scar bright (54–80%)	Avid	Not separable from clear cell
Angiomyolipoma (AML)	Variable	Falls corticomedullary → nephrographic	Macroscopic fat: fat-suppression loss, no OP drop

- Venous thrombus: MRI = CT; gadolinium separates **tumour thrombus (enhances)** from bland thrombus.

5 · MR UROGRAPHY AND BLADDER

- MR urography (MRU): **static** = heavily T2 (urine bright); **excretory** = gadolinium T1. Equal to CT urography for obstruction.
- Stones, tumour, clot and sloughed papillae appear as dark defects in bright urine.
- Upper tract urothelial carcinoma (UC): homogeneous on DWI, ADC ≤451 (vs heterogeneous central RCC).
- Bladder: **detrusor is the T2-dark line**; muscle invasion = intermediate-signal tumour disrupting it. Underdistension → false negatives.
- **Vesical Imaging–Reporting and Data System (VI-RADS)**: 5-point score for muscle invasion from T2, DWI and DCE.
- †**Urethral diverticulum**: T2 MRI is the best study — defines neck, circumferential/saddlebag extent, tumour or stone.
- †**Fistulae** (vesicovaginal, colovesical, urethral): T2 shows the fluid-filled tract.

6 · PROSTATE MRI

Sequences

- 3 tesla without endorectal coil is standard (coil equivalent for Grade Group [GG] ≥3; useful at 1.5 T).
- **T2** (triplanar): anatomy, extraprostatic extension (EPE), seminal vesicle invasion (SVI). Normal peripheral zone (PZ) is bright; low T2 is non-specific (prostatitis, scar, hyperplasia).
- **DWI/ADC**: b-value ≥1000; cancer bright on DWI, dark on ADC. **ADC falls as Gleason score rises**.
- **DCE** (temporal resolution ≤15 s): type 1 persistent rise = normal; type 2 plateau = indeterminate; type 3 washout = most suggestive (overlaps inflammation). DCE is secondary to T2 and DWI.

	Multiparametric (mpMRI)	Biparametric (bpMRI)
Sequences	T2 + DWI + DCE	T2 + DWI; no gadolinium
Detection	Standard	No significant difference
Staging	Better	Limited
Role	Standard of care	Low-risk screening, expert readers

Quality and reporting

Score	PI-RADS v2.1 (clinically significant cancer)	PRECISE (active surveillance)	PI-QUAL (quality)
1	Very low	Resolution	All sequences non-diagnostic
2	Low	Reduction	One acceptable
3	Equivocal	**Stable**	Rules in, **not out**
4	Likely	Increase	Rules in and out
5	Highly likely	Stage progression	All optimal

- Prostate Imaging–Reporting and Data System (PI-RADS): at PI-QUAL 1–2, give no score.
- mpMRI negative predictive value >90%; sensitivity **0.91**, specificity 0.37 for GG ≥2 — rules out, not in.

Fusion biopsy

- MRI before biopsy for raised prostate-specific antigen (PSA) (American Urological Association, European Association of Urology); misses insignificant cancer (less overdiagnosis).
- Reaches anterior/transition-zone lesions missed by systematic 12-core biopsy.
- Fusion beats cognitive targeting; yield depends on scan quality and reader, not platform.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
PROMIS	mpMRI vs template biopsy, raised PSA	MRI highly sensitive for significant cancer	MRI before biopsy (level 1b)
PRECISION	randomised controlled trial (RCT), biopsy-naive: MRI ± targeted vs systematic biopsy	More significant, less insignificant cancer	MRI pathway standard
Göteborg-2	Screening RCT	Targeted-only biopsy reduced overdiagnosis	MRI-targeted screening

8 · MCQ ONE-LINERS

1. NSF threshold historically GFR <30; group II GBCAs are safe even in dialysis.
2. OP signal drop = microscopic fat = adenoma.
3. India ink artefact and fat-suppression loss = myelolipoma.
4. Clear cell RCC metastasis can show OP drop — mimics adenoma.
5. ⅓ of pheochromocytomas are T2-dark; "light bulb" sign unreliable.
6. Most important sign of malignancy in renal mass: enhancement (≥15% SI rise).
7. Absent pseudocapsule predicts pT3a.
8. Papillary RCC: T2-dark, hypovascular, lowest ADC.
9. Detrusor = T2-dark line; VI-RADS scores muscle invasion.
10. ADC inversely correlates with Gleason score.
11. Type 3 DCE curve: early enhancement + washout.
12. PI-QUAL 3 rules in but not out.
13. PRECISE 3 = stability, not equivocation.
14. Prostate MRI: sensitivity 0.91, specificity 0.37 for GG ≥2.
15. Fluid: dark T1, bright T2.
16. Lipid-poor adenoma (CSI indeterminate) → CT washout.
17. Calcium is a signal void — MRI sees enhancement in calcified cysts.

8. Tumour thrombus enhances; bland thrombus does not.
19. bpMRI omits DCE; detection similar to mpMRI.
20. Endorectal coil unnecessary at 3 T.

7 Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine

Reading time ≈ 16 min (slow pace)

1 · PRINCIPLES

- Radiotracers in subpharmacological doses image **function**; the patient is the energy source.
- **Single-photon**: gamma photon (100–300 keV) → collimator (direction) → sodium iodide crystal (light) → photomultiplier tubes → image. Planar = 2D; **single-photon emission computed tomography (SPECT)** = rotating camera, 3D, with low-dose computed tomography (CT) for localisation and attenuation correction.
- **Positron emission tomography (PET)**: positron + electron → annihilation → two **511-keV** photons 180° apart → coincidence detection. Rejects unpaired photons (less noise); quantifies **standardised uptake value (SUV)**; best resolution (≈ 5 mm); costly.

Radionuclide	Half-life	Note
Technetium-99m (^{99m}Tc)	**6 h**	Single 140-keV photon — workhorse
Indium-111 (^{111}In), gallium-67 (^{67}Ga)	67 h, 78 h	Delayed infection imaging
Carbon-11 (^{11}C)	**20.3 min**	Needs on-site cyclotron
Fluorine-18 (^{18}F)	**110 min**	Lowest positron energy, shortest range → sharpest image; distributable
Gallium-68 (^{68}Ga)	68 min	Generator-produced; longest range

2 · RENAL PHYSIOLOGY AND TRACERS

- Renal blood flow ≈ 1 L/min (20% of cardiac output); plasma flow ≈ 600 mL/min. Plasma clearance ≈ 20% filtration, 80% tubular secretion.

	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	**^{99m}Tc-MAG3 (mercaptoacetyltriglycine)**	**^{99m}Tc-DMSA (dimercaptosuccinic acid)**
Handling	Glomerular filtration only	Tubular secretion (protein-bound)	Filtered, then retained in proximal tubule (megalin, cubilin)
Study	Dynamic	Dynamic	Static cortical, 2 h
Use	Glomerular filtration rate (GFR); renovascular hypertension	**Split function, obstruction — first choice**	Pyelonephritis, scarring
Renal impairment	High background, poor images	Good images	Delay imaging

> **KEY POINT** Why MAG3 is preferred: secretion is independent of GFR, extraction is higher, so images stay good in poor function; also cheaper.

3 · DIURETIC RENOGRAPHY

Preparation and protocol

- **Hydrate** well. Bladder outlet obstruction or neurogenic bladder → **Foley catheter**. **Cap nephrostomy** to test ureteric patency.
- MAG3/DTPA 5–10 mCi (185–370 MBq) intravenous bolus. MAG3: cortical peak **3–5 min**, bladder by 10–15 min.

- **Perfusion phase**: 1–2-s frames for 1 min; regions of interest over aorta, kidneys and background → time–activity curve (TAC). Slow upstroke = poor flow.
- **Split function** = area under each TAC at **1–3 min** (before tracer leaves the kidney); normal up to 45:55.
- **Furosemide** 0.5 mg/kg intravenously (adult max 40 mg; more in renal impairment) separates a patulous system from true obstruction.
- Timing varies: F+15 (authors), F0 or F−15 (furosemide 15 min before tracer, maximal diuresis at tracer arrival). †F+20 is the classic "standard" protocol. Always check whether and when diuretic was given.

Post-furosemide half-clearance time (T½)	Interpretation
<10 min	Not obstructed (patulous)
10–20 min	Indeterminate — further evaluation
>20 min	Obstruction (flat/rising curve)

- †Pitfalls of false-positive obstruction: dehydration, full bladder, poorly functioning kidney (cannot mount diuresis), very capacious pelvis.

4 · OTHER DYNAMIC APPLICATIONS

Captopril renography (renovascular hypertension)

- A stenotic kidney maintains GFR by angiotensin II–mediated **efferent arteriolar constriction**; angiotensin-converting enzyme (ACE) inhibition removes it, so filtration falls in that kidney only.
- DTPA preferred (pure filtration).

ALGORITHM
Day 1: oral captopril → renogram (slow uptake, low peak) → if abnormal, repeat 1–2 days later off ACE inhibitors and calcium channel blockers → improvement **≥10%** = high probability of renovascular hypertension → likely benefit from revascularisation

- Largely replaced by CT angiography and Doppler.

Transplant

	Acute rejection	Delayed graft function
Perfusion	Decreased	Maintained
Serial studies	Worsen	Improve

- Biopsy is definitive. Urine leak = tracer outside kidney on delayed images (avoids iodinated contrast).

Vesicoureteral reflux (VUR)

- Direct nuclear cystography (^{99m}Tc-sulfur colloid into bladder) or indirect (void after MAG3/DTPA).
- **Initial study = fluoroscopic voiding cystourethrography (VCUG)** (shows posterior urethral valves, diverticula); nuclear cystography for **follow-up** — more sensitive, lower radiation.

5 · DMSA AND INFECTION IMAGING

- DMSA: 50 µCi/kg (child), images at **2 h** with pinhole or SPECT. Pyelonephritis and scar are both photopenic.
- Pyelonephritis: ill-defined defect, normal size, **resolves**. Scar: sharp, cortical thinning, small kidney, **persists** at **≈ 6 months**.
- **^{111}In-oxine leucocytes**: image at 24 h; no urinary excretion → preferred for genitourinary/abdominal infection; highly specific.

- **^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes**: better images, paediatric dosimetry, but renal/biliary excretion obscures the tract.
- Labelled leucocytes **not usable in transplant** (accumulate in graft). ^{67}Ga-citrate: transferrin-bound, non-specific, 48–72 h.

6 · ONCOLOGICAL PET

- **^{18}F-fluorodeoxyglucose (FDG)**: GLUT uptake → hexokinase phosphorylation → trapped (Warburg effect). Urinary excretion obscures the tract, so FDG is for **distant disease**; prostate cancer is poorly FDG-avid.
- **Bladder**: high nodal specificity; use only when conventional imaging is equivocal.
- **Kidney**: not routine; clear cell renal cell carcinoma (RCC) poorly FDG-avid. **Girentuximab** (anti-carbonic anhydrase IX, ^{89}Zr/^{124}I) lights up clear cell RCC; **^{99m}Tc-sestamibi** lights up oncocytoma (mitochondria-rich) — mirror images.
- **Penile**: FDG confirms metastasis in palpable nodes, guides neoadjuvant chemotherapy.
- **Testis**: FDG for residual **seminoma >3 cm** post-chemotherapy; **not for non-seminoma** (teratoma is not FDG-avid) → retroperitoneal lymph node dissection if >1 cm.

7 · PROSTATE: BONE SCAN AND PET

- **Bone scan** (^{99m}Tc-methylene diphosphonate [MDP]): †binds hydroxyapatite at osteoblastic activity; †non-specific (trauma, arthritis); †flare on early therapy; †"superscan" = diffuse uptake with absent renal activity.
- **Prostate-specific membrane antigen (PSMA)**: transmembrane glycoprotein, rises with grade; normal uptake in salivary/lacrimal glands.
- National Comprehensive Cancer Network: **PSMA-PET preferred** at staging and biochemical recurrence; conventional imaging not a prerequisite.
- Approved: ^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3.
- Staging: nodal sensitivity 30–40%, specificity **>90%** — positive reliable, negative does not exclude micrometastases.
- Choline (^{11}C) and **fluciclovine** (^{18}F, amino acid via LAT1/ASCT2) take up in inflammation and benign prostatic hyperplasia → poor locally.
- PSMA-PET mandatory before ^{177}Lu-PSMA therapy (target loss in neuroendocrine change).

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Calais	^{68}Ga-PSMA vs fluciclovine, biochemical recurrence	Detection 56% vs 26%	PSMA superior
CONDOR	^{18}F-DCFPyL, biochemical recurrence	Detection 61.5% at prostate-specific antigen (PSA) 0.8 ng/mL; management changed 64%	Detects at low PSA
EMPIRE-1	randomised controlled trial (RCT): fluciclovine-guided vs conventional salvage radiotherapy	12.5% absolute 3-year event-free gain	PET-guided planning improves outcome
VISION	RCT, metastatic castration-resistant, after androgen receptor pathway inhibitor and taxane	^{177}Lu-PSMA-617 approved	Radioligand therapy

9 · MCQ ONE-LINERS

1. PET photons: two 511 keV at 180°.
2. ^{99m}Tc half-life 6 h; photon 140 keV.
3. ^{18}F: shortest positron range, best PET resolution.
4. DTPA = pure filtration; measures GFR.

5. MAG3 = tubular secretion; agent of choice for obstruction.
6. DMSA retained by proximal tubule via megalin/cubilin.
7. Split function measured at 1–3 min.
8. Post-furosemide T½ >20 min = obstruction; <10 min = non-obstructed.
9. Catheterise neurogenic bladder during renography.
0. Captopril renography: ≥10% improvement off ACE inhibitor = renovascular.
1. Initial reflux study: fluoroscopic VCUG; follow-up: nuclear cystography.
12. DMSA defect persisting at 6 months = scar.
13. Labelled leucocytes unusable in renal transplant.
14. ^{111}In leucocytes preferred for GU infection — no urinary excretion.
15. Sestamibi hot = oncocytoma; girentuximab hot = clear cell RCC.
16. FDG-PET for residual seminoma >3 cm.
17. Teratoma is FDG-negative.
18. Prostate cancer is poorly FDG-avid.
19. PSMA-PET: nodal specificity >90%, sensitivity 30–40%.
20. †Superscan: absent renal activity.

8 Evaluation and Management of Haematuria

Reading time ≈ 27 min (slow pace)

1 · DEFINITIONS AND CLASSIFICATION

- **Gross haematuria (GH)** (frank, visible, macroscopic): seen by the naked eye. About half have a demonstrable cause.
- **Microhaematuria (MH)**: a laboratory sign, not a symptom: **≥3 red blood cells (RBCs) per high-power field (HPF)** on microscopy of a **single** urinalysis (UA) (American Urological Association (AUA)/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) 2020). Prevalence ≈ 6.5% of adults.
- One positive UA suffices because bleeding from cancer is intermittent. Specificity is gained by **risk-stratifying the work-up**, not by raising the threshold.
- **Dipstick is insufficient** (false positives: myoglobin, semen, bacterial peroxidases). Exclude **pigmenturia** (beetroot, rhubarb, phenazopyridine, porphyrins) and vaginal bleeding (catheterised specimen if needed).

Timing of GH in stream	Usual source
Initial	Urethra (distal to the external sphincter)
Terminal	Trigone, bladder neck, prostate
Total	Bladder or upper tract
Cyclic (female)	Endometriosis

Glomerular vs non-glomerular

Feature	Glomerular (medical renal)	Non-glomerular (urological)
RBC morphology	Dysmorphic	†Isomorphic
Casts / protein	RBC casts, proteinuria	†Absent; clots possible
Action	Nephrology referral ± renal biopsy, **and** risk-based urological evaluation	Risk-based evaluation

> **KEY POINT** Medical renal disease does not exempt the patient from urological work-up: bladder cancer may coexist.

2 · CAUSES

- One-third to two-thirds of MH evaluations find a cause. Commonest: **benign prostatic enlargement (BPE), calculi, urethral stricture**.
- **Malignancy ≈ 3%** of MH evaluations; >85% of these are bladder cancer. Upper tract urothelial carcinoma (UTUC) and renal cell carcinoma (RCC) are each ≈ 1 in 1,000.
- Others: UTUC (think Lynch syndrome), infection (including tuberculosis, schistosomiasis), immunoglobulin A (IgA) nephropathy, strictures, diverticulum, fistula, papillary necrosis (sickle cell, analgesics), arteriovenous malformation (AVM); exercise-induced (exclusion diagnosis: must clear after abstaining).
- A benign explanation is a **hypothesis to be proven**. If you treat it (e.g. urinary tract infection (UTI)), repeat the UA to confirm resolution.

Risk factors for urothelial cancer

In AUA risk stratification	Additional (can lift only to intermediate)
Age, male sex, smoking, degree of MH, persistence, history of MH/GH	Irritative lower urinary tract symptoms; prior pelvic radiotherapy; cyclophosphamide/ifosfamide; family history of urothelial cancer or Lynch; aromatic amine/benzene exposure (rubber, dyes, petrochemicals); chronic indwelling foreign body

- **Anticoagulants/antiplatelets**: haematuria prevalence and cancer yield are the same as in non-users. They may unmask lesions earlier, so evaluate exactly as for non-users.
- **Women** (male:female 4:1) present at higher stage because MH is misattributed to UTI or gynaecological causes.

3 · EVALUATION OF MICROHAEMATURIA (AUA/SUFU 2020)

Initial assessment

- History and examination (pelvic examination in women) for risk factors, renal disease and benign causes.
- Measure **blood pressure and serum creatinine**. Both screen for medical renal disease; creatinine also guides the choice of contrast imaging.
- Suspected UTI → culture → treat → repeat UA; persistent MH → evaluate.

Risk stratification

Factor	Low (ALL)	Intermediate (ANY)	High (ANY)
Age/sex	Women <50; men <40	Women 50–59; men 40–59	All **≥60**
Smoking (pack-years)	Never or <10	10–30	>30
RBCs/HPF	**3–10**	**11–25**	**>25**
History of GH	No	No	**Yes**
Persistent MH (previously low risk, unevaluated)	—	3–25 on repeat UA	>25 on repeat UA
Malignancy risk	0–0.4%	1–2.5%	≈ 6%
Evaluation	Shared decision: repeat UA within 6 months **or** cystoscopy + renal ultrasound (RUS)	**Cystoscopy + RUS**	**Cystoscopy + computed tomography (CT) urogram**

KEY POINT Asymmetry: low risk requires every criterion; intermediate or high needs only one. Men become intermediate risk a decade earlier than women (40 vs 50), but both are high risk at 60. Any GH history means high risk. Cyclophosphamide or radiotherapy alone does **not** make a patient high risk.

ALGORITHM
MH ≥3 RBCs/HPF → history, examination, blood pressure, creatinine → benign cause? treat → repeat UA (negative → release) → stratify → stratum-matched evaluation → positive: treat; negative: consider repeat UA within 12 months → negative: release; positive: shared decision on re-evaluation → re-evaluate if GH, rising MH or new symptoms

Why the work-up is stratified

- Bladder cancer dominates yield, so **white-light cystoscopy** is near-universal. Upper tract cancer is rare, so CT urography is reserved for high risk, where its yield justifies the radiation, contrast and cost.
- RUS detects renal cortical tumours but is poorly sensitive for UTUC; no radiation or contrast.

- **CT urogram** has the highest sensitivity/specificity. Phases: non-contrast (stones) → nephrographic (renal masses) → excretory (urothelium). Dose ≈ 20–30 mSv; a split-bolus protocol reduces it. Hydrate if glomerular filtration rate (GFR) <30 mL/min/1.73 m².
- CT contraindicated → magnetic resonance (MR) urogram. Nephrogenic systemic fibrosis is exceedingly rare at GFR ≥30; below that, use group II/III gadolinium. magnetic resonance imaging (MRI) also contraindicated → non-contrast CT or US + **retrograde pyelography**.
- Von Hippel–Lindau or similar syndrome → upper tract imaging regardless of stratum.
- **Cytology and urinary biomarkers are not recommended initially** (none adds value over cystoscopy + imaging in MH). MCQ UroVysion fluorescence in situ hybridisation (FISH) targets chromosomes 3, 7, 17 and 9p21. Consider cytology after a negative work-up if carcinoma in situ is suspected (irritative symptoms).

After a negative evaluation

- Repeat UA within 12 months; if negative, discontinue. **GH, a significant rise in MH or new symptoms → re-evaluate** (add cytology; CT if previously RUS only).

4 · GROSS HAEMATURIA

- Cancer yield is several-fold higher than in MH. In DETECT I, cancer was found in 13.8% with GH vs 3.1% with MH, and in 3.5% of GH patients under 45. **No age threshold**, no stratification.
- GH without trauma or culture-proven UTI → **cytology + cystoscopy + CT urography**. Culture-proven UTI → treat, then repeat UA.
- Acute: haemodynamics, blood count, coagulation, post-void residual.

Clot retention

- Large-bore (†22–24 Fr three-way) catheter → **complete manual clot evacuation** → continuous bladder irrigation (CBI) until clear. Why complete evacuation: residual clot keeps the bladder distended (so vessels cannot contract) and fuels fibrinolysis.
- Failure → cystoscopy under anaesthesia: clot evacuation (†Ellik evacuator), fulguration, biopsy of suspicious areas.
- †Decide on stopping or reversing anticoagulation together with the prescriber, weighing thrombotic risk; never attribute the bleeding to the anticoagulant without full evaluation.

5 · HAEMORRHAGIC CYSTITIS

Diffuse bladder mucosal bleeding after a full GH work-up excludes another source. In frail patients, weigh current risk against the risk of escalation.

Cause	Key points
Bacterial	Common; resolves with antibiotics
Viral	**BK polyomavirus** (commonest), adenovirus 11 and 35; after bone marrow or renal transplant and in children. Supportive care
Oxazaphosphorines (cyclophosphamide, ifosfamide)	Liver produces **acrolein** → excreted in urine → urothelial sloughing → oedema, fibrosis. Dose-dependent; onset within 48 h. Prophylaxis: **mesna** (binds acrolein), hyperhydration, CBI
Radiation	≈ 5% moderate–severe; onset 6 months–10 years. **Obliterative endarteritis** → ischaemia → necrosis, sloughing
Others	Busulfan, thiotepa, penicillins, danazol; amyloid

> **KEY POINT** Cyclophosphamide and pelvic radiation both cause bladder cancer. Exclude a tumour (cystoscopy, biopsy, cytology) before labelling the bleeding as haemorrhagic cystitis.

Intravesical agents

Agent	Mechanism / use	Precaution
Alum (first-line intravesical)	Astringent protein precipitation. **1%** (50 g in 5 L water) at 200–300 mL/h; no anaesthesia	Aluminium toxicity / encephalopathy in renal failure; variable durability
Silver nitrate 0.5–1%	Chemical coagulation	Upper tract precipitation: cystogram first
Aminocaproic acid	Inhibits plasminogen activation (urokinase) → blocks fibrinolysis	Clot-free bladder only (otherwise hard, organised clots); thromboembolism
Formalin 1–4%	Protein precipitation, capillary occlusion; most effective (80–90%)	Needs anaesthesia; fibrosis, small bladder, ureteric stricture. **Cystogram first**

- **Formalin technique**: cystogram to exclude reflux or perforation (if reflux, occlude ureters with catheters) → general/spinal anaesthesia → instil by gravity, catheter ≤15 cm above the pubis, up to 300 mL or bladder capacity → dwell 10–15 min with catheter on traction; protect urethra and skin. Use the lowest concentration first.
- **Hyperbaric oxygen therapy (HBOT)** for **stable** patients (radiation, cyclophosphamide): 100% O_2 at 2–3 atmospheres, ≈ 90 min × 30–40 sessions. It raises tissue oxygen → neovascularisation. Response 80–90%, but most relapse long-term. Bleeding during HBOT → bilateral nephrostomies ± ureteric occlusion, which diverts urokinase-rich urine away from the bladder.
- **Unstable or intractable** bleeding → **internal iliac (anterior division) angioembolisation**, selective and bilateral. Avoid the posterior division (superior gluteal artery → gluteal claudication/pain). Less reliable in radiation cystitis (ischaemic).
- Failure → **cystectomy + diversion**. Diversion without cystectomy leads to pyocystis and readmissions, so remove the bladder if feasible. Weigh high morbidity against deconditioning from delay.

ALGORITHM
Supportive care + clot evacuation/fulguration + CBI → alum → bleeding at 48 h → HBOT (stable) / embolisation (unstable) → formalin 1% → 4% → cystectomy + conduit

6 · PROSTATIC HAEMATURIA

- A diagnosis of exclusion after full GH work-up. **Benign prostatic hyperplasia (BPH)** is the commonest cause of GH in men >60. Mechanism: increased microvessel density driven by vascular endothelial growth factor (VEGF).
- **5α-reductase inhibitor (5-ARI)** (finasteride, dutasteride) ↓VEGF and microvessel density: ≈ 90% improve, including anticoagulated men; onset 2 weeks–9 months. Fewer recurrences than watchful waiting in randomised trials.
- Escalation: fulguration → transurethral resection of the prostate (TURP), photoselective vaporisation of the prostate (PVP), holmium laser enucleation (HoLEP) or simple prostatectomy (removes the friable transition zone) → **prostatic artery embolisation** (poor-risk or post-TURP bleeding).
- **Prostate cancer** bleeding usually means T4 invasion: androgen deprivation therapy (ADT) ± palliative radiotherapy → fulguration/channel TURP → nephrostomy → cystoprostatectomy.

KEY POINT Prostatic bleeding is vascular, so treatment is anti-vascular: 5-ARIs/ADT shrink vessels, surgery removes friable tissue, embolisation cuts the supply.

7 · URETHRAL AND UPPER TRACT BLEEDING

- **Urethrorrhagia**: blood at the meatus independent of voiding, or initial haematuria → lesion distal to the sphincter. Mainstays: **retrograde urethrogram (RUG)** and cystourethroscopy. Trauma is commonest in men; suspected injury or penile fracture with blood at the meatus → RUG before instrumentation. In women: caruncle, diverticulum.

- **Renal AVM** (≈ 75% acquired: biopsy, partial nephrectomy, PCNL, trauma) → **GH after a renal procedure = expeditious angiography** and selective embolisation.
- **Ureteroiliac artery fistula** (radiation, vascular surgery, chronic stents) → endovascular stenting.
- **Nutcracker syndrome**: left renal vein compressed between aorta and superior mesenteric artery → venous hypertension → bleeding.
- **Lateralising essential haematuria**: cystoscopy during bleeding lateralises → ureteropyeloscopy (gentle guidewire, low-pressure irrigation, every calyx) → biopsy/fulgurate.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
DETECT I	Prospective, UK, 3,556 referrals	Cancer 13.8% GH vs 3.1% MH; 3.5% in GH <45 years	Investigate GH at any age
RICH-ART	Randomised HBOT vs standard care, radiation cystitis	Improved urinary symptom score	Supports HBOT

9 · MCQ ONE-LINERS

1. MH threshold (AUA/SUFU 2020): ≥3 RBCs/HPF on a single microscopic UA.
2. Dipstick alone cannot diagnose MH: myoglobin and semen give false positives.
3. Malignancy in ≈ 3% of MH evaluations; most are bladder cancer.
4. Low risk: women <50, men <40, <10 pack-years, 3–10 RBCs/HPF, no GH.
5. High risk: age ≥60, >30 pack-years, >25 RBCs/HPF or any GH history.
6. High risk evaluation: cystoscopy plus CT urogram.
7. Cytology and biomarkers are not recommended in initial MH evaluation.
8. Anticoagulated patients with haematuria are evaluated exactly like non-users.
9. GH: cytology + cystoscopy + CT urography, no age threshold.
0. CT contraindicated: MR urogram, or non-contrast imaging plus retrograde pyelography.
1. Commonest viral haemorrhagic cystitis: BK polyomavirus.
2. Cyclophosphamide cystitis is caused by acrolein; mesna neutralises it.
3. Radiation cystitis pathology: obliterative endarteritis with ischaemia.
4. First-line intravesical agent: 1% alum; avoid in renal failure (aluminium toxicity).
5. Cystogram before formalin to exclude vesicoureteric reflux.
6. Stable radiation cystitis refractory to alum: hyperbaric oxygen.
7. Embolise the anterior division of the internal iliac; posterior division causes gluteal pain.
8. BPH is the commonest cause of GH in men over 60; mediated by VEGF.
9. GH after renal biopsy or partial nephrectomy: angiography for AVM/pseudoaneurysm.
20. Nutcracker: left renal vein between aorta and superior mesenteric artery.
21. Traumatic blood at meatus: urethrogram before catheterisation.

9 Occupational Health in Urology

Reading time ≈ 12 min (slow pace)

1 · PHYSICAL ERGONOMICS

- Work-related musculoskeletal disorders (MSDs) affect **46–90%** of urologists.
- Site follows technique: **open** → neck, shoulder, low back; **robotic/endoscopic** → eye strain, neck, hand/wrist.
- Non-modifiable risks: inexperience, short stature, female sex, arthritis.
- MSD is a **cumulative-load** problem from static, awkward, repetitive postures. Every fix either restores neutral posture or interrupts the load — which is why cheap measures work.

Domain	Key mitigation
Intraoperative	Monitor straight ahead for gaze-down; table height for modality; postural resets; microbreak stretches; antifatigue mats
Equipment	Lightest lead, removed between cases; **two-piece or belted lead** (weight to hips); light steep-declination loupes (neck <20°)
Patient handling	≥2 people, coordinated lift, mechanical aids; pivot, never twist under load
Practice	Mix case length and modality; alternate operating days

KEY POINT The problem is shoulder loading, not one-piece lead per se.

Modality	Height reference	Posture targets
Open	Working area **5–10 cm below elbow**	Neck flexion <25°; torso–arm <30°
Laparoscopic	Table ≈ surgeon's **pubic symphysis** (handles at/below elbow)	Monitor ahead, top at eye level; wrist neutral
Endourology	Target between symphysis and umbilicus	Endoscope monitor ahead, top at eye level
Robotic console	Adjustable chair	Forearm parallel to floor; back flexion <15°; knee 90–120°

Constant in all: **elbow 90–120°**, gaze-down monitor, relaxed shoulders.

- **Microbreaks** reduce pain and improve focus without lengthening cases; exercise and neck-strength training are protective.
- **Presbyopia** (≈ a decade into practice): general multifocals force neck extension → use **workspace lenses** (near + intermediate), or lower the monitor.
- **Desk**: monitor top at/below eye level (gaze-down ≈ 20°), 50–100 cm away; laptops need a riser.

2 · OCCUPATIONAL EXPOSURES

Common hierarchy: remove/reduce the source → interpose a barrier → protect the individual.

Infection and sharps

- Hand hygiene, personal protective equipment (PPE), immunisation. Centers for Disease Control and Prevention (CDC): vaccinate against **hepatitis B, influenza, measles–mumps–rubella**.
- Bloodborne risk from sharps injuries: human immunodeficiency virus (HIV), hepatitis B and C viruses. Use safety-engineered devices and post-exposure protocols.

Smoke and chemicals

- Surgical smoke (particulates, gases, bioaerosols) → evacuation, ventilation.
- **CO_2 laser for condylomata**: N95 mask + evacuator tip within **5 cm** of the site (human papillomavirus).

- **Intravesical chemotherapy** in theatre: two pairs of chemotherapy gloves, impermeable gown, face shield; closed-system transfer device protects against spills.

Ionising radiation

- Fluoroscopy → cataract, infertility, cancer. Lead apron, **thyroid shield**, lead glasses and gloves; As Low As Reasonably Achievable (ALARA).
- Institutions provide **dosimeters** and training.
- **Pregnant surgeon**: fetal dosimeter at waist *under* the apron; maternity or double-thickness lead; minimise radiation cases.

Principle	Strategy
Less beam	Low-dose default; pulsed fluoroscopy at lowest rate (1 pulse/s); collimate; last-image hold; minimise magnification
Less time	Spot not continuous screening; surgeon foot-pedal control
Geometry	Source far from patient, **image intensifier close** (less scatter); maximise own distance

> **KEY POINT** Magnification feels harmless but increases dose.

Distraction and noise

- Protect critical steps: pause music, no non-urgent pages, no breaks — state it at the briefing.

3 · PHYSICAL HEALTH OF SURGEONS

- **Atherosclerotic cardiovascular disease** is the most threatening modifiable cause of death; suicide and injury also top-five.
- Surgeons use primary care least (time, culture, self-diagnosis, fear of judgement). Treat the surgeon as a patient: lipids, blood pressure, age-appropriate cancer screening.
- **Sleep**: **7–9 h** (age 25–64).
- **Fitness (CDC)**: **150 min moderate or 75 min vigorous/week + strength on 2 days**.

4 · MENTAL AND EMOTIONAL HEALTH

- **Burnout** (Freudenberger): emotional exhaustion, depersonalisation, low personal accomplishment. Urology ≈ **50%**. Risks: more hours, mid-career, chronic MSD pain.
- Levers: autonomy, efficiency, less administration, peer support; **≥20%** time on meaningful work lowers burnout.
- **Second victim / traumatised caregiver**: post-traumatic stress, guilt, shame after an adverse event. Response: psychological first aid and **peer support**.

Stress continuum zone	Features	Coping
Ready (green)	Balanced, focused	Exercise, sleep, mindfulness
Reacting (yellow)	Irritable, poor concentration	Self-care, time management, delegate
Injured (orange)	Anxiety, substance misuse, **burnout**	Support, less workload, time off
Ill (red)	Panic, severe depression, dependence	Mental-health professional, sick leave

- **Cognitive behavioural therapy (CBT)**: identify and challenge distorted thoughts; with mindfulness it builds resilience.
- **Imposter syndrome** is a cognitive distortion (not a trait), commonest in trainees, responds to CBT.
- Tactics: overgeneralisation → self-compassion; catastrophising → double standard; labelling, tunnel vision → exploratory questions.

5 · MCQ ONE-LINERS

1. MSD prevalence in urologists: 46–90%.
2. Open surgery: working area 5–10 cm below elbow.

3. Laparoscopic table height ≈ pubic symphysis.
4. Elbow angle target in all modalities: 90–120°.
5. Robotic console: back flexion <15°.
6. Loupes: steep declination, neck <20°.
7. Presbyopic surgeon: workspace lenses, not progressives.
8. CDC vaccines for staff: hepatitis B, influenza, measles–mumps–rubella.
9. CO_2 laser condylomata: N95, evacuator within 5 cm.
0. Intravesical chemotherapy: double chemotherapy gloves, closed-system transfer.
1. Pregnant surgeon: fetal dosimeter at waist under apron.
12. Image intensifier close to patient; source far.
13. Magnification increases radiation dose.
14. Top modifiable killer of surgeons: atherosclerotic cardiovascular disease.
15. Recommended sleep, age 25–64: 7–9 h.
16. CDC exercise: 150 min moderate weekly + strength twice.
17. Burnout coined by Freudenberger.
18. Burnout sits in the orange (Injured) zone.
19. Imposter syndrome responds to CBT.
20. Catastrophising tactic: double standard.

10 Ethics and Informed Consent

Reading time ≈ 8 min (slow pace)

1 · PRINCIPLES OF MEDICAL ETHICS

- **Hippocratic Oath**: beneficence, non-maleficence, confidentiality, accountability; "I will not cut persons labouring under the stone" is the urological line. Criticised as paternalistic.
- **Bioethics** coined by Potter (1970); **Beauchamp and Childress (1979)** → four principles.

Principle	Meaning and consequence
Autonomy	Deliberated self-rule; patient may choose against medical advice. Founds consent and confidentiality; bounded by justice and others' autonomy
Beneficence	Optimise benefit, seen through the patient's values
Non-maleficence	Primum non nocere; be trained to deliver benefit and know risk magnitude
Justice	Fair distribution; equal treatment ≠ just treatment

All four are equal and prima facie binding; a **dilemma** is a conflict between them requiring prioritisation.

Jonsen's Four-Box method

Box	Principle(s)	Core question
Medical indications	Beneficence, non-maleficence	Diagnosis, prognosis, goals, chance of success; if risk–benefit equivocal, defer to patient
Patient preferences	Autonomy	Informed, capacitated, voluntary choice; advance directive; surrogate
Quality of life	All three above	Patient's values, not provider bias
Contextual features	Justice	Family, finance, culture, law, conflicts of interest

2 · INFORMED CONSENT

Legal milestones

Case	Principle
Mohr v Williams, 1905	Consulting a doctor ≠ consent
Schloendorff, 1914	Patient may limit consent (examination under anaesthesia ≠ surgery)
Salgo, 1957	Term "informed consent" coined
Canterbury v Spence, 1972	**Reasonable-patient** disclosure standard

Trajectory: battery → negligence → patient-centred standard.

Components of valid consent

1. Problem, proposed treatment and alternatives, including **no treatment**.
2. Risks and benefits versus alternatives, with uncertainty.
3. **Assessment of understanding** — the most commonly missed element.
4. Explicit consent or refusal, voluntarily given.

- Lay language; present options before recommending. "Material risk": proposed ≈ **1%** cut-off, rarer risks by severity.
- Disclosing uncertainty **improves** trust.
- **Decision aids** improve knowledge and reduce decisional conflict; **repeat-back** works but is slow.

- **Shared decision making** is the only bidirectional model: clinician gives options and recommendation; patient gives values.

3 · CAPACITY, EXCEPTIONS AND SPECIAL SITUATIONS

- **Emergency**: consent presumed for an unconscious patient. A **conscious, capacitated patient may refuse** even life-saving treatment.
- **No capacity**: legal surrogate uses **substituted judgement** (what the patient would want), then best interests. Surrogate not following wishes → ethics committee → court-appointed conservator.
- **Minors** (<18 yr): parents consent; child participates as able (assent).
- **End of life**: early goals-of-care talks, do-not-resuscitate (DNR) orders, advance directives. Patient Self-Determination Act (1990): institutions must inform patients of the right to an advance directive.
- **Research**: written, understandable consent covering purpose, risks, alternatives, voluntariness; institutional review board may waive for minimal-risk or de-identified data. Vulnerable groups need extra protection.

> **KEY POINT** Consent law protects autonomy by requiring disclosure; its weak point is comprehension.

4 · MCQ ONE-LINERS

1. Four principles: Beauchamp and Childress, 1979.
2. "Bioethics" coined by Potter.
3. Autonomy is bounded by justice.
4. Quality-of-life box draws on three principles.
5. Contextual-features box maps to justice.
6. Salgo coined "informed consent".
7. Canterbury v Spence: reasonable-patient standard.
8. Schloendorff: patient may limit consent.
9. Commonest missed consent element: assessing understanding.
0. Proposed material-risk threshold ≈ 1%.
11. Disclosing uncertainty improves trust.
12. Only bidirectional model: shared decision making.
13. Surrogate standard: substituted judgement first.
14. Conscious patient may refuse life-saving treatment.
15. Decision aids reduce decisional conflict.
16. Repeat-back is effective but time-consuming.
17. Patient Self-Determination Act mandates advance-directive information.
18. Research consent may be waived for de-identified data.
19. Children give assent; parents consent.
20. "Not cut for stone" is the Oath's urological line.

11 Perioperative Care in Urologic Surgery

Reading time ≈ 25 min (slow pace)

1 · PREOPERATIVE EVALUATION AND TESTING

- Goal: detect undiagnosed or undertreated disease that changes the operative course.
- Routine testing predicts morbidity less well than American Society of Anesthesiologists (ASA) status → test **selectively**.
- Electrocardiogram (ECG): age >40 y or cardiac history. Coagulation tests: anticoagulation or bleeding history only. Chest radiograph: cardiopulmonary disease only. Pregnancy test: every woman of childbearing age.

ASA class	Definition
I / II	Healthy / mild systemic disease
III	Severe systemic disease, **limits activity**, not incapacitating
IV	Incapacitating, **constant threat to life**
V	Moribund, not expected to survive 24 h
VI	Brain-dead organ donor
E	Emergency *suffix* (I–V, never VI)

ASA grades physical status, not surgical risk, yet independently predicts mortality.

2 · CARDIAC RISK ASSESSMENT

Three independent axes: **clinical markers** (how sick the heart is), **functional capacity** (reserve) and **surgery-specific risk** (stress). Good reserve + low-stress surgery → no further testing whatever the history.

Major predictors	Intermediate	Minor
Myocardial infarction (MI) <1 month; unstable angina; **decompensated** heart failure; significant arrhythmia; severe valve disease	Mild angina; **prior** MI; **compensated** heart failure; diabetes; renal insufficiency	Age; abnormal ECG; prior stroke; uncontrolled hypertension

- **Revised (modified) Cardiac Risk Index (RCRI)** — 6 predictors, **1 point each (unweighted)**: ischaemic heart disease, heart failure, cerebrovascular disease, high-risk surgery, preoperative insulin, creatinine **>2 mg/dL**. Each added predictor raises risk.
- **Goldman index** — 9 *weighted* factors; heaviest S3/jugular venous distension (11).
- **Functional capacity** in metabolic equivalents (METs): **4 METs** ≈ climbing two flights of stairs; ≥4 METs → no invasive cardiac work-up. Duke Activity Status Index (DASI) quantifies it.
- Surgical risk: **high** — major emergency, big fluid shifts; **intermediate** — intraperitoneal, laparoscopic, robotic; **low** — endoscopic, superficial.
- After MI: old 3–6-month rule abandoned; waiting **4–6 weeks** before elective surgery is reasonable.
- †**Frailty** (weight loss, weakness, slow gait, exhaustion, low activity) predicts complications, institutional discharge and mortality independent of age — screen elderly patients.

3 · PULMONARY, HEPATIC AND SPECIAL POPULATIONS

- **Pulmonary**: forced expiratory volume in 1 s (FEV_1) **<0.8 L or <30%** predicted = high risk. Restrictive fares better than obstructive disease — preserved expiratory flow allows effective cough.
- **Liver**: Child class mortality A 10%, B 30%, C 76–82%; Model for End-Stage Liver Disease (MELD) is more objective.
- **Elderly**: age independently predicts morbidity; assess cognition, function, mood preoperatively.

- **Obesity**: examination underestimates cardiac dysfunction; >3 coronary risk factors → non-invasive cardiac testing.
- **Pregnancy** (usually stones): fetal malformation threshold **10 cGy** → ultrasound first (computed tomography ~1 cGy). **Second trimester safest**; first — teratogenesis; third — preterm labour. Opioids acceptable; **avoid non-steroidal anti-inflammatory drugs (NSAIDs)** (premature ductus closure).
- **Nutrition**: >9 kg loss in 3 months = severe malnutrition. Enteral preferred (preserves mucosal barrier); parenteral nutrition 7–10 days preoperatively helps only the severely malnourished — in the well-nourished it adds sepsis.

4 · OPTIMISATION OF COMORBIDITY

- **Cardiac**: only β-blockade (not nitrates or calcium-channel blockers) improved outcome.
- **Smoking**: stop **≥8 weeks** before. Stopping <8 weeks may *increase* risk — cough reflex and pulmonary toilet fall before mucociliary function recovers. >6 months → non-smoker risk.
- **Chronic obstructive pulmonary disease (COPD)**: bronchodilators, treat infection, physiotherapy.
- **Diabetes**: hyperglycaemia impairs healing; hypoglycaemia is masked under anaesthesia. Stop long-acting oral agents; short-acting/sliding-scale insulin; target **80–250 mg/dL**; insulin pump — continue basal rate. †Withhold metformin on the day of surgery and sodium-glucose cotransporter-2 (SGLT2) inhibitors 3 days before (euglycaemic ketoacidosis).
- **Thyroid**: defer elective surgery until euthyroid. Hyperthyroid: continue antithyroid drugs and β-blocker (storm risk).

Corticosteroids (hypothalamic–pituitary–adrenal axis)

- Prednisone **≤5 mg/day**: not suppressed — usual dose only. **>20 mg/day for >3 weeks** or cushingoid (or suppressive dose within 1 year): presume suppressed → supplement.

Stress dose: hydrocortisone **50–100 mg intravenously** before induction → 25–50 mg 8-hourly for 24–48 h. Minor procedures under local anaesthesia need none.

5 · ANTITHROMBOTIC MANAGEMENT

- **Warfarin**: half-life 36–42 h → stop **5 days** before, aim international normalised ratio (INR) **<1.5**.
- Bridging (American College of Chest Physicians [ACCP]): moderate- and high-risk patients → therapeutic low-molecular-weight heparin (LMWH) or intravenous unfractionated heparin (UFH). †Last LMWH dose 24 h before. Direct oral anticoagulants need no bridging.

Thrombotic risk	Mechanical valve	Atrial fibrillation ($CHADS_2$ score)	Venous thromboembolism (VTE)
High	**Any mitral**; caged-ball/tilting-disc aortic; stroke <6 months	5–6; stroke <3 months; rheumatic valve	<3 months; severe thrombophilia (protein C/S, antithrombin, antiphospholipid)
Moderate	Bileaflet aortic + risk factor	3–4	3–12 months; recurrent; active cancer
Low	Bileaflet aortic, no risk factors	0–2	Single VTE >12 months

Drug	Stop before surgery
Aspirin, clopidogrel (irreversible)	**7–10 days**
†Prasugrel / ticagrelor	7 / 5 days
†Apixaban, rivaroxaban	48 h (72 h if high bleeding risk)
†Dabigatran (renally cleared)	48–96 h depending on creatinine clearance

Coronary stents

- Premature antiplatelet withdrawal → **25–50%** risk of significant MI (stent thrombosis).
- **Bare-metal**: dual therapy 6 weeks; delay elective surgery ≥30 days. **Drug-eluting**: dual therapy 12 months; delay >1 year.
- Surgery cannot wait (cancer) → continue antiplatelets; liaise with cardiology.

6 · BOWEL PREPARATION, ILEUS AND ERAS

- **Mechanical bowel preparation**: no reduction in leak, abscess or mortality; surgical site infection (SSI) may be higher; cystectomy series show equivalent SSI without it. Polyethylene glycol (non-absorbable) if renal, cardiac or hepatic disease.
- Oral antibiotics (neomycin + erythromycin) reduce wound infection; risk *Clostridioides difficile*.
- **Ileus** >72–96 h → look for obstruction or retroperitoneal haemorrhage. Reduce with fewer opioids, selective nasogastric tubes, electrolyte correction, gum chewing, **alvimopan** (peripheral μ-opioid antagonist; shortened cystectomy stay by >2 days).

ALGORITHM

Enhanced recovery after surgery (ERAS): †counselling, carbohydrate loading, clear fluids to 2 h, no bowel preparation → normothermia, goal-directed fluids, opioid-sparing analgesia → no nasogastric tube, early feeding, alvimopan, early mobilisation.

7 · INTRAOPERATIVE ENVIRONMENT AND POSITIONING

- **Hypothermia**: induction vasodilatation redistributes core heat; then radiation/conduction. Normothermia = **36–38 °C**. Even 1–2 °C loss → more blood loss (impaired clotting) and SSI (vasoconstriction → tissue hypoxia).
- Preventable injuries: **retractor** (blades on psoas → **femoral neuropathy**), **thermal** (light cables), **positioning** (stretch, compression, ischaemia).

Position	Key rules
Supine	Arm abduction ≤90° (brachial plexus); pad ulnar groove and spiral groove (radial)
Lithotomy	Move legs together; hips flexed 80–100°, abducted 30–45°; pad fibular head (**common peroneal**); hamstring stretch → sciatic; transient neuropathy usually resolves by 6 months
Flank	Axillary roll just caudal to axilla; arms <90°; tilt table before draping to test security
Steep Trendelenburg	Raised intracranial/central venous pressure; avoid shoulder braces (plexopathy)

8 · ANTIBIOTIC PROPHYLAXIS

- Need = patient susceptibility × wound class × consequence of infection.
- Wound class: clean → clean-contaminated (controlled urinary/genital/gut entry — **most urology**) → contaminated (**transrectal biopsy**) → dirty-infected.
- **Timing**: within **60 min** before incision; giving it at or after incision doubles SSI. **Duration** ≤24 h (longer → *C. difficile*, resistance). Redose if >4 h.
- Hair removal: clippers, not razors.

Procedure (American Urological Association, AUA)	First choice	Duration
Cystoscopy, sterile urine	Only with host risk factors; trimethoprim-sulfamethoxazole (TMP-SMX)	Single
Transurethral resection, ureteroscopy, nephrectomy, prostatectomy, small-bowel conduit	Cefazolin (or TMP-SMX endoscopically)	Single
Transrectal biopsy	Fluoroquinolone or cephalosporin + aminoglycoside (resistance)	Single

Procedure (American Urological Association, AUA)	First choice	Duration
Percutaneous renal surgery	1st/2nd-generation cephalosporin	**≤24 h**
Colon conduit	Cefazolin + metronidazole; cefoxitin	Single
Prosthesis (sphincter, penile, sacral)	Aminoglycoside + cephalosporin or vancomycin	**≤24 h**
Shock-wave lithotripsy	Only with risk factors	Single
Penile (circumcision)	Likely not required	—

KEY POINT Prophylaxis works only if tissue levels exist at the moment of contamination — hence timing, single dose and redosing. Penicillin allergy → clindamycin.

9 · VTE PROPHYLAXIS

- VTE is the leading **non-surgical** cause of death after urological surgery; highest after radical cystectomy.
- Pharmacological prophylaxis raises minor, not major, bleeding (transurethral prostate resection and partial nephrectomy bleed more).
- Agents: pneumatic compression (no bleeding; less effective in high risk); LMWH enoxaparin **40 mg subcutaneously daily** (30 mg if creatinine clearance <30 mL/min; not reversible); UFH **5000 U 8-hourly** (reversible, safe in renal failure).

AUA risk	Definition	Prophylaxis
Low	Minor surgery, <40 y, no risk factors	Early ambulation
Moderate	Minor surgery + risk factors; 40–60 y	UFH 12-hourly, enoxaparin, or compression if bleeding risk
High	>60 y; 40–60 + risk factors	UFH 8-hourly, enoxaparin, or compression if bleeding risk
Highest	Multiple factors (cancer, prior VTE)	**Drug + compression**

†**Caprini score**: weighted points (age, major surgery, cancer 2; prior VTE, thrombophilia 3; stroke, arthroplasty 5); ≥5 = highest risk. Extended 28-day LMWH after radical pelvic cancer surgery (e.g. cystectomy) is recommended.

10 · ANAESTHESIA

- **Spinal**: hypotension (sympathetic block) 10–40%. **Epidural** with general anaesthesia reduces mortality and organ morbidity.
- Nitrous oxide expands closed gas spaces (avoid in obstruction, pneumothorax). Residual neuromuscular block: train-of-four ratio must reach **0.9** before extubation.

11 · TRANSFUSION

- Haematocrit **<21%** → transfusion often indicated; 21–30% individualise (coronary, lung disease); >30% rarely.
- Platelets if **<50,000** before surgery (<100,000 for high-risk surgery); fresh frozen plasma for warfarin reversal, INR >1.5 with bleeding.
- Transfusion-related acute lung injury (TRALI) = commonest transfusion death; platelets carry the highest bacterial risk.

12 · POSTOPERATIVE FLUIDS, ANALGESIA AND DELIRIUM

- †**Fluids**: euvolaemia with balanced crystalloid — overload causes oedema and ileus; stop once drinking.

- **Analgesia — opioid-sparing**: NSAIDs block prostaglandin-mediated nociception (avoid in renal insufficiency, peptic ulcer); epidural beats parenteral opioids; pre-incision local infiltration reduces sensitisation.
- Opioids: ileus, respiratory depression, persistent use; ureteroscopy can be opioid-free.
- **Delirium**: never dismiss as "sundowning" — may be the first sign of sepsis or metabolic derangement → look for the cause. Prevent: quiet, familiar surroundings, sleep, regional anaesthesia, minimal opioids; †avoid benzodiazepines.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Mangano	Randomised trial, atenolol	Better cardiac outcomes	β-blockade the only proven cardiac optimisation
Nichols–Condon	Mechanical prep ± oral antibiotics	Sepsis 43% → 9%	Oral neomycin + erythromycin
Kurz	Randomised, 200 colorectal	Hypothermia 3× wound infection	Maintain normothermia

14 · MCQ ONE-LINERS

1. ASA III vs IV: limits activity vs constant threat to life.
2. "E" suffix never attaches to ASA VI.
3. RCRI: six equally weighted predictors; creatinine cut-off 2 mg/dL.
4. Stopping smoking <8 weeks preoperatively may increase pulmonary risk.
5. Safest trimester for surgery: second.
6. NSAIDs in pregnancy: premature ductus arteriosus closure.
7. Warfarin stopped 5 days before; target INR <1.5.
8. Any mitral prosthesis = high thrombotic risk.
9. Drug-eluting stent: defer elective surgery >1 year.
10. Femoral neuropathy: retractor on psoas, not hip position.
11. Lithotomy: common peroneal nerve at fibular head.
12. Transrectal prostate biopsy is a contaminated wound.
13. Antibiotic within 60 min of incision; redose after 4 h.
14. Only percutaneous renal surgery and prosthetics extend prophylaxis to ≤24 h.
15. Highest AUA VTE tier: drug plus mechanical prophylaxis.
16. Alvimopan: peripheral opioid antagonist shortening ileus after cystectomy.
17. Commonest cause of transfusion death: TRALI.
18. Train-of-four ratio ≥0.9 for safe extubation.

12 Fundamentals of Intraoperative Patient Safety

Reading time ≈ 12 min (slow pace)

1 · BACKGROUND

- Institute of Medicine, *To Err Is Human* (1999): 44,000–98,000 deaths/year from medical error.
- ~50% of adverse events involve surgery, ~50% of those preventable; the surgeon leads and is ultimately responsible.

2 · CHECKLISTS AND THE UNIVERSAL PROTOCOL

A checklist is a memory aid borrowed from aviation: it standardises steps because complexity, stress and fatigue make error almost inevitable. World Health Organization (WHO) Surgical Safety Checklist (2008, 19 items): death **↓47%**, major complications **↓36%**.

Sign in (before induction)	Time out (before incision)	Sign out (before leaving theatre)
Identity, site, procedure, consent; site marked; anaesthesia check; pulse oximeter; allergy; difficult airway; blood loss **>500 mL** (7 mL/kg child)?	Team introductions; confirm patient, site, procedure; anticipated critical events; antibiotic within **60 min**; imaging displayed	Procedure recorded; instrument, sponge, needle counts; specimen labelled; equipment problems; recovery concerns

ALGORITHM
Joint Commission Universal Protocol (2004): preprocedure verification → site marking → time-out

- **Wrong-site surgery** (wrong side, patient or part) is a never event, ~1 in 112,000; up to 30% of root causes arise before admission.
- **Verification**: ≥2 identifiers (name, date of birth, record number); consent names surgeon, procedure and contingent procedures; professional interpreter, never family.
- **Marking**: where laterality exists (nephrectomy, orchidectomy); permanent, on skin, by someone present at the time-out, patient awake; mark only the correct side (never "X"). Lower-tract endourology: mark wristband, not meatus.
- **Time-out**: all participate, no distractions; repeat at every change of surgeon or procedure. Failure mode: "going through the motions".

KEY POINT Swiss cheese model: each barrier has holes; harm occurs when holes align (latent failures — hierarchy, staffing, time pressure — plus active errors). High-risk settings: emergencies, unusual anatomy, multiple surgeons or procedures. Checklists do not delay starts and help most in urgent cases.

3 · POSITIONING INJURIES → *CH 11*

- †**Lithotomy**: common peroneal nerve compressed at the fibular head by stirrups → foot drop; sciatic stretch with hip flexion and knee extension; femoral/lateral femoral cutaneous with excessive hip flexion/abduction.
- †**Well-leg compartment syndrome**: prolonged lithotomy (>4 h), elevated legs and Trendelenburg reduce calf perfusion → reperfusion swelling. Prevent by limiting time or lowering legs periodically; suspect pain out of proportion, raised creatine kinase → urgent fasciotomy.
- †**Flank**: axillary roll protects brachial plexus; pad dependent peroneal nerve and pressure points; rhabdomyolysis with long cases and obesity.

4 · RETAINED SURGICAL ITEMS (RSI)

- RSI = item found **after skin closure**, even in theatre; never event (1 in 2800–10,000).
- Commonest: **sponges** (~two-thirds) → drains (Penrose) → needles.

- Risks: emergency, unplanned change of procedure, obesity, multiple teams, long case, incorrect count.
- Count discrepancy → **~100-fold** risk; delay closure until resolved. A correct count does not exclude RSI → radiography; radiofrequency-tagged sponges are highly accurate.

5 · SHARPS SAFETY

- About one-third of injuring sharps recontact the patient.
- **Double gloving** (inner layer rarely breached), **blunt needles** for fascia, **neutral zone** (no hand-to-hand passing).

6 · BURNS AND FIRE

Fire triangle	Owner	Examples
Oxidiser	Anaesthesia	Oxygen; nitrous oxide (equally oxidising)
Fuel	Nursing	Alcohol prep — dry **≥3 min**, avoid pooling; drapes, hair
Ignition	Surgeon	Monopolar diathermy (~90%); light cables; cracked laser fibres

- Most fires: head/neck, under monitored sedation. Silverstein score (open oxygen + ignition + site above xiphoid): 3 = high.
- Buccal graft harvest: inspired oxygen ≤30%, stop nitrous oxide, warn anaesthesia before diathermy.
- †Diathermy burns: full return-pad contact over muscle away from metal implants; prefer bipolar with pacemakers.

ALGORITHM
Fire: stop → saline/water → remove drapes → extinguisher → **RACE** (Rescue, Alert, Confine — gases off, Extinguish/evacuate)

7 · DISTRACTIONS AND COMMUNICATION

- Alarms, music, phones cause errors; "sterile cockpit" during critical steps (e.g. hilar clamping).
- Communication failure underlies up to **70%** of adverse events, especially at handoffs. Non-technical skills predict outcome as much as technical skill.
- **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety): structure, communication (**SBAR** — situation, background, assessment, recommendation; call-outs; check-backs), leadership (briefs, huddles, debriefs), situation monitoring (STEP — status of patient, team, environment, progress; I'M SAFE self-check), mutual support (**two-challenge rule**, **CUS** — concerned, uncomfortable, safety issue).
- Simulation-based team training changes behaviour; didactic teaching alone does not.

8 · MCQ ONE-LINERS

1. WHO checklist: three pauses — induction, incision, exit.
2. Sign in asks about blood loss >500 mL.
3. Universal Protocol: verification, marking, time-out.
4. RSI definition: discovered after skin closure.
5. Commonest RSI: surgical sponge.
6. Count discrepancy raises RSI odds ~100-fold.
7. Commonest fire ignition source: monopolar diathermy.
8. Silverstein score 3 = high fire risk.
9. RACE: Rescue, Alert, Confine, Extinguish.
10. Lithotomy foot drop: common peroneal nerve at fibular head.
11. Well-leg compartment syndrome: prolonged lithotomy; treat by fasciotomy.
12. CUS and two-challenge rule belong to mutual support.
13. STEP monitors situation; I'M SAFE monitors oneself.
14. Didactic team training alone does not improve outcomes.
15. Double gloving: inner glove rarely breached.
16. About one-third of injuring sharps recontact the patient.

13 Incisions, Exposure and Drains

Reading time ≈ 11 min (slow pace)

1 · PRINCIPLES

- Open versus minimally invasive: decided by pathology size and complexity, patient preference and surgeon skill.
- Stay in avascular planes (linea alba, muscle-splitting) to limit bleeding and nerve injury. Transverse and muscle-splitting incisions trade a smaller field for less pain and fewer hernias.

2 · ABDOMINAL INCISIONS

Incision	Indications	Notes
Lower midline	Radical/simple prostatectomy, pelvic lymph node dissection (PLND), cystectomy, diversion, specimen extraction	Commonest urological incision
Upper / complete midline	Retroperitoneal lymph node dissection (RPLND), bilateral nephrectomy (polycystic), nephroureterectomy, trauma	Most versatile; poor in obesity, multiple laparotomies
Pfannenstiel	Pelvic surgery, donor/laparoscopic specimen extraction	Rectus separated not cut → least hernia, pain
Gibson	Renal transplant, iliac vessels, lower ureter	Muscle-splitting, extraperitoneal; hockey-stick extension → more hernias

Abdominal wall closure

- Midline: through linea alba (avoids rectus vessels); closed in **one layer**, under more tension than Langer-line incisions.
- Fascia: running or interrupted figure-of-eight, strong slowly absorbable (polydioxanone or braided polyglycolide); bites **1 cm from edge, 1 cm apart** → *Ch 14 small-bites*.
- No technique is superior for dehiscence/hernia, but **avoid continuous rapidly absorbable** suture (more hernia). †Suture length : wound length ≥ **4 : 1**.
- Other closures: Smead-Jones (near-far figure-of-eight), mass closure.
- **Pfannenstiel**: transverse incision two fingerbreadths above pubis → anterior sheath opened transversely → rectus freed above and below → midline opened; extend laterally or into a Gibson for the ureter.
- **Gibson**: external oblique → internal oblique → transversus → peritoneum swept medially → ureter found crossing iliac artery; two-layer running absorbable closure.

3 · APPROACHES TO THE KIDNEY

Incision	Indications	Trade-off
Midline transperitoneal	Trauma, inferior vena cava (IVC) thrombus, bilateral disease, horseshoe	Fast, bilateral, early vascular control; poor upper retroperitoneum
Subcostal / chevron	Radical nephrectomy, bilateral tumours, IVC thrombus, invasion	Early vascular control; liver/spleen, bowel injury
Thoracoabdominal (9th–10th rib)	Level III thrombus, T4 or large upper-pole/adrenal mass	Best exposure; pain no worse than flank
Flank – 12th supracostal / 11th transcostal	Partial/simple nephrectomy, xanthogranulomatous pyelonephritis (XGP)	Extraperitoneal; **pleura** at risk; transcostal → flank bulge

Incision	Indications	Trade-off
Transverse / paramedian	Paediatric Wilms, pyeloplasty / avoiding stoma	Paramedian: nerve injury, hernia

> **KEY POINT** Anterior = bilateral reach and early pedicle control at the cost of bowel; flank = extraperitoneal at the cost of pleura. Rib-sparing (supra-11th, infra-11th) variants avoid rib-resection pain.

- **Pleural tear**: red rubber catheter through tear → purse-string → end under water, sustained inflation → withdraw as bubbling stops; large tear → chest tube. Chest radiograph only if tear repaired/suspected.
- **Thoracoabdominal closure**: chest tube (omit if parietal pleura intact) → costal cartilage reapproximated → diaphragm two layers, running absorbable → abdominal wall.
- **Reoperation**: extend into virgin tissue, enter peritoneum there, then take down adhesions under vision.

4 · INGUINAL, SCROTAL, PENILE, PERINEAL

- **Radical orchiectomy**: inguinal, 5–7 cm, 2 cm cephalad and lateral to pubic tubercle; protect ilioinguinal nerve; do not close new external ring tightly. Scrotal incision contraindicated in suspected malignancy.
- Inguinal lymph node dissection (ILND): subinguinal (commonest) → flap necrosis, infection, lymphocele; S-/L-shaped for inguinal + pelvic; node burden chooses incision.
- **Scrotum/penis**: rich supply → close dead space or Penrose; haemostasis more important than incision.
- **Adult circumcision** (sleeve): incisions 5–10 mm from glans and at coronal sulcus; 3-0/4-0 chromic interrupted.
- **Penile fracture**: degloving (no localisation) vs direct longitudinal (needs imaging; better sexual function).
- **Perineal**: 8–10 cm vertical; inverted-Y (Lambda) → more wound complications; close Colles fascia, mattress skin.

5 · RETRACTORS

- Balfour (retropubic), Finochietto (rib spreader), Bookwalter/Omni-Tract (cystectomy, caval thrombus), Lone Star (perineal).
- Table-fixed injury (bowel, vessels, liver/spleen, nerves): pack with moist pads, minimum force.

6 · DRAINS

Type	Examples	Use
Passive	Penrose	Scrotum
Active closed suction	Jackson-Pratt, Blake	Most other surgery

- Purpose: prevent urine, blood or pus collection. Separate stab incision avoiding inferior epigastrics and bowel; secure with non-absorbable suture.
- Routine drains questioned after robotic prostatectomy, partial nephrectomy, cystectomy; drain when findings demand (e.g. XGP).

7 · MCQ ONE-LINERS

1. Most versatile incision: midline.
2. Commonest urological incision: lower midline.
3. Midline fascial bites: 1 cm from edge, 1 cm apart.
4. Avoid continuous rapidly absorbable fascial closure — hernia.
5. Pfannenstiel: lowest hernia rate; rectus separated.
6. Gibson: renal transplant incision.

7. Hockey-stick Gibson: higher hernia rates.
8. Thoracoabdominal: level III thrombus; pain no worse than flank.
9. Flank incision organ at risk: pleura.
0. Anterior incision organ at risk: bowel.
1. Only transcostal adds flank bulge; supracostal spares rib.
2. Routine post-nephrectomy chest radiograph: unnecessary.
3. Transverse upper abdominal: paediatric Wilms.
14. Suspected testis cancer: never scrotal incision.
15. Subinguinal ILND complications: flap necrosis, lymphocele, infection.
16. Lambda perineal incision: more 90-day wound complications.
17. Circumcision sleeve: first cut 5–10 mm from glans.
18. Penrose drain: passive; Jackson-Pratt: active.
19. Drains anchored with non-absorbable suture.
20. Reoperation: enter through unviolated tissue.

14 Sutures and Suturing

Reading time ≈ 9 min (slow pace)

1 · CHOOSING A SUTURE

- Three determinants: **tissue type**, **anticipated tension**, **duration of tensile strength needed**.
- Natural (gut, silk) → more inflammation than synthetic.
- **Braided**: handles well, little memory, secure knots; harbours bacteria. **Monofilament**: passes easily, less inflammation; more memory, less knot security.

Suture	Construction	Strength retained	Absorbed by
Plain gut	Natural mono	7–10 d	70 d
Chromic gut	Natural mono	21–28 d	90 d
Caprosyn	Mono	~10 d	56 d
Monocryl (poliglecaprone 25)	Mono	2 wk 20–40%	91–119 d
Biosyn	Mono	3 wk 40%	110 d
Vicryl (polyglactin 910)	Braided	3 wk 40–50%	**56–70 d**
PDS (polydioxanone)	Mono	6 wk 35–60%	**182–238 d**
Maxon (polyglyconate)	Mono	6 wk 25%	180 d
Prolene (polypropylene)	Mono	Indefinite	Never
Mersilene (polyester)	Braided	Indefinite	Never
Nylon / silk	Mono / braided natural	Lost gradually	Never

> **KEY POINT** Strength is lost weeks to months before absorption. Match how long the wound needs support to how long the suture gives it: skin ~1–2 wk (Monocryl, gut), soft tissue 2–4 wk (Vicryl), fascia ≥6 wk (PDS, Maxon), indefinite (Prolene).

- **Chromic gut**: circumcision, scrotal skin; very inflammatory.
- **Monocryl/Caprosyn/Biosyn**: subcuticular skin. Caprosyn absorbs fastest; Biosyn strongest.
- **Vicryl**: workhorse for fat, muscle, ligation, anastomoses.
- **Silk**: marks tissue for re-entry (e.g. cord at orchidectomy before possible retroperitoneal lymph node dissection (RPLND)). **Nylon**: drain sutures.

Urinary tract

- **Never non-absorbable (Prolene, Mersilene) in contact with urine** — stone nidus.
- †Urothelium and bowel–urinary anastomoses: fine absorbable (Vicryl, Monocryl, chromic, PDS).

2 · BARBED SUTURES

- V-Loc (cut barbs, loop anchor): knotless laparoscopic/robotic suturing; cutting weakens filament → **not for fascia**.
- **Stratafix Symmetric PDS**: moulded barbs → the only barbed suture strong enough for fascia.

3 · NEEDLES

Point	Behaviour	Use
Taper	Stretches tissue; atraumatic; commonest	General, delicate tissue
Blunt	Prevents needlestick	Fat
Cutting	Edge on concave inner curve	Skin, dense tissue
Reverse cutting	Edge on convex outer curve; no tear-out toward wound edge	Retention sutures

- Taper size rises with toughness: RB (**17 mm**, vessels/mucosa) → SH (26 mm) → CT (36 mm). **UR/GU = 5/8 circle** for deep pelvis (urethral anastomosis).
- **Swaged** (no eyelet → less trauma; permanent or pop-off) vs **free** (eyelet, bigger hole).

4 · KNOTS AND STAPLES †

- †Square knot is the basic secure knot; surgeon's knot (double first throw) holds tension. Knot is the weakest point.
- Prolene knots unravel → extra throws.
- †Metal staples: skin and bowel (GIA/TA); avoid exposed metal staples in the urinary reservoir lumen (stones); absorbable staples are used for pouches.

5 · TECHNIQUES

- Running is fast, but one break opens the whole line.
- **Vertical mattress**: eversion; **figure-of-eight**: haemostasis; **subcuticular/deep dermal**: buried knots, better cosmesis, less infection.
- Epidermis seals in **24–48 h** but needs suture until dermis strengthens.
- **Fascia**: heals slowly → long-lasting (PDS) or permanent; running is safe only with slowly absorbable suture. Close defects ≥1 cm; ports >8 mm with suture passer.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
STITCH	randomised controlled trial (RCT), midline: small (5 mm) vs large (1 cm) bites	Hernia at 1 yr 13% vs 21%	Small bites; never >1 cm spacing

7 · MCQ ONE-LINERS

1. Commonest needle point: taper.
2. Reverse cutting: edge on outer curve.
3. Deep-pelvis needle: UR/GU, 5/8 circle.
4. Vicryl absorbed by 56–70 days.
5. Swaged needle: less trauma.
6. Smallest taper needle: RB.
7. Mersilene: braided polyester.
8. Monocryl: poliglecaprone 25.
9. Barbed V-Loc: not for fascia.
0. PDS absorbed by about 6–8 months.
1. Most inflammatory absorbable: chromic gut.
12. Non-absorbable in urine → stones.
13. Nylon, silk: lose strength gradually.
14. Silk marks cord for later RPLND.
15. Fastest-absorbing: Caprosyn.
16. Only fascia-strength barbed suture: Stratafix Symmetric PDS.
17. Prolene knots need extra throws.
18. STITCH: small bites reduce incisional hernia.
19. Vertical mattress gives eversion.
20. Running fascial closure needs slow-absorbing suture.

15 Lower Urinary Tract Catheterisation

Reading time ≈ 11 min (slow pace)

1 · ANATOMY

- Male urethra **18–20 cm**: anterior (fossa navicularis, penile, bulbar) and posterior (membranous, prostatic, bladder neck). Female ≈ **4 cm**.
- Calibre: meatus **24 Fr** (narrowest), bladder neck 28, prostatic 32; female 22.

> **KEY POINT** Lapides' clean intermittent catheterisation (CIC): infection arises from stasis and overdistension, not bacteria alone — regular emptying, not sterility, prevents infection.

2 · INDICATIONS AND CONTRAINDICATIONS

- Aims: **empty** (retention, sepsis), **clot retention**, **protect upper tracts** (high-pressure bladder), **monitor** output, **heal** after surgery.
- **Absolute contraindication**: suspected urethral injury (pelvic fracture, meatal blood, perineal bruising, high-riding prostate) → retrograde urethrogram (RUG) first.
- Relative: artificial urinary sphincter (AUS) — deactivate first; stricture.

3 · TECHNIQUE

- **Men**: 10–15 mL lubricant (2% lidocaine gel), penis straight and taut → advance **to the hub** → urine → inflate with sterile water → withdraw → **replace foreskin** (paraphimosis).
- Urine may flow from the prostatic urethra; inflating there injures it. No urine beyond bladder neck → irrigate (lubricant plug).
- **Women**: frog-leg, labia spread anteriorly; inflate on urine return.
- **Children**: never force physiological phimosis; steady pressure against sphincter spasm.

Age	Straight (Fr)	Balloon (Fr)
Newborn	4–6	6
1–23 months	6–8	6–8
2–6 y	10	8–10
7–12 y	10–12	12–14
>12 y	14	14–16

4 · CATHETER TYPES

- **French (Fr)**: †= outer circumference in mm; **diameter (mm) = Fr ÷ 3**. Larger = stiffer.

Type	Feature	Use
Coudé	Curved tip; knob marks orientation	Benign prostatic hyperplasia (BPH) — tip up
Councill	Open tip	Over a wire
Two-way Foley	Drainage + balloon	Commonest
Three-way	+ irrigation; stiff wall	Clot retention, continuous bladder irrigation
Malecot	Winged, no balloon	Nephrostomy, suprapubic (SP), continent channel
Lapides / Davis, Trattner	Ringed / double balloon	Female urethral length / diverticulum

- **Materials**: latex, silicone, PVC (polyvinyl chloride). Silicone: fewer urethral symptoms and less urethritis than latex. Silver-alloy/antibiotic coatings reduce bacteriuria but **not symptomatic**

infection.

5 · DIFFICULT CATHETERISATION

ALGORITHM

Fails → BPH suspected: **18 Fr coudé**, tip anterior → fails → cystoscopy → wire → Councill over wire

Not BPH: smaller catheter → hydrophilic glidewire → confirm (5 Fr catheter, aspirate urine) → stiff wire → dilate one size above catheter → Councill over wire

Wire fails / disruption / dense stricture → **SP catheter**

- Blood at port/tip → **false passage** → stop blind attempts; cystoscopy, camera angled anteriorly. Membranous strictures most prone.
- Posterior urethral valves: 6–8 Fr (coils) → 5 Fr over glidewire → SP/vesicostomy.
- Obesity: exposure problem — Trendelenburg, retract pannus, vaginal finger guide. Buried penis: stricture likely.
- Neobladder: SP hazardous (intraperitoneal) → image-guided. Continent reservoir: kinked channel → scope + wire.
- **AUS**: deactivate, 12–14 Fr, remove early; prolonged → SP.

> **KEY POINT** Escalation: blind → wire → vision → suprapubic. Repeated forceful blind passage is never a step.

6 · SUPRAPUBIC CATHETERISATION

- Indications: inaccessible urethra (trauma, stricture); long-term drainage (comfort, avoids urethral erosion).
- Percutaneous contraindications: violated retropubic space, bowel over bladder, no ultrasound.

ALGORITHM

Ultrasound: full bladder, no bowel → Trendelenburg → 1–2 cm incision 1–2 fingerbreadths above symphysis → spinal needle aspirates urine, local anaesthetic → trocar + sheath (16 Fr catheter) or Seldinger → do not over-advance

- Open cystotomy/Lowsley retractor if percutaneous unsafe.
- Complications: **bowel injury**, wall cellulitis/abscess. First change at **4–6 weeks**, by experienced staff.

7 · COMPLICATIONS

- **Catheter-associated urinary tract infection (CAUTI)**: urinary tract infection (UTI) is the commonest healthcare-associated infection; **75%** of hospital UTIs are catheter-related.
- †Prevention: catheterise only when indicated; remove early; aseptic insertion; closed system; bag below bladder; prefer CIC; do not treat asymptomatic bacteriuria.
- **Trauma** ≈ 0.3% inpatients; training prevents it.
- Spinal cord injury: bladder cancer risk ↑; no surveillance cystoscopy.

Retained catheter	Management
Balloon won't deflate	†Cut valve/arm; stiff wire end through channel to puncture, or †ultrasound-guided needle → cystoscopy to retrieve fragments
Knotted	Cystoscopic needle puncture of balloon
Calcified	Cystoscopic laser lithotripsy

8 · MCQ ONE-LINERS

1. Narrowest male urethra: meatus, 24 Fr.
2. Diameter (mm) = Fr ÷ 3.

3. Foley balloon: Foley 1929; first self-retaining: Reybard.
4. CIC introduced by Lapides.
5. Blood at meatus: RUG before catheter.
6. Inflate male balloon only at hub.
7. BPH retention: 18 Fr coudé.
8. Clot retention: three-way catheter.
9. Councill catheter: open tip, over wire.
0. Coatings reduce bacteriuria, not symptomatic UTI.
1. Silicone: fewer urethral symptoms.
2. Dilate one size above planned catheter.
3. Pelvic fracture urethral injury: SP tube.
4. Key SP complication: bowel injury.
5. Coudé tip points anteriorly.
6. First SP change at 4–6 weeks.
7. AUS patient: deactivate, 12–14 Fr.
8. Bladder neck closure: no urethral access.
9. 75% hospital UTIs are catheter-associated.
20. Non-deflating balloon: puncture with wire.

16 Fundamentals of Upper Urinary Tract Drainage

Reading time ≈ 19 min (slow pace)

1 · INDICATIONS

- **Obstruction** (commonest): acute — stones, clots, sloughed papilla, fungal balls; insidious — tumours, ureteropelvic junction obstruction (UPJO), strictures; lower tract — bladder tumour at orifice, outlet obstruction (urethral catheter usually suffices).
- **Diagnosis**: Whitaker test — constant-rate infusion into kidney; renal pelvic minus bladder pressure **>12 cm H_2O** = obstruction. Antegrade pyelogram for strictures after "ureteral rest".
- **Perioperative**: pre-placed access for large stones/tumours; pre-stenting passively dilates ureter before retrograde intrarenal surgery (RIRS); post-op stent for oedema.
- **Trauma**: diversion prevents urinoma; **instillation**: chemotherapy for upper tract urothelial carcinoma, stone dissolution.

Obstructed infected kidney †

- †Urological emergency: **urgent decompression** plus antibiotics; definitive stone treatment deferred until sepsis resolves. Send urine from the drained kidney for culture.
- †Retrograde stent and percutaneous nephrostomy (PCN) are equally effective; choose by availability, anatomy, coagulopathy and stone (PCN in very ill patients avoids general anaesthesia; stent avoids external tube). Stent can also be placed awake under local.

2 · ANATOMY CHOOSES THE ROUTE

Situation	Preferred approach
Horseshoe kidney	High anterior ureteral insertion hampers retrograde; percutaneous via **upper-pole** calyx, more medial; retrorenal bowel common → computed tomography (CT) first
Pelvic/ectopic	Supine anterior, CT or ultrasound guidance, laparoscopy-assisted; transgluteal option
Transplant	Anterior, ultrasound-guided; fibrosis → fascial incising needle or metal dilators
Complete duplication	Either; confirm target moiety
UPJO before pyeloplasty	Antegrade (spares ureter); stent after endopyelotomy
Calyceal diverticulum / infundibular stenosis	Antegrade
Transureteroureterostomy, ileal ureter, ureterocalicostomy	Antegrade
Urinary diversion	Antegrade more reliable; retrograde loopogram opacifies refluxing anastomoses

3 · RETROGRADE STENTING

- Urinalysis; American Urological Association (AUA): prophylaxis for all cystoscopy with ureteral instrumentation. Critically ill → stent under local, awake.
- Retrograde pyelogram delineates the system. Catheters 5–6 Fr open-ended; hydrophilic wire best past impacted stones; exchange for stiff wire in a tortuous ureter or dense stricture (prevents stent buckling).
- **Placement**: under cystoscopic vision (no overpassing, less fluoroscopy, stability) or fluoroscopy only — pusher marker at mid-symphysis (male) / lower symphysis (female); invalid after prostatectomy or prolapse.

Stents — design, sizes, materials

- Ideal: easy insertion, anti-migration memory, drainage, radio-opacity, encrustation resistance, few symptoms. Self-retaining **double-pigtail (double-J)** commonest.
- †Adult sizes typically 4.7–8 Fr, 22–30 cm (6 Fr, 24–26 cm commonest).
- Polymers (polyurethane, silicone blends) resist encrustation/biofilm. **Metal** (Resonance, coiled alloy) resists **extrinsic malignant compression**; hybrid metal-reinforced polymers — limited data.
- Single-pigtail: exits via ostomy. Open-ended ureteral catheters: identify ureter at surgery.

Stent symptoms

- Cause: **intravesical stent** irritating trigone/bladder plus **reflux** up the stent. Consent: urgency, frequency, dysuria, haematuria, flank/groin pain.
- Ureteral Stent Symptom Questionnaire (USSQ): validated, six domains, 38 questions.

Factor	Effect
Diameter	No association
Length	No difference in large randomised controlled trial (RCT)
Distal curl crossing midline	Not confirmed
Intraureteral placement (only string in bladder)	Better tolerated
Hydrocoated silicone	Less pain, urinary symptoms
α-blockers, antimuscarinics	Improve symptoms
Novel loop distal end	No benefit

Stent complications

- **Misplacement** (below obstruction, submucosal — suspect when force needed) → repeat pyelogram, renavigate. Migration (too short).
- **Failure**: extrinsic compression, infection, debris, encrustation; risk ↑ with bilateral obstruction, severe hydronephrosis, recurrent urinary tract infection (UTI), intrinsic obstruction, radiation. Obstructive symptoms with stent → image. Failed stent → nephrostomy.
- **Encrustation**: ↑ with dwell time, colonisation, metabolic risk → failure, fracture, ureteral avulsion. Monitor all long-term stents.

> KEY POINT **†Forgotten/encrusted stent**: plain radiograph (kidney, ureter, bladder — KUB) and non-contrast CT map encrustation. Never pull against resistance. Treat bladder coil (cystolitholapaxy) first, then ureteroscopy and/or percutaneous nephrolithotomy (PCNL) for renal coil, in one or staged sessions. Stent registers and patient counselling prevent it; †exchange routinely every 3–6 months.

4 · PERCUTANEOUS NEPHROSTOMY (PCN)

Preparation

- **Imaging**: ultrasound for simple decompression; CT before PCNL; renogram if function uncertain.
- **Antibiotics**: AUA — prophylaxis for **all** percutaneous renal surgery. Culture if prior UTI, struvite stone, anatomic abnormality, indwelling tubes, reconstruction. Cephalosporin; aminoglycoside (aztreonam in renal failure) + metronidazole; ampicillin/sulbactam; fluoroquinolone.

Agent	Stop before access
Aspirin (primary prevention only)	1 week; continue if cardiovascular indication
Warfarin	1 week (bridge if needed)
Clopidogrel	5 days
Non-steroidal anti-inflammatory drugs (NSAIDs)	3–5 days

- Prone → general anaesthesia; supine → laryngeal mask/spinal possible. Tract local anaesthetic or paravertebral block reduces pain.

Calyx choice

- **Posterior calyx**, puncture **end-on through the papilla** (fornix), never directly into the pelvis. †This follows **Brödel's avascular line** — the plane between anterior and posterior segmental arteries, just posterior to the lateral convex border — avoiding infundibular and interlobar vessels. †Posterior calyces point posterolaterally, so prone access lands end-on.
- Sepsis → any easily reached calyx; staghorn → calyx serving later PCNL.

Calyx	Reach	Cost
Upper	Pelvis, ureter, lower pole	Supracostal → pleura (above 12th acceptable; above 11th high risk)
Interpolar	Pelvis, upper ureter	Least versatile
Lower	Most of kidney	No lung risk; iliac crest limits

- **Prone vs supine**: prone — wide field, any pole, upper pole closest; airway harder, ocular/pressure injury. Supine — easier airway, cardiopulmonary benefit, no repositioning, combined retrograde access; limited puncture field, low intrarenal pressure → poorer vision.
- **Guidance**: fluoroscopy (rotate C-arm for 3D configuration; needs retrograde catheter for contrast/air) vs ultrasound (no radiation, shows viscera and vessels, separates anterior/posterior calyces; ideal after diversion, children). CT for ectopic kidneys.

Technique

ALGORITHM
Seldinger: needle (21 G → 0.018-inch wire, must upsize; or 18 G → 0.035-inch) → J-tipped PTFE (polytetrafluoroethylene) wire → angled catheter steers wire **down the ureter** → Amplatz super-stiff wire + safety wire → skin incision to full sheath size → dilate → tube

- Dilate short, never beyond target calyx (pelvic/UPJ perforation).
- Dilators: metal Alken (best in scar), Amplatz (serial), **balloon** (commonest; less bleeding; poor in scar), single-step (shorter access/fluoroscopy). Only four-arm RCT: no outcome difference.
- Scarred kidney → cutting balloon, fascial incising needle.

Drainage tubes

Tube	Size	Notes
Foley/Councill	≤24 Fr	Partly fill with saline, not contrast
Cope loop	6–14 Fr	Loop in polar calyx; obesity → dislodgement
Nephroureteral	8–12 Fr	Rarely dislodges; re-access easy
Malecot	16–30 Fr	Stiffener to place/remove
Circle	—	Two tracts; secure, irrigable

- Smaller tubes → less pain and leak, no more bleeding. Clamp before removal to test distal patency. Exchange every **3 months**.
- **Tubeless** (stent only): shorter stay vs large tubes, not vs 8–9 Fr; more stent symptoms; lose second-look tract. Keep nephrostomy for pyonephrosis, bleeding, collecting-system injury.

5 · COMPLICATIONS

- Major complications ≈ 10% of percutaneous surgery.
- **Acute haemorrhage** (most significant; PCN transfusion 0.5–4%): occlude tube to tamponade → irrigate next day → angiography; arterial → embolisation.
- **Delayed haemorrhage** (~1 week): pseudoaneurysm or arteriovenous fistula (AVF) → angiography (>90% diagnostic) + **selective embolisation**.

- **Sepsis** 0.5–2.5% after PCNL.
- **Collecting system injury**: stent; nephrostogram before removal; urinoma → drain.
- **Colon** ≈ 0.1%, left > right; retrorenal colon 10% prone vs 2% supine. Extraperitoneal → separate colonic and renal drainage, conservative; peritonitis → open repair.
- Liver/spleen: supracostal access, organomegaly. Gallbladder → cholecystectomy.
- **Pleural**: needing intervention ≈ 5% above 12th rib, 25% above 11th; small Cope tube on Heimlich valve.
- **Positioning**: prone → ocular injury, brachial plexus (abduct ≤90°).
- Venous thromboembolism (VTE) ≤3%: early ambulation. **Air embolism**: sudden end-tidal CO_2 fall, mill-wheel murmur → 100% oxygen, head down, right side up.

> **KEY POINT** Timing sorts bleeding: immediate = venous/parenchymal → tamponade; ~1 week = arterial (pseudoaneurysm/AVF) → angioembolisation.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
†Stent vs PCN, obstructing infected stone	Small RCT	Equal time to defervescence, recovery	Either method; decompression is what matters

7 · MCQ ONE-LINERS

1. Whitaker test obstruction: pressure gradient >12 cm H_2O.
2. Commonest renal fusion anomaly: horseshoe.
3. Horseshoe PCN: upper-pole calyx.
4. UPJO before pyeloplasty: antegrade drainage.
5. Ileal ureter: antegrade access.
6. Commonest stent: double-pigtail.
7. Metal stent: extrinsic malignant compression.
8. Stent diameter: no effect on symptoms.
9. α-blockers improve stent symptoms.
0. USSQ: 38 questions, six domains.
1. Encrustation rises with dwell time.
2. Puncture posterior calyx through papilla.
13. Never puncture renal pelvis directly.
14. Brödel's line: avascular plane.
15. Supracostal above 11th rib: high pleural risk.
16. Lower-pole access: no pulmonary morbidity.
17. Retrorenal colon commoner prone.
18. Colon injury commoner on left.
19. Commonest dilator today: balloon.
20. 21 G needle takes 0.018-inch wire.
21. Clopidogrel stopped 5 days before.
22. Nephrostomy exchange every 3 months.
23. Delayed bleed: pseudoaneurysm/AVF → embolise.
24. Air embolism earliest sign: falling end-tidal CO_2.
25. AUA: antibiotics for all percutaneous surgery.

17 Principles of Urologic Endoscopy

Reading time ≈ 39 min (slow pace)

1 · EVOLUTION OF THE ENDOSCOPE — THE MCQ MILESTONES

Year	Person / device	Contribution
1853	Desormeaux	Kerosene lamp; excised urethral papilloma → **first therapeutic endoscopy** MCQ
1877	Nitze	Light source at the tip (platinum filament) + relay lenses
1966	**Hopkins rod-lens**	Glass rods with short air gaps: more light, smaller calibre; basis of all rigid scopes; first rigid ureteroscope 1979
1854 / 1964	Tyndall / Marshall	Total internal reflection (principle) / first flexible ureteroscopy (URS)
1970	Boyle and Smith	Charge-coupled device (CCD) sensor → later "chip-on-tip"
2005 / 2006	ACMI DCN-2010 / Olympus Invisio	First digital cystoscope / first digital ureteroscope
2011 / 2016	PolyScope / **LithoVue**	First single-use ureteroscope / **first digital single-use** ureteroscope

- Flexible URS became therapeutic only after a **working channel and active tip deflection** were added (early 1980s).

2 · OPTICS AND VIDEO SYSTEMS

Fibreoptic bundles

- Each glass fibre is coated with **cladding** (glass of different refractive index) → better internal reflection, transmission, durability.
- **Incoherent (random) fibres** = light bundle (illumination only). **Coherent (ordered) fibres** = image bundle; lenses at both ends make a telescope.
- More tightly packed fibres → better image, smaller scope, larger channel.

Signal pathway

- **Traditional**: rod-lens/fibre image → eyepiece → external camera CCD (analogue waveform) → video processor (digital) → monitor. Four components: light source, camera, processor/recorder, monitor.
- **Digital ("chip-on-tip")**: CCD or complementary metal oxide semiconductor (CMOS) chip at the tip converts image to electrical signal at once; no internal optics, light cable or camera head. Better resolution and durability; **no focusing or white balance** needed.

Light source and enhanced imaging

- **Narrow-band imaging (NBI)** (2007): blue **415 nm** and green **540 nm** light, absorbed by haemoglobin → capillaries, small papillary tumours and carcinoma in situ (CIS) stand out. **No instillation needed.**
- **Blue-light cystoscopy** (Food and Drug Administration (FDA) approved 2010): **hexaminolevulinate** instilled **1 h** before → accumulates in tumour-cell mitochondria (washed out of normal urothelium) → blue light **360–450 nm** → porphyrins fluoresce **red**; normal urothelium blue. → *bladder cancer chapters*

3 · IRRIGATION FLUIDS AND TRANSURETHRAL RESECTION (TUR) SYNDROME

- **Normal saline** is the default irrigant. **Monopolar** diathermy needs an **electrolyte-free** fluid (glycine, mannitol) because saline disperses the current. **Bipolar** diathermy permits saline, avoiding the complications of electrolyte-free fluids.

- **Sterile water** is avoided — absorption, electrolyte disturbance, haemolysis.
- **Ureteroscopy: always normal saline** — prevents absorption of a hypotonic fluid and TUR syndrome.

Fluid	Tonicity	Use	Specific hazard
Normal saline 0.9%	Isotonic	Diagnostic, URS, bipolar	Volume overload only (no hyponatraemia)
†Glycine 1.5%	Hypo-osmolar (≈200 mOsm/L)	Monopolar transurethral resection of prostate (TURP) / bladder tumour (TURBT)	Hyponatraemia; transient visual loss; hyperammonaemia
†Sorbitol 3% / mannitol 5% (or mixture)	Sorbitol hypo-osmolar; mannitol iso-osmolar	Monopolar resection	Hyponatraemia (dilutional); osmotic diuresis / hyperglycaemia (sorbitol)
Sterile water	Hypotonic	Avoid for resection	Intravascular haemolysis, hyponatraemia

KEY POINT **†TUR syndrome** = absorption of hypotonic irrigant through open venous sinuses → dilutional hyponatraemia, fluid overload. Signs: hypertension and bradycardia → hypotension; confusion, nausea, visual disturbance, seizures (serum sodium usually <125 mmol/L).
Management: stop/finish quickly, check sodium, furosemide for overload; **3% hypertonic saline** if severe/symptomatic, correcting slowly (≤8–10 mmol/L in 24 h) to avoid osmotic demyelination.
Prevention: bipolar saline resection, limit resection time, low irrigation height/pressure.

4 · CYSTOURETHROSCOPY

Indications

- Commonest office procedure. Haematuria evaluation (cytology, retrograde pyelography if contrast cannot be given); **urothelial cancer surveillance**; office biopsy/fulguration of small lesions.
- Upper-tract surveillance (catheterisation, washings, brush biopsy); recurrent infection, voiding dysfunction, pelvic pain; office therapy of strictures, stones, foreign bodies, stent exchange.

Equipment

- **French (Charrière) gauge**: **3 Fr = 1 mm** diameter (1 Fr = 1/3 mm). Choose the **smallest scope that does the job**.

	Rigid	Flexible
Optics	Hopkins rod-lens — clearer	Fibreoptic or digital chip
Channel	Larger; better flow; more instruments	Single 6.4–7.5 Fr, shared by irrigation and instruments
Handling	One-handed; other hand free	Active deflection (120–220°) — negotiates high bladder neck, median lobe; full inspection
Position	Lithotomy / frog-leg	Supine; more comfortable → **office scope of choice**
Size	Diagnostic sheath 15–17 Fr; larger for therapy	14–17 Fr; 37–40 cm long; field ≈120°

- **Rigid set** = lens, bridge, sheath, obturator. Lenses: **0°/12°** urethra; **25°/30°** therapeutic/general; **70°/120°** dome, anterior wall, bladder neck.
- Bridge: diagnostic (no channel), therapeutic (1–2 channels), **Albarran bridge** — lever deflects catheters/wires to cannulate orifices; for elevated bladder neck, large median lobe, reimplanted ureter.
- Obturator: blunts the sheath tip for blind passage.
- Fibreoptic vs digital (in vitro): high-definition (HD) better resolution and depth of field; **illumination better with fibreoptic** MCQ. Randomised: no durability difference; damage mostly

occurred placing scopes in **storage cases**. Commonest repair: rubber of the distal bending segment.

Preparation

- Consent; urinalysis/culture; **treat urinary tract infection (UTI) before instrumentation** (bacteraemia, sepsis).
- **Prophylaxis**: **none for diagnostic cystoscopy** unless patient risk factors (American Urological Association (AUA) and European Association of Urology (EAU) agree) — patient-driven, not procedure-driven.
  - Shared risk factors: older age, malnutrition, immunodeficiency, catheters/colonised material. AUA adds smoking, steroids, distant infection, prolonged stay, anatomical anomaly; EAU adds diabetes, weight extremes, prior UTI, previous instrumentation.
  - **TURP**: both recommend prophylaxis (AUA <24 h). **TURBT**: AUA yes; **EAU only if high sepsis risk** — the examinable divergence.
  - AUA agents: **cefazolin or trimethoprim-sulfamethoxazole (TMP-SMX)**; alternatives aminoglycoside ± ampicillin, other 1st/2nd-generation cephalosporin, amoxicillin-clavulanate. EAU: local antibiogram decides.
- **Antisepsis**: aqueous povidone-iodine for genitalia (any age). Chlorhexidine and alcohol-based preparations damage mucosa — not for genitalia.
- **Lidocaine gel** reduces pain in men (meta-analysis) only if dwell **>5 min, ideally 10–20 min**; equivocal in women.
- Comfort (men; membranous urethra hurts most): **"bag squeeze"**, asking the patient to void (relaxes sphincter), watching the monitor (men only), music.

Technique

ALGORITHM
Inspect genitalia → dilate meatal stenosis to ≥2 Fr above scope size → penis on stretch (rigid: whole hand; flexible: pinch at corona between 4th–5th fingers, thumb/index drive) → anterior urethra with penis 45–90° → beyond membranous urethra direct anteriorly (flexible: deflect up; rigid: drop eyepiece) → advance under full irrigation

- Female: rigid with obturator, direct anteriorly; flexible passed like a Foley. Obese: retract pannus, enter like a Foley, then transfer hands.
- Urethra: strictures; glands of Littré (dorsal); verumontanum and utricle; prostatic length and lobes.
- Bladder with 30°: trigone, orifices (number, position, efflux — blood?), then stones, trabeculation, cellules, diverticula, erythema, papillary/sessile lesions. Rotate scope keeping camera fixed for lateral walls; **70°/120°** for dome and anterior wall.
- Empty the bladder before withdrawal — **unless a Foley is to be placed** (leave partly full).

Special circumstances

- **Suprapubic catheter**: higher risk of infection, stones, cancer, but **no surveillance cystoscopy** for chronic catheters (AUA/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) 2021). Avoid an **immature tract** (weeks); if unavoidable, pass a **wire through the tube before removing it**. Ureteric access via tract: angled catheter/Albarran → flexible cystoscope → semirigid ureteroscope per urethra. Oedematous orifices: IV indigo carmine/methylene blue.
- **Continent diversion**: get the operative note (bowel segment, ureteroenteric anastomosis, continence mechanism, afferent limb). Orthotopic: rigid per urethra; anastomotic contracture without obstruction → **smallest scope, do not dilate/incise** (incontinence). Cutaneous pouch: flexible scope through channel for diagnosis; therapy **percutaneously** (fragile continence mechanism; stomal stenosis) after computed tomography (CT)/ultrasound to avoid bowel and pedicle. Irrigate mucus first, then judiciously — too little shows folds; overdistension blocks afferent limb and retroflexes the scope.

5 · UPPER-TRACT ENDOSCOPY — INDICATIONS

Stone	Treatment (AUA/Endourological Society)
Ureteric	Shock wave lithotripsy (SWL) or URS (URS higher stone-free, more morbidity). EAU: **URS preferred >10 mm**
Renal, non-lower pole	<20 mm SWL or URS; **>20 mm percutaneous nephrolithotomy (PCNL)** (URS if PCNL unsafe)
Lower pole	<10 mm SWL or URS; >10 mm consider PCNL (EAU: SWL 10–20 mm if no unfavourable factors)

- **Unfavourable for SWL**: calcium oxalate monohydrate, brushite, cystine; steep infundibulopelvic angle; lower-pole calyx >10 mm long; infundibulum <5 mm; skin-to-stone distance >10 cm. → *stone chapters*
- **Upper tract urothelial carcinoma (UTUC)**: diagnostic URS if imaging equivocal or kidney-sparing therapy planned. Biopsy predicts grade in ≈**90%**. Does not worsen survival but **raises intravesical recurrence** → bladder surveillance. Low-grade disease: elective endoscopic treatment acceptable. Prefer URS over percutaneous (tract seeding).
- **Endopyelotomy** (ureteropelvic junction (UPJ) obstruction): ureteroscopic is outpatient; go percutaneous if **concomitant stones**; **crossing vessel → pyeloplasty**.
- **Endoureterotomy** fails more with stricture **>1.5 cm**, allograft, irradiated ureter, poorly functioning kidney, ureteroenteric stricture. Balloon dilation alone ≈50% durable at 1 year.
- Diagnostic URS: persistent positive cytology, filling defects, unilateral recurrent UTI; foreign bodies.
- **Benign essential haematuria**: unilateral gross haematuria, negative imaging; commonest finding a small **haemangioma** → fulgurate.

6 · URETEROSCOPES

Semirigid

- Fibreoptic; tip ≤**7 Fr**, channel >3 Fr. **Two working channels preferred**: irrigate through one while an instrument fills the other; fire a lithotripter through one at a stone held in a basket in the other.
- In-line eyepiece standard; **offset eyepiece** gives a straight channel for rigid instruments (pneumatic probe, rigid forceps) — less needed with flexible laser fibres.

Flexible

- Components: optics, deflection, channel. Deflection: lever → control wires through metal rings → fixed at tip. Deflection in one plane (up/down), marked by the **reticle** notch; **rotate** the scope to align plane with target.
- Modern deflection **>270°**; ureteroinfundibular angle averages 140° (max 175°). Visualising the lower pole ≠ working there. **Passive (secondary) deflection** off the upper pelvic wall lengthens the effective tip; fails in **gross hydronephrosis** (no wall to buttress).
- Tip 4.5–7.5 Fr; working channel **3.6 Fr** (devices ≤3 Fr). Pressurised irrigation needed with instruments in situ. Cobra: two 3.3 Fr channels.
- **Digital**: now preferred; larger diameter → more access-sheath use and sheath complications; same stone-free rate, **shorter operating time**.
- **Single-use** (LithoVue: tip 7.7 Fr, CMOS, light-emitting diode (LED), 3.6 Fr channel, 270° both ways): avoids repair costs (reusables may need repair after 3–5 cases); useful for large lower-pole stones, low-volume centres.

Care, damage and sterilisation

- **Laser fibre** passed through a deflected scope scrapes, then **perforates the channel** → fluid ingress at sterilisation destroys the optics.

- Fire with fibre tip in the **centre of the field**, never inside the channel; stop when repositioning.
- Clean channel with warm water and non-abrasive detergent; sterilise by **ethylene oxide gas, glutaraldehyde soak, or Steris (peracetic acid)**.

7 · ACCESSORIES

Guidewires

- Uses: access, straightening ureter, guiding dilation, stent placement. Standard: stainless steel core + wrapped outer wire, PTFE (polytetrafluoroethylene)-coated.
- **Nitinol core**: kink-resistant, stiffer; with polyurethane cover (Zebra) → transmits push, less friction. **Hydrophilic ("glide")**: best past impacted stones, tortuosity, strictures, but **too slippery for a safety wire** → exchange via open-ended catheter. **Hybrid** (hydrophilic tip, PTFE shaft) does both.
- Diameter 0.018–0.038 inch (usually **0.035/0.038**); length most useful **145 cm**; floppy tip 1–3 cm. Stiff wires straighten tortuous ureters. Workhorse: 0.038-inch straight flexible-tip PTFE wire.

Dilators

- Need falls with scope size (<1% for 7.4 Fr). Passive (stent) or active (dilating catheter, balloon); rarely >15 Fr.
- **Zero-tip balloon**: dilates right up to an impacted stone. Ureteroscopic 3 Fr balloon (to 12 Fr) for infundibula and diverticular necks — removed with the scope.

Intraluminal lithotripsy

Energy	Key points
Electrohydraulic	Spark shock wave; cheap; poor on cystine/calcium oxalate monohydrate; poorly focused → ureteric and scope damage; **AUA: not first line in ureter**
Pneumatic	"Jackhammer"; safe, good fragmentation; **retropulsion**; rigid probe needs rigid scope
Holmium:YAG	Standard; **2100 nm**; absorbed within **0.4 mm** of water; photothermal; **all stone types**; fibres ≥200 µm
Thulium fibre laser	1940 nm; penetration 0.15 mm; up to 2000 Hz; 0.025 J (dusting); fibres to 50 µm; small generator. Randomised trial: no clear advantage over holmium

†Source quotes holmium absorption "4 mm" and thulium "2.5 mm"; standard values are ≈0.4 mm and ≈0.15 mm respectively — learn the relative point: thulium penetrates less.

Retrieval and other devices

- Ureteroscopic devices **≤3 Fr**. **Three-prong grasper**: safest — releases an oversized stone. Never basket an irretrievably large stone through a single-channel flexible scope.
- Basket sheaths: polyimide (durable, stiff) vs PTFE (flexible, stretches). Helical (round wires, safe to rotate; open above, pull down, rotate); flat-wire (larger gaps, percutaneous, tumour biopsy). **Nitinol tipless baskets**: soft, kink-resistant, open fully in a calyx, **disengage reliably**.
- Retropulsion (depends on stone size/site, ureteric dilation, energy): Stone Cone (3 Fr coil above stone), NTrap (2.8 Fr mesh).
- **Ureteral access sheath** (11–16 Fr): repeated access, fragment retrieval, **lower intrapelvic pressure** (less pyelovenous backflow and sepsis); risk of ureteric injury; unnecessary when one pass suffices.
- **Fluoroscopy**: mobile C-arm preferred (mobility, image, source below → less scatter); last-image hold, collimation, pulsed mode, foot pedal.

8 · URETEROSCOPY TECHNIQUE

Preparation

- Treat UTI with culture-specific antibiotics. **All URS patients get prophylaxis** (AUA): TMP-SMX or 1st/2nd-generation cephalosporin (alternatives aminoglycoside ± ampicillin, amoxicillin-clavulanate). EAU: per local pathogens.
- Anaesthesia: endotracheal (ET) intubation lets tidal volume/apnoea control renal movement; laryngeal mask → less efficient lasing. Meta-analysis: neuraxial = general for outcomes.
- Theatre readiness = **redundancy**: spare scopes, several wire types, dilating catheters and balloons (incl. zero-tip), torque catheters, baskets ≤3 Fr, stents.

Access

ALGORITHM
Cystoscopy → **safety wire** (0.038 PTFE) → impacted stone: angled hydrophilic wire ± torque catheter, or semirigid scope up to stone → exchange hydrophilic for PTFE wire → suspected obstruction with infection: aspirate pelvis; **turbid → stent and postpone** → drain bladder → ureteroscopy

- **Semirigid below the iliac vessels; flexible above.**
- **"Riding the rails"**: scope alongside the safety wire, lift wire to open orifice; add a second wire through the scope and sit between both wires.
- Orifice will not admit scope: **stent and return in 1–2 weeks** (safest), or 4 mm balloon. **Never dilate a segment containing a stone** (perforation, extravasation). Stricture below stone: zero-tip balloon.
- Impacted stone: push it proximally out of oedematous segment; else **"drill and core"**; place wire past it once visible. Endpoint: dust to **≤2 mm** or basket fragments.
- Flexible access: 10 Fr dual-lumen catheter dilates orifice and places a working wire → scope over taut wire; hang-up → rotate 90–180° (eccentric channel), dilate, or stiffer nitinol wire. Formal dilation ≈8–25% (registry: ⅓). Safety wire remains standard although its necessity in uncomplicated cases is debated.

Flexible technique

- Movements: deflect, **rotate** (commonest novice error is under-rotation; pinch near meatus), advance/withdraw. Over-rotation damages fibreoptic bundles.
- Irrigation controlled, only as needed; **saline**.
- Pass laser fibre with scope **straight** (except ball-tip fibre). 200 µm fibre when deflection needed (more flow, less stiffness).
- **Lower pole**: **relocate stone** before lasering (randomised: higher stone-free); T-tilt (15° Trendelenburg + 15° airplane away) helps; disposable scope for large immovable lower-pole stones.
- **Golden rule**: no pedal unless the fibre tip is seen touching the stone; no firing without visible aiming beam (broken fibre).
- Post-op: non-steroidal anti-inflammatory drugs (NSAIDs) ± alpha-blocker.

9 · NEPHROSCOPY AND COMPLICATIONS

- †**Nephroscopy**: rigid nephroscope (rod-lens, offset eyepiece, straight channel for ultrasonic/pneumatic probes, ≈24–26 Fr through a 26–30 Fr Amplatz sheath) plus flexible nephroscope for calyces out of line. Low-pressure saline irrigation through the sheath. → *PCNL chapter*
- †**URS complications and management**: sepsis (high intrapelvic pressure; prophylaxis, access sheath, drain infected systems first); mucosal injury/perforation (stop, stent 2–4 weeks); **ureteric**

avulsion (never drag an oversized basketed stone; open/robotic reconstruction); stricture (follow-up imaging); stent symptoms.

10 · MCQ ONE-LINERS

1. First therapeutic endoscopic procedure: Desormeaux, urethral papilloma, 1853.
2. Rod-lens system: Hopkins, 1966.
3. First digital single-use ureteroscope: LithoVue, 2016.
4. 3 Fr = 1 mm; French gauge devised by Charrière.
5. NBI uses 415 nm and 540 nm, needs no instillation.
6. Blue light: hexaminolevulinate 1 h before; tumour fluoresces red.
7. Monopolar resection needs electrolyte-free irrigant; bipolar permits saline.
8. Coherent fibres transmit images; incoherent fibres only light.
9. Fibreoptic flexible cystoscope illuminates better than digital models.
0. Albarran bridge deflects catheters for difficult orifice cannulation.
1. Lidocaine gel works only with ≥5 min (ideally 10–20 min) dwell.
2. No prophylaxis for diagnostic cystoscopy without risk factors.
3. EAU: TURBT prophylaxis only if high sepsis risk.
14. All ureteroscopy patients receive antibiotic prophylaxis.
15. No surveillance cystoscopy for chronic indwelling catheters (AUA/SUFU).
16. Endoureterotomy poor if stricture >1.5 cm or irradiated.
17. Benign essential haematuria: commonest cause a small haemangioma.
18. Flexible ureteroscope working channel: 3.6 Fr.
19. Passive deflection fails in gross hydronephrosis.
20. Holmium wavelength 2100 nm; thulium 1940 nm.
21. Electrohydraulic lithotripsy: not first line in the ureter (AUA).
22. Three-prong grasper: safest retrieval device.
23. Access sheath lowers intrapelvic pressure.
24. Turbid aspirate above stone → stent, defer URS.
25. Never balloon-dilate a ureteric segment containing a stone.
26. Laser fibre passed only through a straight flexible scope.
27. T-tilt: 15° Trendelenburg plus 15° airplane away.

18 Fundamentals of Laparoscopic and Robotic Urologic Surgery

Reading time ≈ 35 min (slow pace)

1 · PATIENT SELECTION AND CONTRAINDICATIONS

- History of previous surgery and a **map of every abdominal scar**. Routine tests as for any major operation under general anaesthesia.
- Severe chronic obstructive pulmonary disease (COPD) → arterial blood gases and pulmonary function tests (CO_2 load). Treat arrhythmias first: hypercarbia and acidosis destabilise the myocardium.

Absolute contraindications	Relative (caution, specific consent)
Uncorrectable coagulopathy · intestinal obstruction (unless being treated) · abdominal wall infection · massive haemoperitoneum/haemoretroperitoneum · generalised peritonitis · suspected malignant ascites	Morbid obesity · extensive previous surgery · pelvic fibrosis · organomegaly · benign ascites · pregnancy · hernia · iliac/aortic aneurysm · surgeon inexperience

Condition	Problem → management
Morbid obesity	Short instruments, higher pressures; renal/adrenal laparoscopy still better than open. Prostatectomy/cystectomy with body mass index (BMI) >30 → later in learning curve
Previous surgery	Adhesions → **Palmer point** Veress, open Hasson, or retroperitoneal start
Pelvic fibrosis	Peritonitis, endometriosis, hip-prosthesis cement leak → expect a hard node dissection
Organomegaly	Veress away from the organ, or Hasson
Benign ascites	Watertight closure, firm dressing (leak)
Pregnancy	Enter cephalad of fundus (Palmer); **8–12 mm Hg**; second trimester; harder after 20 weeks
Hernia	Diaphragmatic → mediastinal CO_2; never insufflate through an abdominal-wall hernia
Aneurysm	Vascular review; left upper quadrant Veress or Hasson; secondary trocars under vision

> **KEY POINT** Every relative contraindication is a problem of **safe entry** or **tolerance of pneumoperitoneum**. The answers recur: enter elsewhere (Palmer point, retroperitoneum), enter openly (Hasson), or insufflate less.

2 · PREOPERATIVE PREPARATION AND SETUP

- **No mechanical bowel preparation** for retro/extraperitoneal or transperitoneal surgery not using bowel (no benefit even emerging for cystectomy).
- **Enhanced recovery after surgery (ERAS)**: solids until 6 h, carbohydrate drink 2 h pre-op; balanced fluids; early drain/tube removal, mobilisation and feeding; thromboprophylaxis, normothermia, ileus prevention → stay ≈30% shorter, complications halved.
- Blood: group and screen for nephrectomy, nephroureterectomy, prostatectomy; crossmatch for partial nephrectomy, cystectomy, caval thrombectomy early in experience.
- Checklist before insufflation: suction–irrigation, diathermy, full CO_2 tank + spare, white-balanced camera, insufflator flow and kink response, Veress flow and tip spring; **open laparotomy tray** ready. Robot: components on **separate circuits**, homing, endoscope calibrated.
- Venous thromboembolism: pneumatic compression ± heparin 5,000 units in high-risk patients. Foley catheter (bladder decompressed); orogastric tube (especially left-sided upper-tract work).
- **Positioning**: pad all prominences; lateral — lower leg flexed 45°, upper straight, axillary roll 3–4 fingerbreadths below axilla. **Rhabdomyolysis** risk highest in men, BMI ≥25, **kidney rest raised**

(avoid >20–30 min) and full flexion; plain mattress as good as gel. Pelvic: split legs/Allen stirrups; **never shoulder braces** (brachial plexus).

3 · PHYSIOLOGY OF PNEUMOPERITONEUM

> **KEY POINT** Two drivers: **pressure** (caval, renal, diaphragmatic compression) and **absorbed CO_2** (acidosis, sympathetic drive). Each recommendation lowers one of them; each high-risk patient (COPD, hypovolaemia, renal impairment) tolerates one badly.

Insufflants

Gas	For	Against / use
CO_2 (standard)	Non-combustible, **highly soluble → low embolism risk**, cheap, rapidly cleared	Hypercarbia, acidosis, arrhythmia; COPD cannot blow it off
Nitrous oxide	Less peritoneal irritation, fewer acid–base effects	**Supports combustion** — only without electrosurgery
Helium / argon	Inert, no hypercarbia	**Poorly soluble → embolism risk**; switch to it to salvage intraoperative hypercarbia or in lung disease
Air, oxygen	—	Embolism, combustion — abandoned

- Hypercarbia peaks **after extubation** as stored CO_2 is mobilised — watch lung-disease patients in recovery.

Pressure

- 15 mm Hg commonest; **12 mm Hg** more physiological, less pain; **≤10 mm Hg** markedly reduces oliguria; 8 mm Hg possible with AirSeal. 20 mm Hg may reduce venous bleeding. **Never >20 mm Hg for long.**

Effect (supine)	5	10	20	40 mm Hg
Mean arterial pressure (MAP), systemic vascular resistance (SVR)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓
Glomerular filtration rate (GFR), urine	↔	↓	↓↓	↓↓
Arterial pH	↔	↔/↓	↓	↓

Renal function is the first casualty (from 10 mm Hg); afterload rises at every pressure; output collapses at 40.

Systems

- **Cardiovascular**: hypovolaemic → caval compression, venous return falls (≤20 mm Hg); hypervolaemic → venous return rises. Hypercapnia → tachycardia, ectopics; peritoneal stretch → **vagal bradycardia**. Arrhythmia may herald pneumothorax, hypoxia or embolism. **Central venous pressure (CVP) unreliable** (raised intrathoracic pressure).
- **Respiratory**: diaphragm splinted, functional residual capacity falls, peak airway pressure rises (dead space unchanged); positive end-expiratory pressure (PEEP) in lung disease. Head-down: endobronchial tube migration, facial oedema → restrict fluids.
- **Renal**: oliguria from **renal vein and parenchymal compression** (not hormones or ureteric compression); cortical flow falls. Manage: pressure ≤10 mm Hg; if needed furosemide, mannitol 12.5–25 g, dopamine 2 µg/kg/min — **not fluid boluses** (overload).
- **Splanchnic**: flow falls (rare mesenteric thrombosis); less ileus than open.
- **Acid–base**: respiratory acidosis from transperitoneal (and extraperitoneal) CO_2 absorption, rising over 30–60 min then plateauing. Rising end-tidal CO_2 → anaesthetist increases ventilation, surgeon lowers pressure/desufflates. COPD: blood gases if insufflation >1 h.
- **Position**: head-down raises cardiac output (better venous return) despite respiratory cost; head-up lowers it. **Transperitoneal** causes larger haemodynamic swings; extraperitoneal pelvic insufflation

gives higher arterial CO_2.
- Less stress, metabolic and immune response than open surgery; no proven oncological benefit.

4 · ACCESS

Closed (Veress) access

- 14-gauge, 2 mm; check spring-back, flow (≤2 mm Hg at 2 L/min), no leak. Pass perpendicular, feeling **two pops** (fascia, peritoneum).

Site	Advantage	Danger / adjustment
Umbilicus	Thinnest wall	Great vessels: angle to pelvis if thin; perpendicular if obese (umbilicus migrates caudally)
Lower quadrant / lateral (2 fingers medial and superior to anterior superior iliac spine)	Less vascular risk	Colon
Palmer point (left subcostal, midclavicular)	Adhesions	Deep → liver, spleen

ALGORITHM
Aspirate (blood/bowel?) → inject saline freely → re-aspirate (nothing returns) → **drop test** → **advancement test** (1 cm free) → insufflate; opening pressure **<8–10 mm Hg** confirms free flow → 15 mm Hg

- Do not lift the fascia — it widens the fascia–peritoneum gap, not the abdominal space.

Open (Hasson) access

- Safer early in experience and after previous surgery; larger incision, more leak.
- Steps: 2 cm infraumbilical incision → open fascia and peritoneum separately → confirm by sight and touch → two 0 silk stay sutures → blunt cannula, sutures on struts → insufflate.
- **Hand-port first**: safest for the inexperienced; secondary trocars guided by the intra-abdominal hand.

Retroperitoneal and extraperitoneal access

- **Retroperitoneal**: 2 cm incision below 12th rib tip, midaxillary line → split muscle, open thoracolumbar fascia → finger feels **psoas behind, Gerota-covered lower pole in front** → balloon dilation anterior to psoas, posterior to Gerota (cephalad, then caudad).
- Balloon: commercial (up to 800 mL, scope inside to confirm) or Gaur glove-on-catheter (cheap; latex fragments on rupture). Misplacement → intramuscular dilation (hernia) or peritoneal tear.
- **Extraperitoneal pelvis**: infraumbilical incision, anterior sheath opened, finger tunnel behind recti to pubis → prevesical balloon.
- Standard Hasson in retroperitoneum → heavy emphysema and hypercarbia → use **balloon blunt-tip port** (internal balloon + foam cuff).

Transperitoneal vs retroperitoneal

	Transperitoneal	Retroperitoneal
Space, landmarks	Large space, familiar anatomy	†Small space; psoas the key landmark; direct hilar (artery-first) access
Physiology	Larger haemodynamic swings	More emphysema, pneumothorax; CO_2 absorption similar with sealed port
Best for	†Large tumours, reconstruction	Previous transperitoneal surgery; †posterior tumours
Vascular injury	Upper-tract (retroperitoneal-organ) surgery ≈5× pelvic rate	

5 · TROCARS AND PORT PLACEMENT

- Trocar = cannula + obturator. **Blunt (dilating) trocars** have replaced bladed: ≈5× less epigastric injury; bleeding 3% vs 9%; port-site hernia 1.83% → 0.19%.
- Retention devices prevent dislodgement and emphysema. AirSeal: valve-free, stable pressure despite suction → work at 8–12 mm Hg, continuous smoke evacuation.
- **First trocar**: twisting insertion with braking hand; perpendicular (70° caudad for pelvis); hiss of CO_2 confirms; optical trocar watched with 0° lens. Some raise to 20 mm Hg for entry.
- **Secondary trocars**: inspect the whole abdomen for entry injury first; insert **under vision**, transilluminate wall vessels; aim each port at the target (redirecting widens tract → emphysema). Midline or ≥6 cm lateral.
- **Triangulation**: diamond around target; drivers <55° to horizontal, 25–45° between instruments. Avoid **crossing swords** (move handles together), **striking handles** (withdraw sheaths), **rollover**.
- **Never metal trocar with plastic retaining ring** — capacitive coupling.
- Robot: camera 8 mm (Xi) or 12 mm (Si); arm ports 8 mm, **8–10 cm apart**; assistant port caudal, not between arms; thick black line just inside the wall.

Hand-assisted and single-site

- Hand port (GelPort): incision in cm = glove size; trace footprint, plan ports around it; seal glove–gown junction. Costs surgeon fatigue and hand numbness; wound infection ≈9%, midline hernia 4–7%.
- **Laparoendoscopic single-site surgery (LESS)**: worst clashing → articulating instruments, varied lengths, low-profile scope; add a 5 mm port without hesitation. Da Vinci SP: single port, articulating camera, three wristed instruments.

6 · ROBOTIC SYSTEMS AND INSTRUMENTS

- Da Vinci: **patient-side cart, surgeon console, vision cart**. Si: fixed approach (between legs in pelvis). Xi: boom-mounted, any side, finer arms, camera in any port, no stirrups. SP: single arm, no external collisions.
- EndoWrist: **7 degrees of freedom**, 90° articulation; ≈10-use limit. Allen bolt opens locked jaws after failure.
- Robot failure <1% (≈0.4% intraoperative), mostly before incision; rarely forces conversion.
- Laparoscope: 30° lens sees more; digital chip-on-tip gives better image; 3D binocular optics give depth. Defog: warm, antifog, dry junction; white-balance.

Energy devices

Device	Lateral spread	Key point
Monopolar	5–7 mm	Versatile; coupling and insulation hazards
Bipolar	1–6 mm	No capacitive coupling
LigaSure/vessel sealer	1–3 mm	Impedance feedback; seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current (pacemakers); tissue <80 °C but **jaws >200 °C**, cool slowly → keep off bowel

Clips, staplers, sutures, entrapment

- Staples: **grey 2.0 / white 2.5 mm vascular**; blue 3.8 / green 4.8 mm thick tissue (bowel, bladder, ureter). Never fire across clips.
- **Hem-o-lok**: contraindicated by maker for **donor renal artery**; safe with technique — circumferential dissection, tip seen beyond vessel, feel snap, ≥2 clips patient side, 2 mm cuff. No diathermy near clips.
- Lapra-Ty (absorbable clip-knot), barbed suture (faster vesicourethral anastomosis).

- LapSac most tear-resistant (morcellation); morcellate with ring forceps (least sack puncture).

7 · EXITING THE ABDOMEN

ALGORITHM
Lower pressure to **5 mm Hg** (unmasks bleeding) → inspect operative site and every port → fascial sutures under vision → remove each port under vision, scope last → close hand-port first, then reinsufflate

Trocar	Adult	Child
Bladed	Close all 10–12 mm	Close all ≥5 mm
Blunt	Midline >10 mm or stretched sites	5 mm sites closed

- BMI >30: closure devices (Carter-Thomason, Endo Close, angiocatheter technique). Transverse extraction incisions herniate less than vertical midline.

8 · COMPLICATIONS AND MANAGEMENT

Contemporary rates 13–22%; **vascular, then adjacent-organ** injuries commonest. Complications follow the operation's chronology: entry → insufflation → dissection → exit → recovery; the dangerous ones present days later.

Access and insufflation

Complication	Recognition	Management
Preperitoneal insufflation	High opening pressure, asymmetric distension, scope sees fat	Evacuate → open access
Veress in vessel	Blood in hub	Withdraw **without manipulation**, re-site; inspect track; hidden retroperitoneal haematoma if unstable
Veress in bowel/bladder	Bowel content/urine aspirated	Withdrawal only; re-site
Gas embolism	Sudden collapse, **end-tidal CO_2 abruptly rises then falls**, desaturation, **"millwheel" murmur**	Stop and desufflate → **left lateral decubitus, head down** → 100% oxygen, hyperventilate → aspirate via central line
Subcutaneous emphysema	Crepitus, pneumoscrotum	Purse-string, balloon trocar, lower pressure (12 mm Hg retroperitoneally)
Pneumothorax	High airway pressure, hypotension	Tension: **16-G needle 2nd intercostal space** → chest drain
Pneumopericardium	Recovery chest film	As embolism; pericardiocentesis for tamponade

First trocar injuries

- **Bowel** (commonest): one-wall injury seen at once; through-and-through only by viewing the primary port from a secondary port (do routinely). **Leave trocar in** → access elsewhere → repair; general surgery, 4–5 L antibiotic irrigation.
- **Major vessel** (aorta, common iliac commonest): **leave trocar in situ**, call vascular surgeon, emergency laparotomy along trocar, proximal/distal control, repair. Prevention: review computed tomography (CT), adequate pneumoperitoneum, blunt optical trocars.
- **Bladder**: pneumaturia/haematuria; **always repair**. Catheterise every major case.
- Port-site bleeding: cantilever to tamponade; diathermy, Keith-needle suture over bolster, or closure device.

During dissection

Electrosurgical mechanism	Prevention
Inappropriate direct activation	Instruments out when idle; only surgeon activates
Direct coupling	Activate only with whole metal tip visible
Capacitive coupling	No mixed metal/plastic systems; cut/blend current
Insulation failure	Inspect; **active electrode monitoring**

- **Thermal bowel injury** presents late (up to **18 days**): fever after day 1, pain at nearest port, left shift. Diagnose with **CT with oral contrast** (free gas persists ≈9 days). Monopolar → resect with **6 cm margins**; bipolar → excise visible area; mechanical → laparoscopic repair.
- **Venous injury**: raise pressure to 25 mm Hg, suction, tamponade, sealant, extra port for suturing or Satinsky, hand-assist (pinch side walls), or convert. Hypovolaemia at high pressure → embolism risk. Renal artery → staple; aorta → convert.
- **Nerve injury** (≈2.8%): brachial plexus (abduction >90°, shoulder braces), femoral (lithotomy), sciatic (lateral). Neurology consult, conduction studies, physiotherapy.

Exit and postoperative

- Bowel entrapment at port: ileus, point tenderness day 2–3 → reduce laparoscopically.
- Right shoulder-tip pain = diaphragmatic CO_2; delayed diffuse pain with low-grade fever = occult bowel injury.
- **Port-site hernia**: adults >10 mm, children 5 mm; CT shows bowel above fascia → laparoscopic repair.
- **Rhabdomyolysis**: downside hip pain, brown urine, creatine kinase >5,000 → prevent: kidney rest <1 h, avoid hypotension, case <5 h.
- Deep vein thrombosis: American Urological Association (AUA) — compression for all; add heparin 5,000 units 12-hourly if high risk.

9 · TRAINING

- Box trainers (cheap, proven), virtual reality (VR; metrics, improves novice performance, costly), live animal (pig), cadaver (most realistic). E-BLUS (EAU), Fundamentals of Robotic Surgery (randomised benefit). Robotic surgeons with >20 cases have better outcomes; mentor early cases.

10 · MCQ ONE-LINERS

1. Palmer point: left subcostal, midclavicular line — for adhesions.
2. Veress opening pressure >10 mm Hg suggests preperitoneal placement.
3. Pregnancy: pressure 8–12 mm Hg, second trimester preferred.
4. CO_2 is highly soluble — lowest embolism risk.
5. Helium: no hypercarbia, but embolism risk.
6. Nitrous oxide supports combustion.
7. Hypercarbia peaks after extubation.
8. Oliguria: renal vein and parenchymal compression.
9. Renal effects begin at 10 mm Hg.
0. CVP unreliable during pneumoperitoneum.
1. Peritoneal stretch causes vagal bradycardia.
2. Head-down position increases cardiac output.
3. Gas embolism: end-tidal CO_2 rises then falls; millwheel murmur.
14. Embolism position: left lateral decubitus, head down.
15. Tension pneumothorax: needle 2nd intercostal space midclavicular.
16. Major vessel injury: leave the trocar in.
17. Commonest major vessels injured: aorta, common iliac artery.
18. Blunt trocars: fivefold less epigastric injury.
19. Metal trocar with plastic collar → capacitive coupling.
20. Thermal bowel injury can present up to 18 days.
21. Monopolar bowel burn: resect 6 cm each side.
22. Ultrasonic jaws exceed 200 °C.
23. LigaSure seals vessels up to 7 mm.
24. Grey/white staples vascular; blue/green thick tissue.
25. Hem-o-lok contraindicated on donor renal artery (manufacturer).
26. EndoWrist: 7 degrees of freedom.

27. Hand-port incision (cm) = glove size.
28. Inspect at 5 mm Hg before exit.
29. Rhabdomyolysis: kidney rest, men, BMI ≥25.
30. No shoulder braces in Trendelenburg.

19 Energy Modalities in Urology

Reading time ≈ 54 min (slow pace)

1 · PHYSICS ESSENTIALS

- Five surgical effects: **cutting, ablation (tissue removal), destruction, coagulation, fragmentation**. Every modality is a question of **where the wave's energy is absorbed and in what form it is released**.
- Delivery: directed waves, or direct contact (water vapour, cryoprobe, saline jet).
- **Units**: power (W) = J/s; laser power = pulse energy × frequency (0.5 J × 20 Hz = 10 W). Intensity = W/cm². Pressure: 1 Pa = 1 N/m²; **10 bar = 1 MPa** (trap: not 1 bar).
- Speed = frequency × wavelength. **High frequency = shallow, low frequency = deep** (linear scrotal vs curvilinear renal probe). Absorption distance = 1/α (amplitude falls to 1/e); absorption usually rises with frequency.
- **Photon energy E = hf**: raising laser power sends more photons per second, not more energetic ones.
- **Focusing** (lens/reflector, like a magnifying glass): energy spread at the skin, concentrated at the target. Focal zone ≈ **1 wavelength wide × 10 long** (rice-grain).
- **Acoustic impedance** = sound speed × density. Large mismatch (tissue–air, tissue–bone/stone) → reflection: hence gel/water coupling, gas and rib shadows, stone acoustic shadow.
- Radiation force = power absorbed ÷ wave speed — negligible for light, useful for sound (ultrasonic propulsion); doubled on total reflection.
- **Shock** = abrupt pressure rise from nonlinear steepening (crests travel faster than troughs) — acoustic, not electrical. Short pulses are broadband (SWL "clap"); long bursts narrowband (BWL "note").
- **Cavitation**: negative pressure tears liquid into a vapour bubble (pressure drops); boiling = temperature rises.

	Electromagnetic (EM) waves	Elastic (acoustic) waves
Devices	Lasers, electrosurgery, microwave, radiofrequency (RF)	Shock wave lithotripsy (SWL), ultrasound therapies, intracorporeal lithotripters
Medium	Free space and media	Needs gas, liquid or solid
Wavelength	Microns (light)	Millimetres
Bioeffect	Absorption → heat	Heat, mechanical stress (solids), cavitation (fluids/tissue)
Penetration	Short (absorbed by water/tissue) → fibre to target	Centimetres → transcutaneous, large source at skin

2 · ELECTROSURGERY

Principles

- Ohm's law **V = I × R**; heat = current through resistance — the tissue is the "toaster element". For alternating current (AC) the opposition is **impedance**.
- Generator ≈ **500 kHz** (radiofrequency range): reversal too fast to depolarise nerve/muscle → ions oscillate → heat. Mains 50 Hz causes spasm and arrest.
- **Current density decides the effect**: same current is intense at the small active tip, negligible over a large dispersive pad. The tip stays cooler than tissue.
- First human use 1926 (Bovie with Cushing) MCQ. The theatre "Bovie" is an electrosurgical unit, not electrocautery.

Waveforms

Setting	Waveform	Voltage / current	Tissue effect
Cut	Continuous sinusoid	Lower voltage, higher current	Rapid heating → intracellular water flash-boils → clean vaporisation, little char
Coagulation (contact)	Interrupted; ≈ **6% duty cycle**	Higher voltage, lower current	Tissue cools in off-cycles, heat conducts deeper → denaturation below boiling → desiccation; rising impedance self-limits; spread up to **3 mm**
Spray (fulguration)	Interrupted	Highest	Sparks ionise the air gap → non-contact surface char; diffuse ooze
Blend	Cut base with interruptions	Intermediate	Cutting with added haemostasis

Recommended open settings: cut **50–80 W**, coagulation **30–50 W**. The **waveform, not the power**, chooses cut vs coagulation.

> **KEY POINT** Why does coagulation go deeper yet char less than cutting? In the off-cycles heat conducts outward and tissue never reaches boiling, so cells cook and dry (like egg white) instead of bursting; as desiccated tissue conducts less, current falls and the effect limits itself. Many generators' "cut" is actually a blend with adjustable interruptions.

	Monopolar	Bipolar	Electrocautery
Circuit	Generator → active tip → whole patient → dispersive pad → generator	Only tissue between two equal jaws	Heated wire; current never enters patient
Power	Higher	Lower → less collateral damage	Low (battery, clinic)
Hazards	Pad burns, current leaks, capacitive coupling	Circuit completes if tines touch	Hot tip

Advanced energy devices

- **Advanced bipolar sealers** (e.g. LigaSure): bipolar current + mechanical jaw compression; a **tissue-response generator** adjusts output as clamped tissue desiccates; no pad. Used for prostatic pedicles and cystectomy; cost limits use.
- **†Ultrasonic shears** (harmonic scalpel): blade vibrating ≈ 55 kHz denatures protein by friction at lower temperature; no current passes through the patient, so no stray-current or pad risk; thermal spread a few millimetres → *Ch 20*.

Safety

- **Dispersive pad** — commonest injury site. Partial peeling → smaller area → high current density → burn. Place on clean, well-perfused muscle mass; avoid bony prominences, scars, dense hair, metal implants and pacemakers. Hip prosthesis is not a contraindication — place pad contralaterally. Remove jewellery. Most units detect poor contact and block activation; inspect site post-op.
- **Insulation failure**: current reaches metal, fluid, patient or surgeon (bowel burns outside the visual field).
- **Capacitive coupling**: AC induces current across insulation into an adjacent conductor (capacitor = two conductors separated by insulator). Causes: cords wrapped together or around metal instruments. **Active electrode monitoring** detects insulation failure and coupling.
- **Direct coupling**: activation touching another metal instrument.
- **Surgical fire**: fuel (drapes, alcohol prep) + oxidiser (oxygen-rich gas) + ignition; electrosurgery implicated in most theatre fires (head/upper torso, supplemental oxygen).
- **Inadvertent activation**: holster electrodes; awareness of pedals. Surgeon burns through damaged gloves — use lowest effective current.
- **Pacemakers / implantable defibrillators**: on the implant checklist. †Prefer bipolar; if monopolar, pad placed so the current path is away from the device; short bursts; cardiology review — pacing

set to asynchronous and defibrillator anti-tachycardia function suspended, with external defibrillator available; recheck device after surgery.

Endoscopic electrosurgery: monopolar vs bipolar resection

	Monopolar transurethral resection of prostate / bladder tumour (TURP/TURBT)	Bipolar TURP/TURBT
Circuit	Loop → patient → pad	Completed within instrument (loop → saline → return electrode); plasma designs keep patient out of circuit
Irrigant	**Ion-free**: 1.5% glycine, sorbitol–mannitol	Isotonic saline
Hyponatraemia (TUR syndrome)	↑ — limit resection time, esp. vascular glands	Reduced
Efficacy	Reference	Comparable; safer (moderate-quality, short follow-up)
Obturator reflex	Risk	Thought reduced
Typical settings	Cut 180 W, coagulation 80–120 W	—

- **Why ion-free?** Ions disperse current away from the loop (cutting "under seawater"). Irrigant must also be inert, non-toxic, transparent.
- Loop insulated except tip; damaged sheath insulation → loss of cut or **urethral burn** — inspect before mounting.
- **Obturator reflex**: lateral-wall TURBT stimulates obturator nerve → sudden adduction → **bladder perforation** (under general or spinal anaesthesia; arcing produces low-frequency components). Prevention: neuromuscular blockade timed with anaesthetist, selective obturator nerve block, avoid overdistension, intermittent activation, bipolar resection.

Device-assisted intravesical therapy (non-muscle-invasive bladder cancer, NMIBC)

	Electromotive drug administration (EMDA)	Chemohyperthermia
Energy	Electric field: intravesical electrode + abdominal pads	915-MHz microwave via catheter (Synergo) with thermocouples
Mechanism	Iontophoresis, electro-osmosis, electroporation → mitomycin C (MMC) into lamina propria	Wall at **42 °C** → ↑ permeability, cancer cell heat sensitivity, heat-shock proteins
Evidence	EMDA-MMC + bacille Calmette–Guérin (BCG) randomised controlled trial (RCT): ↑ recurrence- and progression-free survival; meta-analysis: recurrence only; Cochrane: low quality	—

3 · LASERS

Principle

- LASER = light amplification by stimulated emission of radiation. **Gain medium** pumped (flashlamp, current, another laser) inside a **resonator** (total + partial reflector) → photon stimulates identical photon (energy, direction, phase) → **monochromatic, collimated, coherent** beam that can be concentrated down a thin fibre.
- The **wavelength decides the chromophore** (water, haemoglobin, pigment) — the whole reason lasers differ.
- Output: continuous wave (CW); **Q-switched** (shutter, giant ns pulse); mode-locked (femtoseconds). Diodes: semiconductor, small, efficient, 800–980 nm clinically.

- Named by gain medium: holmium:yttrium-aluminium-garnet (**Ho:YAG**), thulium fibre laser (**TFL**), thulium:YAG (Tm:YAG), neodymium:YAG (**Nd:YAG**), potassium-titanyl-phosphate / lithium triborate (**KTP/LBO**, crystals that halve 1064 nm to green 532 nm).
- CO_2 (10,600 nm): very high water absorption; superficial penile lesions; cannot be used intracorporeally (fibre losses).

Laser lithotripsy — pulse duration decides the mechanism

Fragmentation begins with photon absorption; the key dose is **fluence** absorbed by stone and/or water. Manufacturers quote the electrical (pump) pulse duration; the optical pulse duration is the meaningful one.

Optical pulse	Mechanism	Bubble / pressure	Fragments
Ultrashort <500 ns (Q-switched)	Plasma	Spherical; extremely high	Large
Short 0.5–10 µs (pulsed dye; FREDDY = frequency-doubled double-pulse Nd:YAG)	Cavitation + shock (photomechanical)	Spherical, single-locus collapse; **>300 bar**	Large
≈10–100 µs	Mixed	Intermediate	—
Long >250 µs (Ho:YAG, TFL)	**Photothermal**	Asymmetric, multi-locus; **<20 bar**	Small / dust
>20 ms / CW	Photothermal + coagulation, carbonisation	—	Adjacent tissue injured

- Photomechanical stress generators: (1) **thermoelastic expansion** → wave reflects at the free surface as tension → spallation; (2) **recoil** of ejected material → stress fracture deep to the crater; (3) **plasma** (ionised "fourth state of matter") expansion. Shock arises twice per pulse — at bubble expansion and at collapse (the larger).
- Photomechanical: collapse pressure ∝ (bubble radius)^3; needs **stress confinement** (energy delivered faster than stress escapes); also spallation at the far surface. Pulsed dye (504 nm) was safe because haemoglobin is an "energy sink", but failed on hard stones (cystine, brushite, calcium oxalate monohydrate).
- **Photothermal (holmium, thulium)**: hundreds of µs preclude stress confinement → melting, chemical decomposition, micro-explosion of interstitial water. Evidence: thermal breakdown products; warm stones ablate better; crater volume ∝ pulse energy. Ho:YAG transients: **8–20 bar** at <1 J (usual clinical range), 20–40 bar at 2 J — far below photomechanical lasers.
- **Moses effect**: the first ≈ 60 µs boil a vapour channel; vapour absorbs mid-infrared far less than water → rest of the pulse reaches the stone. Hence **fibre close and perpendicular** (0° → most ablation; parallel → no ablation).
- One wavelength can do everything — short pulse cavitation, hundreds of µs ablation, >20 ms collateral burn: **dosimetry, not platform**, sets the mechanism.
- Smaller fibre → narrow deep craters; larger fibre → wide shallow craters.

	Holmium:YAG (Ho:YAG)	Thulium fibre laser (TFL)
Wavelength	2100 nm	1940 nm (diode-pumped fibre)
Water absorption	High (penetration 0.31 mm)	Higher, ≈ 4× (0.08 mm)
Pulse	Up to 120 W; variable pulse; **pulse modulation** (Moses-type double pulse)	50 µs–20 ms; typical 700–800 µs at **low peak power**
Fragments / retropulsion	Small; more retropulsion	Smaller, faster dusting; **less retropulsion**

KEY POINT Why does TFL dust with less kick? Its wavelength is absorbed more by water, so energy is dumped in a shorter distance (finer dust), and it delivers that energy as a long, low-peak-power pulse (gentler push per pulse). Pulse-modulated high-power holmium narrows the gap — hence the discordant trials.

Lithotripsy settings, retropulsion and heat

Parameter	Effect on retropulsion
↑ Pulse energy	↑
↑ Frequency (at low energy)	No increase
↑ Pulse duration / ↓ peak power	↓

- **Dusting** = low energy, high frequency (little kick, fine dust). **Fragmenting** = high energy, low frequency (bigger pieces for basketing; more kick). †Typical Ho:YAG: dusting ≈ 0.2–0.5 J × 50–80 Hz long pulse; fragmenting ≈ 0.6–1.5 J × 6–15 Hz.
- **Heat**: energy not entering the stone heats irrigant; a calyx with poor outflow becomes a "kettle" → cytotoxic temperatures. Limit with: lower power, duty-cycle pauses, ↑ flow (hand pump, pressure bag, small fibre to free the channel), **ureteral access sheath** for outflow, fibre on the stone.
- Why keep the fibre on the stone? A photothermal laser works only if its photons enter the stone; photons absorbed by water need more total energy, prolong lithotripsy and add heat.
- Misdirected Ho:YAG/TFL injures urothelium directly.
- **†Laser room safety**: wavelength-specific eye protection for all staff and patient, warning signage, standby mode when not firing, fibre tip visible beyond the scope before firing (avoids scope damage), no flammable material near a free beam.

Soft-tissue lasers — chromophore and penetration

Laser	λ (nm)	Chromophore	Penetration (mm)	Behaviour / use
KTP/LBO ("GreenLight")	532	Haemoglobin	0.8 (effect 1–3)	Photovaporisation of benign prostatic hyperplasia (BPH); continue anticoagulation
Nd:YAG	1064	Neither water nor haemoglobin (weak)	**10**	Deep coagulation; delayed necrosis/slough. †Forward scatter can injure bowel beyond bladder wall
Diode	940–1490	Water + haemoglobin (variable)	0.5–5	Prostate vaporisation
TFL	1940	Water	**0.08**	Precise vaporisation; consistent char
Tm:YAG	2000	Water	0.2	CW; larger coagulation area; prostate enucleation, tumours
Ho:YAG	2100	Water	**0.31**	Pulsed; deeper incision, no char; enucleation, stones, upper-tract tumours

- Pattern: water-absorbed mid-infrared penetrates least; Nd:YAG most. Trap: KTP is shallow because of haemoglobin, not wavelength per se.
- Strong scatter or high absorption → confined; low absorption + high scatter → large volume, low fluence. Incision zones: vaporisation → carbonisation → irreversible → reversible damage.
- **GreenLight**: Nd:YAG frequency-doubled through KTP (80 W) → LBO crystal (120/180 W), 70° side-firing. Best ablation at **4 mm/s sweep, ≤2 mm distance**; gain plateaus beyond 180 W; 180 W shortens operating time vs 120 W.
- **Photodynamic therapy** (photochemical): photosensitiser accumulates in tumour → specific wavelength → singlet oxygen; whole-bladder urothelial treatment with one intravesical application.

4 · ACOUSTIC (ELASTIC-WAVE) THERAPY

Tissue interaction — heating vs cavitation

Heating	Cavitation
↑ with frequency (absorption)	↑ as frequency falls (longer negative phase)
Dose-dependent	**Threshold** effect

- Nucleation at interfaces, stone surfaces, gas → lung and bowel vulnerable; **no SWL within 24 h of microbubble contrast**. Doppler **twinkling artefact** = microbubbles on stone.
- Bubble collapse near a solid → micro-jet and rebound shock that pit the stone. Secondary effects: sub-injurious shocks cause protective renal vasoconstriction; too many → ischaemia–reperfusion injury.
- Acoustic waves are non-ionising; repeated safe exposures do not accumulate. Operator risk only at the focus or a ballistic tip.

Generators (transducers)

Type	Mechanism	Focusing
Electrohydraulic	Underwater spark vaporises water → spherical shock at source	Ellipsoidal reflector: spark at F1, stone at F2
Electromagnetic	Coil pushes metal membrane against fluid; shock forms en route	Acoustic lens or parabolic reflector
Piezoelectric	Poled ceramic thickness vibration; many elements	Spherical bowl (focus at centre of curvature); only type producing both impulses and narrowband bursts
Pneumatic	Projectile strikes probe	Rigid waveguide to stone (intracorporeal)

- Piezoceramic is most efficient where half a wavelength fits the element thickness; low frequency needs thicker elements, higher voltage and larger sources. **Phased arrays** steer the beam electronically but are costly; most sources are single-element. Typical focus <1 cm transverse, a few cm axial.

Shock wave lithotripsy (SWL)

- Shock source + table + fluoroscopy (± ultrasound). Coupling: water bath → water cushion + gel; small **air pockets** at the interface disproportionately reduce focal pressure.
- Pulse: peak positive **30–120 MPa** (~1 µs), peak negative ≈ **10 MPa** (~5 µs); centre ≈ 300 kHz; ≈ 3 J/pulse; ≈ 2000 shocks at 0.5–3 Hz.
- **Why stones break, kidneys survive**: stones are brittle (fail in tension), tissue is elastic/ductile; energy concentrated at focus.
- **Fracture sequence** (beam must be wider than stone): wave inside stone outruns fluid wave → surface wave generates shear waves converging distally → reflection and inversion at the distal face (**spallation**) → maximal tension distally → crack starts, grows by fatigue over successive shocks; surface cavitation grinds fragments smaller.
- **Bubble life cycle**: long negative phase tears water → bubble grows by inertia to ≈ 1 mm long after the 5-µs pulse → violent collapse → rebound shock; near a stone, asymmetric collapse drives a **micro-jet** into it. A 1-µm nucleus grows to ≈ 30 µm and needs ≈ 1 s to dissolve.
- **Rate**: each shock leaves bubbles needing ≈ 1 s to dissolve; faster firing → growing cavitation cloud (rectified diffusion) that **shields the stone** and injures the kidney. **60/min (1 Hz) or slower** beats 120/min.
- Poor fragmentation: calcium oxalate monohydrate, cystine, brushite; stone density **>1000 HU** (Hounsfield units). Lowest morbidity of SWL/ureteroscopy/percutaneous nephrolithotomy (PCNL), but renal trauma and re-treatment occur.

	SWL	Burst wave lithotripsy (BWL)	Ultrasonic propulsion
Status	Established	Investigational	Investigational
Waveform	Single broadband shock	Narrowband, 5–100 cycles, 200–1000 kHz	~1500 cycles, ≤3-s bursts
Peak pressure	30–120 MPa	5–8 MPa	≤2.4 MPa
Rate	≤1 Hz best	≥10 Hz (bubbles dissolve in 1/16 s)	50% duty
Guidance / anaesthesia	Fluoroscopy; sedation or general anaesthesia	Real-time ultrasound; none	Ultrasound; none
Action	Break	Break; fragments uniform, ∝ wavelength	Move (radiation force)

- BWL: internal reflections amplify negative pressure >5×; operator pauses if a white cavitation cloud appears. Propulsion: RCT — repositioning residual fragments reduced relapse at 4 years.

> **KEY POINT** Acoustic therapy is management of the negative-pressure phase: tension cracks brittle stones and spares ductile tissue, but the same tension nucleates bubbles that either destroy tissue deliberately (histotripsy) or shield the stone and injure the kidney (SWL fired too fast). Frequency is the lever — raise it for heat, lower it for cavitation.

- **Shock wave therapy** (Peyronie disease): minor injury → more pliable repair.

Intracorporeal lithotripters

Device	Mechanism	Notes
Pneumatic (ballistic)	Projectile → rigid probe → jackhammer impact; stress maximal at contact	Rigid only; cheap, safe on wall; †high retropulsion in the ureter
Ultrasonic	Piezo-driven probe vibration via waveguide (± cavitation)	†Hollow probe with suction clears fragments (PCNL); needs irrigation to cool
†**Combined** ultrasonic–pneumatic	Ballistic cracking + ultrasonic suction in one probe	Faster PCNL clearance of large/hard stones
Electrohydraulic (EHL)	Spark → cavitation, water hammer; fragments proximal face	Flexible; bubble jets strain and shear the ureter — narrow safety margin

Direct tip contact with the ureteric or pelvic wall can tear it.

Soft-tissue ablation

	High-intensity focused ultrasound (HIFU)	Cavitation histotripsy	Boiling histotripsy
Mechanism	Absorption → coagulative necrosis (>100 °C in seconds) + some cavitation	Large negative pressure → bubble cloud tears tissue	Shock heating boils focus within one burst (<100 ms)
Intensity	1500–10,000 W/cm²; ~1-s pulses	>10,000 W/cm²; one or few bursts	
Lesion	Rice-grain units, "painted"	Submicron liquefaction (mechanical, not thermal)	
Ultrasound visibility	Poor — treated looks like untreated → gaps	Echogenic, real-time	
Status	Approved: BPH, prostate cancer (transrectal)	Investigational (kidney trials; prostate fibrous tissue resists)	

Frequency trade-off: higher frequency heats the focus more but is lost along the path; nonlinear shocks forming near the focus add high-frequency components → millisecond heating. HIFU risks:

necrosis along beam path; bone or gas in path → local heating; many exposures needed. Histotripsy risks: off-target bubbles, shielding.

5 · THERMAL AND OTHER ABLATION

	Radiofrequency ablation (RFA)	Microwave ablation (MWA)	Cryoablation
Energy	AC **375–900 kHz**; Joule heating in tissue (probe itself not hot)	**915 MHz / 2.45 GHz**; dielectric hysteresis (water dipoles realign)	**Joule–Thomson** argon expansion → <−100 °C at probe
Target	**60 °C**; avoid 150 °C (char ↑ impedance → incomplete)	Higher temperatures, larger zones; shaft cooling	**<−40 °C**; **two freeze–thaw cycles**, passive thaw
Heat sink	Susceptible	Less susceptible	—
Monitoring	Temperature probes	—	Visible ice ball (ultrasound or computed tomography)
Use	Small renal masses (oncological outcomes RFA ≈ cryo; MWA ≈ RFA); cryo also focal prostate therapy		

- **Freeze–thaw injury**: extracellular ice → hypertonicity → cell dehydration → intracellular ice → membrane rupture → on thaw, water rushes in → swelling and bursting; delayed apoptosis and microvascular ischaemia. Two cycles repeat both insults.
- RFA: probe at lesion centre; heat conducts outward spherically; temperature probes follow a time–temperature plan. MWA: generator → coaxial line → antenna in tumour; early concerns (collecting-system absorption, inconsistent zones) not borne out — safety and efficacy ≈ RFA.
- **Heat sink**: perfused kidney and adjacent large vessels carry heat away → incomplete ablation; MWA heats faster and wider.
- †**Irreversible electroporation (IRE)**: non-thermal; short high-voltage direct-current pulses create permanent membrane nanopores → apoptosis; spares vessels, ducts and urethra (collagen scaffold) — no heat sink. Needs general anaesthesia with full neuromuscular blockade and electrocardiogram (ECG)-synchronised pulses (arrhythmia risk). Focal prostate cancer.
- **Water vapour thermal therapy** (Rezum, BPH): RF stores energy in steam (103 °C) injected into the transition zone via 18-G needle; releases **latent heat** on condensing, spreads by convection → tissue ≥70 °C in a 9-s injection; no char; stays within the zonal compartment.
- **Aquablation**: robot-guided high-velocity saline jet, ultrasound-mapped; mechanical, non-thermal; thin rim of swollen cells.

Prostate ablative energies compared

Modality	Energy	Mechanism	Distinguishing feature
KTP/LBO photovaporisation	532-nm laser	Photothermal vaporisation	Haemoglobin absorption; anticoagulation continued
Holmium/thulium enucleation	Mid-infrared laser	Photothermal incision	Shallow penetration, precise
Monopolar / bipolar TURP	RF electrosurgery	Loop resection	Irrigant decides hyponatraemia risk
Rezum	RF-generated vapour	Convective and latent heat	No char; zone-confined
Aquablation	High-pressure saline	Mechanical, non-thermal	Robotic, ultrasound-mapped
HIFU	Focused ultrasound	Thermal (+ cavitation)	Transrectal; BPH and cancer
Histotripsy	Focused ultrasound	Mechanical liquefaction	Investigational

KEY POINT Viva synthesis: current density explains electrosurgery; pulse duration explains laser lithotripsy; penetration depth (chromophore) explains soft-tissue lasers; negative pressure (frequency, rate) explains SWL and histotripsy; and in direct-contact modalities the tissue boundary, not absorption depth, limits the lesion.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
TFL vs low-power Ho:YAG	RCT, ureteroscopy for renal stones	TFL: higher stone-free rate, shorter time, less bleeding	TFL beats 30-W holmium
TFL vs high-power Ho:YAG	RCT vs 120-W pulse-modulated Ho:YAG	Stone-free equal; TFL less retropulsion; Ho:YAG less energy	Modern holmium ≈ TFL; comparator explains discordance
Ultrasonic propulsion of residual fragments	RCT	Fewer relapses (growth, urgent visit, surgery) at 4 years	Clearing residual fragments matters

7 · MCQ ONE-LINERS

1. 10 bar = 1 MPa.
2. Electrosurgical generator frequency ≈ 500 kHz; too fast to depolarise nerve and muscle.
3. Coagulation waveform: interrupted, ~6% duty cycle, higher voltage; spread up to 3 mm.
4. Cut waveform: continuous, low voltage — clean vaporisation, little char.
5. Commonest electrosurgical injury site: partially detached dispersive pad.
6. Capacitive coupling: current induced across intact insulation; detected by active electrode monitoring.
7. Monopolar TURP irrigant must be ion-free: 1.5% glycine.
8. Obturator jerk risk greatest during lateral-wall TURBT; prevent with paralysis or obturator block.
9. Ho:YAG 2100 nm; TFL 1940 nm; KTP 532 nm; Nd:YAG 1064 nm.
0. Optical penetration: TFL 0.08 mm, Ho:YAG 0.31 mm, KTP 0.8 mm, Nd:YAG 10 mm.
1. Ho:YAG and TFL lithotripsy are photothermal (pulse >250 µs).
2. Moses effect: vapour channel transmits mid-infrared energy to the stone.
3. Retropulsion rises with pulse energy, not frequency; falls with longer pulse.
4. Dusting: low energy, high frequency.
5. Pulsed dye laser 504 nm: haemoglobin acts as energy sink.
6. GreenLight chromophore: haemoglobin; anticoagulation can continue.
7. CO_2 laser cannot be used intracorporeally.
18. SWL rate 60/min fragments better and injures less than 120/min.
19. Electrohydraulic lithotripter: spark at first focus of ellipsoidal reflector.
20. Piezoelectric source: spherical bowl, focus at centre of curvature.
21. SWL stone fracture initiates at the distal face (spallation).
22. Avoid SWL within 24 h of ultrasound microbubble contrast.
23. Stone density >1000 HU predicts poor SWL fragmentation.
24. Heating rises with frequency; cavitation with lower frequency and is a threshold effect.
25. HIFU lesion ≈ 1 wavelength wide × 10 long; poorly visible on ultrasound.
26. Histotripsy: mechanical liquefaction, echogenic in real time.
27. RFA target 60 °C; carbonisation at 150 °C stops current.
28. MWA less affected by heat sink than RFA.
29. Cryoablation: argon Joule–Thomson; <−40 °C, two freeze–thaw cycles.
30. Chemohyperthermia heats bladder wall to 42 °C with mitomycin C.
31. Rezum: latent heat of condensing steam, 103 °C, 9-s injections.
32. Advanced bipolar sealer: tissue-response generator adjusts output as clamped tissue desiccates.
33. Radiation force = power ÷ wave speed; basis of ultrasonic stone propulsion.
34. Best GreenLight vaporisation: 4 mm/s sweep, ≤2 mm working distance.

20 Complications of Urologic Surgery

Reading time ≈ 28 min (slow pace)

1 · CLASSIFICATION AND CONTEXT

- Minimally invasive surgery (MIS: laparoscopic, robotic) carries every open complication plus its own (access, pneumoperitoneum); presentation is often **atypical and late**. Complications rise with complexity and fall with experience, but absolute numbers may rise as surgeons take harder cases.
- Laparoscopic risk factors: complex procedures (partial nephrectomy, cystectomy), operating time **>4 h**, creatinine >1.5 mg/dL.
- **Clavien–Dindo**: recommended uniform system; grades the **treatment required, not the event**. Limitations: no early/late distinction, omits intraoperative events, may not suit paediatrics; poor concordance at grade ≤3 for percutaneous nephrolithotomy (PCNL).

Grade	Definition	Examples / traps
I	Deviation without drugs or intervention	Antiemetics, analgesics, diuretics, electrolytes, physiotherapy; **wound opened at bedside**
II	Other drugs	**Blood transfusion**, total parenteral nutrition (TPN)
IIIa / IIIb	Surgical, endoscopic or radiological intervention	Split by **anaesthesia**: IIIa without, IIIb under general
IVa / IVb	Life-threatening, intensive care	IVa single organ (**dialysis**); IVb multiorgan; central nervous system bleed/stroke counts, transient ischaemic attack does not
V	Death	—
Suffix "d"	Present at discharge	Flags follow-up; not a grade

> **KEY POINT** The same urinoma is grade I if it settles with observation, IIIa if drained by IR under local anaesthesia, and IIIb if a stent is placed under general anaesthesia — the grade reflects the response.

2 · POSITIONAL AND NEUROMUSCULAR COMPLICATIONS

- Pressure × time: nerve compressed against table or body; obesity worsens both. ≈ 1% laparoscopic, up to ≈ 6% robotic (steep Trendelenburg endangers all four limbs). Most recover spontaneously.

Position	Nerve	Deficit
Lithotomy	Common peroneal (stirrup at fibular neck)	Foot drop, weak eversion
Lithotomy	Posterior tibial (knee on stirrup)	Weak plantarflexion, numb sole
Exaggerated lithotomy	Obturator (hip hyperflexion)	Weak adduction
Lateral	Brachial plexus (dependent arm); peroneal (dependent leg)	Arm weakness; foot drop — axillary roll
Supine	Brachial plexus (abduction >90°); radial (supination), median (pronation)	Wrist drop; weak grip
Arms tucked (robotic)	Ulnar	Claw hand — pad hands, wrists straight
Prone	Lateral femoral cutaneous; anterior tibial (plantarflexed foot); ilioinguinal (firm bolster)	Numb thigh; foot drop

Operative nerve injury

- **Genitofemoral** (psoas hitch, ≈ 5%) → "psoas syndrome" (scrotal/medial thigh numbness, pain): place hitch sutures **along** the tendon, not across. Femoral nerve may be mistaken for tendon in thin patients.
- **Obturator** (pelvic lymph node dissection, PLND): clip only with nerve in view; remove clips on it. Transection → **tension-free 7-0 non-absorbable** epineural repair (open or robotic) + early intensive physiotherapy.
- Ilioinguinal (inguinal orchidectomy). Intercostal T10–T12 (flank incision): layered closure, avoid bulky single layer.

Compartment syndrome and rhabdomyolysis

- **Lower-limb**: 1 in 3500 overall, up to **1 in 500** radical cystectomies; lithotomy, long cases, obesity, peripheral vascular disease, hypotension, blood loss, muscular calves. Extreme leg pain → exclude deep vein thrombosis (DVT) by duplex, measure pressures; fasciotomy if **>30 mmHg** and symptomatic; >50 mmHg for 3–4 h → irreversible necrosis. Prevent: lower legs every 2 h, avoid excessive elevation and ankle dorsiflexion.
- **Gluteal**: obese, long operations; buttock tenderness, swelling → orthopaedic consult, early fasciotomy prevents rhabdomyolysis, creatine kinase surge and kidney injury.
- †Rhabdomyolysis management: serial creatine kinase, aggressive isotonic fluids to high urine output, correct hyperkalaemia, avoid nephrotoxins; dialysis for refractory acute kidney injury (AKI).

3 · URINE LEAKS AND URINARY TRACT INJURY

After partial nephrectomy

- Second only to haemorrhage. Risk: size, endophytic, hilar, long warm ischaemia and operating time. Presents ≈ day 13 (ileus, fever, pain after transperitoneal MIS).
- Diagnosis: **drain creatinine** (confirms, not site); **computed tomography (CT) urography with delayed images**. Trap: early high drain output after MIS may be redistributed blood/irrigant.
- Prevention: retrograde dilute methylene blue via ureteric catheter; watertight collecting-system closure before renorrhaphy.

ALGORITHM
Conservative drainage → Foley (lower bladder pressure) → ureteric stent + retrograde pyelogram → drain off suction, withdraw (siphon) → interventional radiology (IR) drainage of collections → nephrostomy / ureteroscopic fulguration + fibrin sealant → nephrectomy (last resort)

Most heal; prolonged leaks may take >80 days. Rule: the leak heals once the path of least resistance runs down the ureter.

Ureteric injury

- Commonest intraoperative consult: hysterectomy <1%, colorectal ≈ 1%; most unrecognised. Late recognition → urinoma, sepsis; ≈ 10× nephrostomy need. Risk: radiotherapy, malignancy, adhesions, extensive retroperitoneal lymph node dissection.
- Danger sites: gonadal vessels/lower pole, **pelvic brim** (best place to find it), under uterine vessels, behind seminal vesicles in robotic prostatectomy. Prophylactic stents: no robust evidence.

Intraoperative injury	Repair
Needle through wall; limited thermal	Remove suture; stent
Partial tear	4-0/5-0 absorbable over stent (4–8 weeks)
Transection, upper two-thirds	Spatulated stented ureteroureterostomy
Transection, lower ureter	Ureteroneocystostomy + psoas hitch
Extensive loss	Boari flap; long upper → ileal ureter; buccal graft

- Principles: tension-free, spatulated, stented, drained. Indocyanine green (ICG) checks end perfusion.
- **Delayed**: CT with delayed images. Endoscopic stent or nephrostomy first; repair **≤7 days** if endoscopy fails, otherwise divert and repair **after 6 weeks**; uncontrolled sepsis → repair now. Drain every urinoma.
- **Ureterovaginal fistula**: continuous leakage; **double-dye test** (oral phenazopyridine + intravesical methylene blue): orange pad = ureterovaginal, blue = vesicovaginal (VVF), both = both. Usual repair: reimplantation.
- Strictures: endoscopy if short; surgery if >2 cm or failure.

Bladder injury and anastomotic leak

- Intraoperative: two-layer absorbable closure, catheter 7–10 days. Postoperative: CT cystography; intraperitoneal → repair; small extraperitoneal → catheter.
- Robotic radical prostatectomy (RARP): posterior bladder-neck "buttonhole"; identify ureteric orifices when difficult (prior resection, large median lobe).
- **Urethrovesical leak** (up to ≈ 15%, days 7–10): rising drain, falling catheter output; cystogram. Risk: bleeding (haematocrit fall >6), obesity, wide bladder neck, salvage surgery.

ALGORITHM
Patent catheter, **never traction** → side-fenestrated catheter → drain away from anastomosis, off suction → IR drainage → low intermittent suction on catheter → nephroureteral stents / nephrostomy → formal repair of major disruption **before day 7** (later, tissue will not hold sutures)

Sequelae: bladder neck contracture, incontinence.

4 · BLEEDING AND VASCULAR INJURY

- Haemorrhage = commonest intraoperative complication (≈ 2%) and commonest reason for conversion (≈ 1.5%).
- **Response**: control stress, alert team (massive transfusion), pressure, pause, state plan. Benchmark: control <1 min.

ALGORITHM
Raise pneumoperitoneum cautiously (gas embolism) → suction, expose → sponge pressure → add ports → identify every injury → clips judiciously (failed clips obstruct rescue) → **suture repair** → low threshold to convert + vascular surgeon

- Venous tears: insufflate to 20–25 mmHg (≤10 min), suture, return to 12–15 mmHg; external iliac vein — stop sequential compression devices.
- **Access injury** (≈ 0.1%): blood in Veress → stop insufflating; **leave trocar in place** (tamponade) and convert via midline. Commonest vessel at umbilicus: **left common iliac vein**. Veress 45° at normal body mass index (BMI), 90° obese; atherosclerotic aorta → left upper quadrant entry.
- **Carbon dioxide (CO_2) embolism**: hypoxia, hypotension, falling end-tidal CO_2, "mill-wheel" murmur → desufflate → 100% oxygen → **Trendelenburg + left lateral decubitus** → central line aspiration.
- Insufflation 15–18 mmHg for trocar insertion, then 12–15 mmHg. Veress injury to aorta or inferior vena cava (IVC) may self-limit (inspect); trocar injuries almost always need repair; through-and-through injuries only open.
- **Epigastric vessels**: lateral ports ≥6 cm from midline, transilluminate; repair with fascial closure device figure-of-eight.
- Major vessels: repair, not ligate (only internal iliac, inferior mesenteric ligatable). Superior mesenteric artery mistaken for left renal artery (lateral position) — mortality up to 75%; check medial/anterior course.
- Stable non-expanding retroperitoneal haematoma → observe; expanding → open repair.

- **Stapler failure** (≈ 1%, renal hilum; thick tissue): see whole jaw, no clips in jaws, leave room for rescue stapler.
- **Partial nephrectomy**: clamp short right renal vein; most haemorrhage after discharge — **pseudoaneurysm / arteriovenous fistula** (≈ 2–3%, 2–3 weeks): haematuria, haematocrit fall → **selective angioembolisation** (≈ 100% success, estimated glomerular filtration rate preserved); high suspicion → direct angiography.

5 · BOWEL INJURY

- ≈ 0.2%; up to two-thirds unrecognised. **Post-laparoscopic presentation**: single trocar-site pain out of proportion, **leucopenia**, diarrhoea, ileus at 24–48 h, often no peritonitis. CT with oral, intravenous ± rectal contrast; free gas normal ≤7 days.
- Most delayed injuries are **thermal**: monopolar spread >20 mm; stray current via insulation defect or capacitive coupling. Use cold scissors for adhesiolysis; energised tip always in view.

Injury	Management
Small bowel tear; limited colon, no spillage	Primary repair
Extensive, mesenteric, doubtful viability	Resection + anastomosis
Colonic with faecal soiling	Colostomy, irrigation, antibiotics
Thermal burn	Lembert oversew — never conservative; extensive → resect (damage exceeds visible)
Rectal (prostatectomy, ≈ 0.5–1%; salvage ≈ 7%)	Debrided multilayer closure, air leak test, omental interposition, drain, antibiotics; irradiated or missed → **diverting colostomy**

- Duodenum (right nephrectomy): even <5 mm injuries perforate late. Pancreatic tail (left): high-amylase drain fluid → drainage, bowel rest; obvious injury → distal pancreatectomy.
- **Port-site hernia**: 12-mm ports, extraumbilical; close fascia ≥10–12 mm.

6 · VENOUS THROMBOEMBOLISM (VTE) AND LYMPHOCELE

- Symptomatic VTE ≈ 1% (nephrectomy, prostatectomy), ≈ **5% radical cystectomy**; most **after discharge**. Each transfused unit ↑ risk ≈ 7%.
- **Caprini** score: 1 point age 41–60, BMI >25; 2 points age 61–74, **malignancy**, major or laparoscopic surgery >45 min; 3 points age ≥75, personal/family VTE, thrombophilia; 5 points stroke, arthroplasty, fracture. Very low 0, low 1–2, moderate 3–4, **high ≥5**.

Risk	Prophylaxis
Very low (transurethral, anti-incontinence)	Early ambulation
Low	Intermittent pneumatic compression (IPC)
Moderate	Low-molecular-weight heparin (LMWH), low-dose unfractionated heparin or IPC
High	Heparin + IPC; high bleeding risk → IPC until haemostasis secure
High-risk cancer surgery	Plus **extended LMWH ≈ 4 weeks** after discharge

- Pharmacological start 2–12 h preoperatively; within 48 h does not increase bleeding.
- **Lymphocele** (commonest PLND complication; extended dissection doubles): asymptomatic <5 cm → observe; symptomatic (leg oedema, frequency) → percutaneous drainage; recurrent → **laparoscopic marsupialisation**. Avoid aspiration (recurs, infects); infected → antibiotics + drainage; exclude urine leak. Prevention: clip or seal visible lymphatics; peritoneal flap after RARP.

7 · DISCLOSURE AND THE SECOND VICTIM

- †**Open disclosure**: promptly tell the patient and family what happened, express regret, explain consequences and corrective plan, document, and report to institutional safety systems.
- **Second victim syndrome**: almost half of providers; distress, isolation, medicolegal fear, impaired cognition for months — itself a safety risk. Support: confidential peer forums, trusted seniors, mindfulness.
- Prevention matters most: selection, positioning, planning, meticulous technique.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Preventer	Randomised, perioperative heparin after prostatectomy	No reduction in symptomatic VTE	Routine heparin not mandatory for low-risk prostatectomy
Extended LMWH meta-analysis	Randomised trials, abdominopelvic cancer surgery	≥1 month LMWH cut VTE, no extra bleeding	Extend after cystectomy

9 · MCQ ONE-LINERS

1. Clavien–Dindo grades the treatment needed, not the event.
2. Transfusion = grade II; bedside wound opening = grade I.
3. Dialysis = IVa; transient ischaemic attack excluded.
4. IIIa vs IIIb split by general anaesthesia.
5. Lithotomy foot drop: common peroneal at fibular neck.
6. Psoas hitch sutures across the tendon → genitofemoral entrapment.
7. Obturator nerve transection: tension-free 7-0 non-absorbable repair.
8. Leg compartment syndrome: up to 1 in 500 cystectomies; fasciotomy >30 mmHg.
9. Drain creatinine confirms urine leak but not its site.
0. Best place to identify the ureter: crossing the common iliac at the pelvic brim.
1. Delayed ureteric injury: repair within 7 days or after 6 weeks.
2. Orange pad = ureterovaginal; blue = vesicovaginal fistula.
3. Never apply traction on catheter for anastomotic leak.
14. Commonest vessel injured at umbilical access: left common iliac vein.
15. CO_2 embolism: left lateral decubitus + Trendelenburg.
16. Suspected trocar vascular injury: leave trocar in place.
17. Ligatable major vessels: internal iliac, inferior mesenteric.
18. Post-partial nephrectomy pseudoaneurysm: angioembolisation.
19. Post-laparoscopic bowel injury: trocar-site pain, leucopenia, no peritonitis.
20. Free gas normal up to 7 days after laparoscopy.
21. Thermal bowel injury: always oversew; never conservative.
22. Radical cystectomy symptomatic VTE ≈ 5%, mostly after discharge.
23. Caprini high risk ≥5; malignancy scores 2.
24. Extended LMWH ≈ 4 weeks after pelvic cancer surgery.
25. Recurrent lymphocele: laparoscopic marsupialisation.

21 Urologic Considerations in Pregnancy

Reading time ≈ 27 min (slow pace)

1 · PHYSIOLOGICAL CHANGES

System	Change	Urological relevance
Cardiovascular	Systemic vascular resistance ↓ (progesterone, placenta); cardiac output ↑ 30–50%; heart rate ↑ up to 30%	Uterus compresses the inferior vena cava (IVC) → **supine hypotension**. Anaesthesia removes compensation → **15–20° left tilt** for every procedure
Respiratory	Diaphragm up 4 cm; functional residual capacity ↓ to 80%; O_2 use ↑; respiratory alkalosis	Rapid desaturation, difficult airway
Haematological	Blood volume ↑ 40–50% → dilutional anaemia; mild thrombocytopenia; hypercoagulable	Venous thromboembolism (VTE) risk ~10×; white blood cell (WBC) count up to 12,200 (30,000 in labour) → **WBC unreliable for infection**
Renal	Glomerular filtration rate (GFR) **↑ 40–65%**; creatinine ≈ 0.8 mg/dL; sodium ↓ 4–5 mmol/L; kidneys ↑ 1 cm	A "normal" non-pregnant creatinine may mean impairment
Urine	Glycosuria common; protein up to 300 mg/24 h; calciuria doubles	Persistent proteinuria → exclude pre-eclampsia, diabetes, renal disease, urinary tract infection (UTI)

- VTE prophylaxis: compression stockings and pneumatic compression; low-dose unfractionated heparin (UFH) in selected high-risk women (reversible with protamine).
- **Microscopic haematuria** on dipstick in up to 20% (uterus elevates trigone, bladder vascularity ↑). Usually benign. Pregnancy-specific causes: **pre-eclampsia** and **placenta percreta**.

2 · PHYSIOLOGICAL HYDRONEPHROSIS OF PREGNANCY

- **Mechanism**: early hormonal (progesterone, relaxin relax smooth muscle; begins at 6–10 weeks, before the uterus is large) then mechanical (uterus compresses the ureter at the pelvic brim). Mechanical compression is probably primary: dilatation is **right-predominant** (right ≈ 15 mm vs left 5 mm) and rare below the brim. †The left ureter is cushioned by the sigmoid colon and the right is crossed by the dextrorotated uterus and right ovarian vein.
- Peaks in second trimester; resolves postpartum (up to 12 weeks).
- Grade correlates poorly with pain. Dilatation alone never proves obstruction.
- **Spontaneous renal fornix rupture**: rare, first pregnancy, third trimester; pain only; diagnosed on imaging; ↑ preterm labour.
- **Symptomatic hydronephrosis without stone**: conservative first (analgesia, positioning, intravenous fluids). In a randomised trial, stenting relieved pain better but caused stent symptoms, and few conservative patients crossed over. Reserve stent or percutaneous nephrostomy (PCN) for infection, failing renal function or genuine obstruction.

	Physiological hydronephrosis	Obstructing calculus
Side	Right-predominant	Left = right
Level	Stops at pelvic brim	Anywhere, including distal ureter
Doppler resistive index (RI)	Unchanged	Raised
Drainage	Only for infection/failing function	Urgent if septic

3 · URINARY TRACT INFECTION

- **Asymptomatic bacteriuria (ASB)**: ≥1 species at **≥10^5 colony-forming units (CFU)/mL** without symptoms; 2–10% of pregnancies.
- Why treat in pregnancy only: stasis in the dilated tract lets bacteria ascend. Untreated ASB → **pyelonephritis 15–45%**; treatment lowers this to **1–4%** and reduces preterm birth and low birth weight.
- **Screen all** (US Preventive Services Task Force, Infectious Diseases Society of America (IDSA)): urine culture at **12–16 weeks** or first antenatal visit. Positive = clean-catch >100,000 CFU/mL of one uropathogen.
- Treat for **4–7 days** (single dose clears less). †Repeat culture after treatment (test of cure) is common practice.
- **Acute pyelonephritis** (mostly third trimester): admit → parenteral antibiotics until improved → image if obstruction suspected or slow response → oral culture-directed agent to complete **≥14 days** → low-dose prophylaxis in selected women. Risks: preterm labour, sepsis, respiratory distress.

Drug	Status in pregnancy
β-Lactams (ampicillin, cephalexin)	**Safe throughout**
Nitrofurantoin	Use if penicillin-allergic; **stop at 35 weeks** (neonatal haemolytic anaemia)
†Fluoroquinolones	†Avoid (fetal cartilage)
†Trimethoprim / sulfonamides	†Avoid trimethoprim in first trimester (folate antagonist); sulfonamides near term (kernicterus)
†Tetracyclines	†Contraindicated (teeth, bone)
Morphine	Colic: frequent small doses
Non-steroidal anti-inflammatory drugs (NSAIDs), ketorolac	**Contraindicated**: third-trimester ductus arteriosus closure, fetal pulmonary hypertension
α-blockers, calcium channel blockers	Off-label; counsel
OnabotulinumtoxinA	Only if benefit clearly outweighs risk (animal toxicity)
Intravesical bacillus Calmette–Guérin (BCG), mitomycin C	Defer until after delivery

- The US Food and Drug Administration **Pregnancy and Lactation Labeling Rule (PLLR, 2015)** abolished categories A, B, C, D, X. Labels now give a narrative risk summary, data and clinical considerations, so each drug needs individual evidence review.

4 · IMAGING IN PREGNANCY

- Fetal risk depends on **gestational age × dose**. First trimester (organogenesis) = teratogenic risk; later exposure mainly raises childhood cancer risk.
- American College of Obstetricians and Gynecologists (ACOG): **ultrasound (US) and magnetic resonance imaging (MRI)** are the techniques of choice. Computed tomography (CT), radiography and nuclear studies are far below harmful doses and **must not be withheld** if needed. Keep cumulative exposure **≤50 mGy**.
- **Gadolinium**: only if it significantly improves diagnosis and outcome. Iodinated contrast crosses the placenta but is not teratogenic in animals. Technetium-99m mercaptoacetyltriglycine (^{99m}Tc-MAG3): fetal dose <5 mGy.
- Fluoroscopy: shield, collimate, pulsed low-dose; place the lead apron **beneath** the patient because the source is below the table.

Modality	Role	Notes
Renal–bladder US	**First line**	No radiation; cannot reliably separate physiological from obstructive dilatation
Doppler RI / transvaginal US	Adjunct	RI rise (unaffected by pregnancy) = acute obstruction; transvaginal US finds distal ureteric stones. Ureteric jets unreliable
MRI (half-Fourier single-shot turbo spin-echo (HASTE)), no gadolinium	Second line	Stone seen as filling defect; small stones missed; shows level of obstruction
Low-dose non-contrast CT	Second line	High positive predictive value; shows other pathology

5 · SURGERY AND ANAESTHESIA

- No anaesthetic agent is teratogenic at standard doses. **Never deny or delay necessary surgery**, in any trimester; postpone elective surgery until after delivery. The **second trimester** is safest (least risk of abortion and preterm contractions).
- Obstetric consultation, fetal heart rate monitoring, corticosteroids if viable preterm, VTE prophylaxis; operate where caesarean and neonatal care are available. Left lateral tilt.
- Laparoscopy: place ports under vision by fundal height; insufflation 10–15 mmHg; monitor end-tidal CO_2.
- Perinatal loss is driven by the disease, not the operation.

6 · UROLITHIASIS IN PREGNANCY

- 1:200–1:1500 pregnancies; **commonest non-obstetric cause of admission**. Presents in the second/third trimester; left = right. Mostly **calcium phosphate** stones.
- **Lithogenic**: GFR ↑ → hypercalciuria, hyperuricosuria; placental calcitriol ↑ gut calcium absorption; urine pH ↑; stasis. **Protective**: ↑ citrate, magnesium, glycosaminoglycans and urine output. Net stone rate is barely changed, but stents **encrust fast**.
- Presentation: flank pain + haematuria; often misdiagnosed as appendicitis or abruption. Always send urine culture.

ALGORITHM
History, urinalysis + culture, blood count, metabolic panel → renal–bladder US → uncertain: MRI or low-dose CT → controlled symptoms: hydration + analgesia (50–80% pass) → septic obstruction: **urgent stent** (or PCN) → failed conservative care: **ureteroscopy (URS) + laser** (non-complex stones) → postpartum definitive care if drained

	Ureteric stent	PCN	URS
Nature	Temporising	Temporising	Definitive
Exchange	**Every 4 weeks** (4–8 weeks reported)	**Every 3 weeks**	—
Problems	Encrustation, UTI, migration, pain	Dislodgement, occlusion, more preterm labour	Preterm labour in third trimester; radiation
Later	Definitive treatment postpartum	Definitive treatment postpartum	Usually none

- Why practice has moved to early URS: repeated exchanges add infection and morbidity, and drainage still leaves a stone. URS complications match non-pregnant rates and URS is acceptable in any trimester. Use neuraxial anaesthesia where possible.
- **Shock wave lithotripsy is contraindicated**. Percutaneous nephrolithotomy (PCNL): defer until after delivery, except in expert centres.

7 · UROLOGICAL MALIGNANCY IN PREGNANCY

- Urological cancers are rare (≈ 13 per million pregnancies). Symptoms are often attributed to pregnancy, so diagnosis is delayed.
- **Renal cell carcinoma**: commonest urological tumour. Characterise with MRI or contrast CT (CT preferred when enhancement is needed, since gadolinium is controversial). Surgery as in non-pregnant women, timed to trimester; laparoscopy is feasible in any trimester; nephrectomy can be done at caesarean. Avoid targeted therapy and immunotherapy.
- **Angiomyolipoma**: pregnancy predisposes to rupture (hormones, vascular volume, abdominal pressure). Stable → selective **embolisation**; unstable → surgery.
- **Bladder cancer**: mostly non-muscle-invasive. Haematuria is mistaken for vaginal bleeding, so obtain a catheterised specimen. Transurethral resection is safe at any gestation. **Defer intravesical therapy** until after delivery. Muscle-invasive disease: chemotherapy and cystectomy timed to trimester.

8 · PLACENTA ACCRETA SPECTRUM AND URINARY TRACT INJURY

Grade	Depth
Accreta	Attached to myometrium
Increta	Invades myometrium
Percreta	Through serosa into bladder, ureters, rectum

- Main risk factor: **prior caesarean** (scar implantation). Blood loss 3–5 L; maternal mortality up to 7%. Haematuria in ≈ 25% of percreta.
- Diagnosis: grey-scale US (standard); MRI maps invasion of bladder and ureters. Pre-delivery cystoscopy: **do not biopsy**.
- Plan: tertiary multidisciplinary team → caesarean hysterectomy at **≈ 34 weeks** with placenta left in situ. Place **retrograde ureteric catheters** to identify the ureters and recognise injury. With detrusor invasion, perform **partial cystectomy** rather than dissecting the placenta–bladder plane (haemorrhage). Genitourinary injury ≈ 20–29%.
- †**Bladder injury at routine caesarean** (usually the dome, risk ↑ after previous caesarean): recognise it intraoperatively (methylene blue instillation, haematuria), repair in two layers, catheter drainage 7–10 days. †Ureteric injury: stent if partial; reimplant if transected near the bladder.

9 · PREGNANCY AFTER UROLOGICAL RECONSTRUCTION

- Fertility is usually normal (myelomeningocele (MMC) ≈ 70%, bladder exstrophy ≈ 68% successful). Risks: febrile UTI, preterm delivery, worsening hydronephrosis. Obtain operative records, urodynamics and upper tract studies, and plan multidisciplinary care.
- **Augmentation / continent diversion**: the augment pedicle stays anterior to the uterus (conduit pedicle displaced laterally). Catheterisable channels may become hard to catheterise as the uterus stretches them (coudé catheter, urethra, or temporary indwelling catheter). Hydronephrosis worsens at **20–28 weeks**. Pyelonephritis up to 60% after augmentation; most treat ASB, prophylaxis for the highest risk. Use **serum human chorionic gonadotrophin (hCG)**: urine tests are falsely positive after contact with bowel mucosa.
- **Mode of delivery**: augmentation with an intact bladder neck → vaginal delivery is not contraindicated. Bladder neck reconstruction or artificial urinary sphincter → caesarean is preferred by most. Ureterosigmoidostomy → vaginal delivery risks anal sphincter damage and incontinence. At caesarean, use a **high midline incision** to avoid the reservoir and its pedicle, with a urologist present.
- **Bladder exstrophy**: early, parity-independent pelvic organ prolapse; **planned caesarean** is standard. Heritability ≈ 1 in 70.

- **MMC**: latex allergy (up to 60%); ventriculoperitoneal shunt malfunction or infection; transmission 1–8% (15% if both parents affected); folic acid **4–5 mg** daily from 3 months before conception to 12 weeks (routine 0.4 mg).
- **Spinal cord injury**: screen for ASB; **autonomic dysreflexia** → decompress the bladder.
- **†Renal transplant**: wait ≥1 year with stable graft function. Stop mycophenolate before conception (teratogenic); tacrolimus, azathioprine and steroids are acceptable. The pelvic graft rarely obstructs labour, so vaginal delivery is usual; caesarean is for obstetric reasons.

10 · MCQ ONE-LINERS

1. GFR rises 40–65% in pregnancy; creatinine falls to ≈ 0.8 mg/dL.
2. Hydronephrosis of pregnancy is right-predominant and stops at the pelvic brim.
3. Doppler resistive index is unchanged by physiological hydronephrosis.
4. Screen ASB by culture at 12–16 weeks.
5. Untreated ASB: 15–45% pyelonephritis; treated: 1–4%.
6. ASB treatment course: 4–7 days.
7. Pyelonephritis: total antibiotic course ≥14 days.
8. Nitrofurantoin stopped at 35 weeks: neonatal haemolytic anaemia.
9. NSAIDs contraindicated: premature ductus arteriosus closure.
0. First-line imaging for suspected stone: renal–bladder US.
1. Second line: MRI without gadolinium or low-dose CT.
2. Maximum cumulative fetal dose (ACOG): 50 mGy.
3. Commonest pregnancy stone: calcium phosphate.
14. Stent exchange every 4 weeks; nephrostomy every 3 weeks.
15. Shock wave lithotripsy is contraindicated in pregnancy.
16. Ureteroscopy acceptable in any trimester; third trimester risks preterm labour.
17. Safest trimester for non-obstetric surgery: second.
18. PLLR (2015) replaced FDA letter categories.
19. Commonest urological tumour in pregnancy: renal cell carcinoma.
20. Defer intravesical BCG/mitomycin until after delivery.
21. Percreta main risk factor: previous caesarean.
22. Percreta delivery: planned caesarean hysterectomy at ≈ 34 weeks.
23. After bowel augmentation, use serum hCG for pregnancy testing.
24. Bladder exstrophy: planned caesarean delivery.
25. Myelomeningocele mothers: folic acid 4–5 mg daily.
26. Caesarean after reconstruction: high midline incision.

22 Intraoperative Consultations

Reading time ≈ 15 min (slow pace)

1 · DEFINITION AND PRINCIPLES

- A call from the primary surgeon to a colleague for unanticipated additional surgery, iatrogenic injury or unexpected findings. Role ranges from advice → operative help → full takeover.
- The consultant has no prior relationship, the patient is unconscious and the specific change of plan was never consented → protect the absent patient without blaming the colleague.
- Injuries recognised **at surgery** are far simpler to manage than those found postoperatively. Calling for help reflects integrity; the consultation is never the time for critique.

2 · ETHICS AND DECISION MAKING

- Accepting the call creates a patient–physician relationship; disclose the reason for consultation fully to family/surrogate.
- Stakeholders: primary surgeon, anaesthesiologist, family or surrogate, urology colleagues — the patient cannot take part.
- Four factors in act-now vs temporise: **patient condition, procedure morbidity, long-term consequences, reversibility**.

	Act now	**Temporise and defer**
Example	Low-grade ureteric injury → stent	Complex proximal high-grade injury
Morbidity / reversibility	Low / reversible	High / often irreversible
Action	Proceed; disclose afterwards	Await shared decision making (SDM)

- Exception: complex repair now if deferral (second anaesthetic, harder reoperation) is the greater harm.
- No surrogate or emergency → act in best interest. Surgeons disagree → default to primary surgeon unless harmful; escalate to department chair. The urologist ultimately decides the urological problem.

3 · CONSULTATION TIME-OUT

ALGORITHM
Assess field first → introductions → patient identity → **current condition** (anaesthesiologist, surgeon) → procedure → instruments and fluids → **estimated blood loss and added time** → concerns → verbal agreement by all

It follows assessment, because the procedure cannot be named until the field is seen.

4 · URETHRAL INJURY

- **Male**: catheter injury — balloon inflated in urethra or false passage; commonest with benign prostatic enlargement plus inexperienced operator; also stricture, prior surgery, radiotherapy.
- Diagnosis: failed catheterisation, blood at meatus → cystoscopy.

ALGORITHM
One blind attempt by urologist → cystoscopy + guidewire, catheter over wire → suprapubic catheter if unsafe

- **Female**: sling trocar/mesh or dissection injury in pelvic reconstruction; found at cystoscopy. Vaginal repair → interpose peritoneum, omentum or labial fat against fistula.

5 · BLADDER INJURY

- Most commonly injured organ in pelvic surgery; retropubic tape **3–9%** (vs transobturator 0.5%) → routine cystoscopy. Laparoscopic hysterectomy and hernia repair exceed open.
- Transurethral resection of bladder tumour (TURBT) is the commonest source: extraperitoneal perforation → catheter drainage; intraperitoneal → usually open repair.

Grade	Bladder injury (AAST)
I	Contusion, haematoma, partial-thickness
II	Extraperitoneal <2 cm
III	Extraperitoneal ≥2 cm or **intraperitoneal <2 cm**
IV	Intraperitoneal ≥2 cm
V	Into trigone/bladder neck/orifice, any size

AAST = American Association for the Surgery of Trauma.

- Signs: visible laceration or Foley, urine in field, blood/gas in catheter bag. Suspected: fill with methylene blue, cystography, cystoscopy. **Suspected but not seen → cystotomy.**

ALGORITHM
Debride → close in 1–2 layers, absorbable suture → fill test → urethral catheter + drain → **7–14 days** → selective cystogram

KEY POINT Three questions: closed? drained? ureters intact? Always exclude ureteric/orifice injury, especially at the trigone — when in doubt, reimplant. No routine suprapubic tube.

6 · URETERIC INJURY

- Iatrogenic = **75–80%** of ureteric injuries; >90% **distal third**; gynaecology >50%. Only ~one-third recognised at open surgery, fewer at laparoscopy.
- Mechanisms: ligation (commonest gynaecological), laceration, crush, thermal, devascularisation, avulsion.
- Signs: urine leak, dilated ureter, dusky wall, no capillary refill. Absent haematuria does not exclude injury.

Grade	Ureteric injury (AAST)
I	Contusion/haematoma, no devascularisation
II / III	<50% / ≥50% transection
IV / V	Complete transection, <2 cm / >2 cm devascularised

Intraoperative assessment

- **Cystoscopy + retrograde pyelography** = gold standard. Efflux alone misses partial injury. Intravenous dye or direct instillation (indigo carmine, methylene blue) — dye in field = explore. One-shot intravenous pyelogram (IVP) negative in up to **60%** of injuries.

Repair by level

Principles: debride → spatulated, tension-free, mucosa-to-mucosa, absorbable suture → watertight → stent (keeps lumen patent, diverts urine) → drain → omentum/peritoneum.

Injury	Repair
Grade I–II (ligation, crush, minor thermal)	Remove ligature + stent 2–6 weeks (less stricture than nephrostomy); image after removal
Grade III–V distal	Refluxing reimplant → **psoas hitch** → **Boari flap**; avoid distal ureteroureterostomy (poor blood supply)
Grade III–V mid/proximal	Ureteroureterostomy over stent

Injury	Repair
Complex grade V	Temporise; defer ileal ureter, transureteroureterostomy, autotransplant for SDM
Ureteroscopic avulsion	Restore continuity (ureteroureterostomy, reimplant, ureterocalicostomy)

- **Unstable** or consultant lacks skill → ligate + percutaneous nephrostomy (or ureterostomy), delayed repair.
- After reimplant: catheter 1–2 weeks; stent 4–6 weeks; image to exclude stricture.
- †**Thermal injury**: damage extends beyond the visible zone → generous debridement, not simple stenting.
- **Prophylactic ureteric catheters** do not prevent injury but aid recognition; raise acute kidney injury (AKI) and cost; arrange preoperatively.

7 · RENAL, BOWEL AND VASCULAR INJURY

- Percutaneous nephrolithotomy (PCNL) is the commonest cause of renovascular injury. Venous bleeding → tamponade, clamp nephrostomy; arterial → **selective angioembolisation**, not exploration.
- †**Bowel**: small serosal/full-thickness tears → two-layer transverse closure; extensive or devascularised → resection and anastomosis; gross faecal contamination or unprepared unstable patient → general surgery, consider diversion; irrigate and give anaerobic cover.
- †**Vascular**: pressure first, call vascular surgeon, inform anaesthesia (blood, massive transfusion); never blind clamping; venous tears → sponge-stick control and fine suture.

8 · UNEXPECTED FINDINGS — PLACENTA PERCRETA

- Accreta → increta → percreta; most with placenta praevia; risk rises with caesarean rate. Posterior bladder commonest invasion site.
- Planned caesarean hysterectomy at 34–38 weeks = gold standard.
- Cystoscopy insensitive; **never biopsy** (exsanguinating haematuria).
- Intraoperative: excise all placental tissue, two-layer closure, omentum, catheter, cystogram. Ureteric involvement up to 7%; prophylactic ureteric catheters *may* reduce injury here.

9 · DOCUMENTATION

- Scrub, assess personally, write a **separate note** of observed facts — even if no injury found.
- Record mechanism only when directly observed and management-changing (cautery vs cold cut). Never "careless" or "avoidable".

10 · MCQ ONE-LINERS

1. Commonest cause of ureteric injury: iatrogenic (75–80%).
2. Commonest site: distal third.
3. Commonest gynaecological ureteric injury: ligation.
4. Gold standard ureteric assessment: cystoscopy + retrograde pyelography.
5. One-shot IVP misses up to 60%.
6. Grade IV vs V ureter: 2 cm devascularisation.
7. Intraperitoneal bladder tear <2 cm: grade III.
8. Trigonal extension: grade V bladder.
9. Bladder injury suspected, not seen: cystotomy.
0. Bladder repair catheter: 7–14 days.
11. Distal ureteric injury: reimplant, not ureteroureterostomy.
12. Reimplant short: psoas hitch, then Boari flap.
13. Prophylactic ureteric catheters: aid recognition, not prevention.
14. Unstable patient: ligate ureter + nephrostomy.
15. Commonest renovascular injury cause: PCNL.
16. Commonest bladder injury source: TURBT.
17. Highest bladder injury rate: retropubic tape.
18. Percreta commonest site: posterior bladder; never biopsy.
19. Consultation time-out follows assessment.
20. Temporise when injury is complex and irreversible.

23 Assessment of Urologic and Surgical Outcomes

Reading time ≈ 22 min (slow pace)

1 · DEFINITIONS AND FRAMEWORK

- **Outcomes research** (Agency for Healthcare Research and Quality): study of the end results of care that patients experience and care about — function and quality of life as well as mortality.
- **Quality of care** (Institute of Medicine): degree to which services increase the likelihood of desired outcomes and match current professional knowledge.

Donabedian	Structure	Process	Outcome
Definition	Context of care	What is done	Effect on patients
Examples	Volume, subspecialty training, nurse:bed ratio	Antibiotic and thromboprophylaxis; guideline-concordant care	Mortality, length of stay, readmission, quality of life
Strength	Easy, cheap	Detects problems before outcomes differ; less affected by case mix	Most meaningful to patients
Weakness	Largely non-modifiable	Valid only if evidence links it to outcome	Confounded → needs **risk adjustment**

KEY POINT Without case-mix adjustment, surgeons treating high-risk patients look worse — a disincentive to treat them.

2 · LONG-TERM END POINTS

Overall mortality

- Most objective end point. Express as a **rate over a defined period**; a crude proportion ignores time.
- **Time-to-event analysis** handles varying follow-up by **censoring**: a patient event-free at last follow-up, at study end, or after a competing event (e.g. unrelated death) contributes time until then. **Kaplan–Meier** curves display survival incorporating events and censoring.

Bias	Mechanism	Example
Lead-time	Earlier diagnosis lengthens apparent survival; death not delayed	Kidney cancer: more computed tomography (CT) detection, mortality unchanged
Length-time	Slow tumours have longer preclinical phase → over-detected by screening	Indolent prostate cancer among screen-detected cases
Stage migration (Will Rogers)	Sensitive test up-stages patients without changing outcome → every stage looks better	Prostate-specific membrane antigen–positron emission tomography (PSMA-PET)
Attribution	Cause of death misassigned	Disease-specific mortality
Detection	One arm imaged more	Retrospective progression

KEY POINT Lead time: clinical diagnosis 70, death 75 = 5 yr; ineffective screening detects at 60, death still 75 → "15 yr survival".

Disease-specific mortality

- Used because overall mortality ignores intercurrent illness. Hard to define: perioperative deaths, cardiac death on androgen deprivation therapy.
- **Attribution bias** is the greatest limitation — a known cancer is often written on the certificate even when unrelated.
- **PSMA-PET** detects recurrence at low prostate-specific antigen (PSA); earlier detection plus up-staging improves **progression-free survival (PFS)**, not overall survival.

Surrogate (proxy) end points

Prentice criteria for a valid surrogate: (1) treatment affects the true end point; (2) treatment affects the surrogate; (3) surrogate is associated with the true end point; (4) **surrogate explains the full treatment effect**. Few urological proxies meet all four.

- **Intermediate Clinical End Points in Cancer of the Prostate (ICECaP)**: PSA-based end points (biochemical failure, doubling time, nadir) did **not** track overall survival; **metastasis-free survival (MFS)** correlated strongly → valid intermediate end point for localised disease.
- **Receipt of secondary therapy** looks objective but is biased: treatment after biochemical recurrence is often discretionary.

Response Evaluation Criteria In Solid Tumours (RECIST 1.1)

- Measurable lesion ≥**10 mm** on CT (20 mm chest radiograph); node ≥**15 mm short axis**. Not ultrasound; not markers alone, except PSA response (Prostate Cancer Clinical Trials Working Group).
- 1–5 largest **target lesions**; sum of diameters followed. Original RECIST (2000) ignored nodes, MRI and PET.

Category	Target lesions
Complete response	All disappear; nodes <10 mm short axis
Partial response	≥**30%** decrease in sum vs **baseline**
Progressive disease	≥**20%** increase vs **nadir** and ≥5 mm absolute, or any new lesion
Stable disease	Neither

> **KEY POINT** Response is measured against baseline, progression against nadir; progression needs both relative and absolute increase. Observer variability → central review in trials.

3 · SHORT-TERM OUTCOMES: COMPLICATIONS

Complication = any deviation from the expected postoperative course. Systems: Common Terminology Criteria for Adverse Events (CTCAE; clinician, organ-based, oncology drugs), patient-reported PRO-CTCAE (tolerability, not dose decisions) and Clavien–Dindo (surgery).

CTCAE grade	Definition
1	Mild; observation only
2	Moderate; minimal intervention; limits **instrumental** activities of daily living (ADLs)
3	Severe; hospitalisation; limits **self-care** ADLs
4	Life-threatening; urgent intervention
5	Death

Clavien–Dindo classification

Grades by the **therapy needed**, not perceived severity — this reduces subjectivity. Dindo (2004) added anaesthesia type and organ failure/ICU.

Grade	Definition	Example
I	No drug/surgical/radiological treatment; antiemetics, analgesics, diuretics allowed; wound opened at bedside	Ileus after cystectomy managed with fluids
II	Other drugs, incl. **blood transfusion**	Transfusion after nephrectomy
IIIa	Intervention **without** general anaesthesia	Percutaneous drainage of pelvic collection
IIIb	Intervention **under** general anaesthesia	Unplanned return to theatre
IVa	Life-threatening, ICU; **single-organ** dysfunction (incl. stroke, not TIA)	Myocardial infarction after percutaneous nephrolithotomy

Grade	Definition	Example
IVb	**Multiorgan** dysfunction	Renal failure + respiratory distress
V	Death	—

TIA = transient ischaemic attack.

- Limitations: residual subjectivity (rectal injury graded I or III); anaesthesia type varies by institution; multiple complications not captured.

4 · PREDICTING COMPLICATION RISK

Three patient factors predict complications across surgery: **functional status, comorbidity, frailty**; adjust for them in every non-randomised comparison.

ECOG	Status	Karnofsky
0	Fully active	100
1	Restricted strenuous activity; light work	80–90
2	Self-care, no work; up >**50%** of waking hours	60–70
3	Limited self-care; bed/chair >50%	40–50
4	Completely disabled, confined	10–30
5	Dead	0

ECOG = Eastern Cooperative Oncology Group; Karnofsky Performance Scale (KPS) is clinician-scored 100–0.

- **Comorbidity** (Feinstein) drives treatment choice → confounder. Classic: administrative data showed higher mortality after transurethral resection of the prostate (TURP) than open prostatectomy; adjusting for comorbidity removed the difference.
- **Charlson Comorbidity Index** (predicts 1-year mortality): weight 1 — myocardial infarction, heart failure, vascular, pulmonary, diabetes; 2 — hemiplegia, renal disease, any malignancy; 3 — moderate–severe liver disease; **6 — metastatic cancer, AIDS**.
- **Frailty**: multisystem loss of reserve. **Fried phenotype**: frail = **≥3 of 5** — weak grip, exhaustion, slow walk, low activity, unintentional weight loss. Essentially sarcopenia. Modified Frailty Index (11 items) validated in urological cancer surgery.
- **Pain**: visual analogue scale = 10 cm line scored 0–100.

5 · PATIENT-REPORTED OUTCOMES (PROS)

- Clinicians consistently **underestimate** urinary, sexual and bowel dysfunction → collect directly from patients. A symptom score counts what is wrong; a health-related quality of life (HRQoL) instrument measures function plus **bother**.
- Items below 8th-grade reading level, one question each; Likert format usual.

Property	Meaning	Threshold
Internal consistency	Items agree on one concept	Cronbach's α **>0.7**; >0.9 = redundancy
Test–retest	Reproducible over weeks	r >0.7
Face/content validity	Experts judge items appropriate	—
Criterion validity	Correlates with gold standard	r ≥0.7
Construct validity	Convergent and divergent behaviour	—

Property	Meaning	Threshold
Responsiveness	Detects meaningful change	Minimal clinically important difference: AUA Symptom Index **3.1**; else ≥½ standard deviation (SD)

Validity is population-specific — revalidate before use in a new group. **Patient-Reported Outcomes Measurement Information System (PROMIS)**: free, computer-adaptive item banks.

Area	Key instruments
Male lower urinary tract symptoms	**International Prostate Symptom Score (I-PSS)**: 7 American Urological Association (AUA) items, 0–35, plus 1 quality-of-life item; BPH Impact Index (benign prostatic hyperplasia)
Incontinence	ICIQ (4 items); IIQ-7/UDI-6; OAB-q (100 = better); King's Health Questionnaire; I-QOL (impact only). Pad tests cannot separate stress from urgency
Male sexual	**International Index of Erectile Function (IIEF)**, 15 items = gold standard; **SHIM** = 5 erection items, no bother
Female sexual	**Female Sexual Function Index (FSFI)**, 19 items, 6 domains — most accepted
Generic HRQoL	SF-36, SF-12, EQ-5D
Cancer	EORTC QLQ-C30, FACT-G; prostate: **EPIC**, UCLA-PCI; bladder: Bladder Cancer Index, FACT-BL
Other	Chronic Prostatitis Symptom Index; O'Leary–Sant (interstitial cystitis)

ICIQ = International Consultation on Incontinence Questionnaire; OAB = overactive bladder; EORTC = European Organisation for Research and Treatment of Cancer; EPIC = Expanded Prostate Cancer Index Composite.

- Costs: direct, indirect (lost wages), intangible (suffering).

6 · STUDY DESIGN AND STATISTICS BASICS †

Level †	Evidence (Oxford-style)
1	Systematic review/meta-analysis of randomised controlled trials (RCTs); high-quality RCT
2	Lower-quality RCT; prospective cohort
3	Case–control; retrospective cohort
4	Case series
5	Expert opinion

- †Randomisation balances known and unknown confounders — hence risk adjustment is critical in non-randomised data.
- †Type I (α) error = false positive (usually 0.05); type II (β) = false negative; power = 1 − β (usually 0.8).
- †Hazard ratio (Cox proportional hazards) compares event rates over time; number needed to treat = 1/absolute risk reduction.
- †Sensitivity = true positives/all diseased; specificity = true negatives/all non-diseased; predictive values depend on prevalence.

7 · MCQ ONE-LINERS

1. Donabedian triad: structure, process, outcome.
2. Censoring handles loss to follow-up in survival analysis.
3. Lead-time bias: survival lengthens, death not delayed.
4. Length-time bias over-detects indolent tumours.
5. Will Rogers phenomenon = stage migration.
6. PSMA-PET improves PFS by stage migration, not overall survival.
7. Greatest limitation of disease-specific mortality: attribution bias.
8. Prentice criterion 4: surrogate explains full treatment effect.

9. ICECaP valid intermediate end point: metastasis-free survival.
0. RECIST partial response: ≥30% decrease from baseline.
1. RECIST progression: ≥20% from nadir plus ≥5 mm.
2. RECIST 1.1 node measurable: ≥15 mm short axis.
3. CTCAE 2 vs 3: instrumental vs self-care ADLs.
4. Clavien II includes blood transfusion.
5. Clavien IIIa vs IIIb: general anaesthesia.
6. Clavien IVa vs IVb: single vs multiorgan dysfunction
7. ECOG 2 ≈ KPS 60–70.
18. Charlson weight 6: metastatic cancer, AIDS.
19. Fried frailty: ≥3 of 5 criteria.
20. Comorbidity confounded TURP vs open prostatectomy mortality.
21. Cronbach's α >0.7 good; >0.9 redundant.
22. AUA Symptom Index minimal important difference: 3.1 points.
23. I-PSS: 7 items scored 0–35.
24. SHIM: 5 IIEF items, no bother.
25. FSFI: 19 items, 6 domains.
26. OAB-q: 100 = better.

★ Putting it together: everyday decisions

Reading time ≈ 6 min (slow pace) · Integrates Ch 1–23 · quick-reference tables for ward, theatre and viva

A · WHICH IMAGING TEST FOR WHICH QUESTION?

Clinical question	First test	Notes
Suspected stone / renal colic	Low-dose non-contrast computed tomography (CT)	Ultrasound (US) in pregnancy and children; plain film (KUB) to follow radio-opaque stones
Haematuria, high risk	CT urography + cystoscopy	Low/intermediate risk: cystoscopy ± renal US (→ *Ch 8*)
Renal mass characterisation	Multiphase renal CT or magnetic resonance imaging (MRI)	Enhancement >15–20 Hounsfield units (HU) = suspicious
Obstruction vs dilatation; split function	Mercaptoacetyltriglycine (MAG3) diuretic renogram	Dimercaptosuccinic acid (DMSA) for cortical scars and split function (→ *Ch 7*)
Prostate cancer before biopsy	Multiparametric MRI	Prostate-specific membrane antigen (PSMA) PET for staging and recurrence
Acute scrotum, scrotal mass	Scrotal US with Doppler	Never delay exploration for suspected torsion
Urethral stricture / trauma	Retrograde urethrogram ± voiding cystourethrogram	Urethral diverticulum: T2 MRI
Pregnant patient	US	Then MRI without gadolinium, then low-dose CT (→ *Ch 21*)

B · BEFORE THEATRE: THE SAFETY CHECKLIST

ALGORITHM

Risk: functional status, comorbidity, frailty; cardiac risk assessment.

Medicines: warfarin stopped 5 days before (INR <1.5), with bridging only for high thrombotic risk; direct oral anticoagulants need no bridging (stop 48 h before, longer if renal impairment or high bleeding risk); metformin withheld on the day of surgery (→ *Ch 11*).

Infection: urine culture before urinary tract instrumentation; procedure-specific antibiotic prophylaxis.

Clots: mechanical ± pharmacological prophylaxis by risk; extended 28-day low-molecular-weight heparin after pelvic cancer surgery.

Theatre: World Health Organization (WHO) checklist; padded positioning (lithotomy → peroneal nerve, compartment syndrome); diathermy pad placement (→ *Ch 12*).

C · IRRIGATION FLUID AND ENERGY SOURCE

Setting	Fluid	Why
Ureteroscopy, cystoscopy, bipolar resection, laser	Normal saline	Isotonic; no hyponatraemia
Monopolar resection	Glycine 1.5% or sorbitol/mannitol	Saline would disperse the current; risk of TUR syndrome (dilutional hyponatraemia) (→ *Ch 17, 19*)
Never for resection	Sterile water	Haemolysis and hyponatraemia

D · DRAINING AN OBSTRUCTED INFECTED KIDNEY

KEY POINT An obstructed, infected kidney is a urological emergency: start antibiotics and decompress urgently. Retrograde ureteric stent and percutaneous nephrostomy are equally effective. Choose by the patient's condition (a nephrostomy avoids general anaesthesia), anatomy, coagulation and stone; send the drained urine for culture; treat the stone only after the infection has settled (→ *Ch 16*).

E · GRADING A COMPLICATION (CLAVIEN–DINDO)

Grade	Treatment needed
I	None beyond antiemetics, analgesics, diuretics, electrolytes, physiotherapy; wound opened at bedside
II	Other drugs, including blood transfusion or parenteral nutrition
IIIa / IIIb	Intervention without / under general anaesthesia
IVa / IVb	Life-threatening, intensive care: single / multi-organ dysfunction
V	Death